

10 Norway pout in North Sea, Skagerrak, and Kattegat

nop.27.3a4 – Trisopterus esmarkii in Subarea 4 and Division 3.a

10.1 General

The September 2025 assessment of Norway pout in the North Sea and Skagerrak is an update assessment based on the August 2016 ICES WKPOUT benchmark assessment (ICES WKPOUT, 2016) and an inter-benchmark assessment in July 2020 (Brooks and Nielsen, 2020). In the benchmark assessment in 2016, a new assessment model was introduced (Seasonal Stochastic Assessment Model SESAM instead of the Seasonal XSA, SXSA), the assessment year was changed (from the calendar year to 1 October to 1 October and accordingly also now including quarter 3 in the assessment year compared to quarter 2 in previous assessments), the overall assessment period was changed (cutting off the original first assessment year 1983), the plus-group in the assessment was changed (from 4+ to 3+), and the assessment tuning fleets have been changed (removing the quarter 1, 3, and 4 commercial tuning fleets and keeping the same survey fleets). The assessment biological parameter settings are the same according to the inter-benchmark assessment in spring 2012 (ICES IBPNorwayPout, 2012b) with respect to the population dynamic parameter settings for natural mortality, maturity-at-age and mean weight at age. The previous settings in the assessment were constant natural mortality by quarter and age fixed at 0.4, 10% maturity for the 1-group and 100 % mature for the 2+ group, and constant MWA assumed in stock. The new settings according to the inter-benchmark (from May 2012 onwards) include constant quarterly and yearly natural mortality, but with varying M by age, 20% maturity for the 1-group, and slightly changed levels of constant mean weight at ages in the stock which were calculated from long-term averages of mean weight at age in the catch. These parameters have impact on the predictions and estimates of the SSB because the stock consists of very few year classes. Due to the introduction of revised IBTS (International Bottom Trawl Survey) quarter 1 (Q1) and quarter 3 (Q3) indices for the full survey time-series for all age groups of Norway pout by ICES in 2020 (https://github.com/ices-tools-prod/DATRAS/tree/master/ALK_substitution), the sustainability of the MSY $B_{trigger} = B_{lim}$ and $F_{cap} = 0.7$ reference points were evaluated in the inter-benchmark in July 2020 in Brooks and Nielsen (2020). Despite only a slight change in B_{lim} of less than 10% from $B_{lim} = 39\,447\text{ t}$ (Benchmark ICES WKPOUT 2016 estimate) to $B_{lim} = 42\,573\text{ t}$ by running the benchmark assessment with the new IBTS indices (Brooks and Nielsen, 2020), the WGNSSK 2020 working group decided to switch to the new B_{lim} reference point, and on this basis to calculate a new B_{pa} reference point. The sustainability of the currently implemented $F_{cap} = 0.7$ was accordingly evaluated with this new B_{lim} reference point (Brooks and Nielsen, 2020). These inter-benchmark evaluations showed that the current F_{cap} was also sustainable with the slightly revised B_{lim} reference point (Brooks and Nielsen, 2020). See also Section 10.9.1 below and the Stock Annex under Long Term Management Strategy Evaluations. The assessment is a “real time” monitoring and management run up to 1 October 2025 and includes new information from the second half year 2024 and for the quarters 1, 2 and 3 in 2025. The assessment includes the new 3rd quarter 2025 survey information also covering the 0-group 2025-year-class information, which is used real time in the 3rd quarter. Consequently, the assessment does not backshift this survey information to the second quarter as done in the SXSA assessment run up to 1 July in the assessment year before the benchmark assessment in 2016.

Furthermore, a short-term prognosis (Forecast) up to 1 November 2025 and 1 November 2026 is given for the stock based on the assessment. The catch projection is based on a changed forecast year from 1 November to 31 October.

10.1.1 Ecosystem aspects

Norway pout is a short-lived species and most likely a one-time spawner. The population dynamics of Norway pout in the North Sea and Skagerrak are very dependent on changes caused by recruitment variation and variation in predation or other natural mortality, and less by the fishery (Nielsen *et al.*, 2012; Stock Annex). Recruitment is highly variable and influences SSB and total-stock biomass (TSB) rapidly because of the short lifespan of the species (Nielsen *et al.*, 2012; Sparholt *et al.*, 2002a, 2002b; see review in Nielsen, 2016; Stock Annex). Furthermore, 20% of age 1 is estimated mature and is included in the SSB (Lambert *et al.*, 2009). Therefore, the recruitment in the year after the assessment year influences the SSB in the following year. Also, Norway pout is to a limited extent exploited from age 0. Only limited knowledge is available on the influence of environmental factors, such as temperature, on the recruitment (Kempf *et al.*, 2009; see review in Nielsen, 2016, Section 7; Stock Annex). On this basis, Norway pout should be managed as a short-lived species.

Stock definition: So far, there is no evidence for separating the North Sea component, as well as the North Sea and Skagerrak component, into smaller stock units (Lambert *et al.* 2009; Nash *et al.* 2012; Nielsen 2016; Munch-Petersen *et al.*, 2019; Stock Annex). Details on stock identity and delineation according to stock distribution and density patterns as well as maturity, spawning, spawning distribution and migration can be found in the Stock Annex, section A.1).

Ecological role: The population dynamics of Norway pout in the North Sea and Skagerrak are very dependent on changes caused by high recruitment variation and variation in predation mortality (or other natural mortality causes) due to the short lifespan of the species (Nielsen *et al.*, 2012; ICES WGSAM, 2011; ICES WGSAM, 2014; Sparholt *et al.*, 2002a, b; Lambert *et al.*, 2009). Norway pout natural mortality is likely influenced by spawning and maturity having implications for its age specific availability to predators in the ecosystem and the fishery (Nielsen *et al.*, 2012).

With present fishing mortality levels in recent years the status of the stock is more determined by natural processes and less by the fishery, and in general the fishing mortality on 0-group Norway pout is low (Nielsen *et al.*, 2012; ICES WGNSSK Reports; see review in Nielsen, 2016, Section 5; Stock Annex). There is a need to ensure that the stock remains high enough to provide food for a variety of predator species. This stock is among other an important food source for the species saithe, haddock, cod, whiting, and mackerel and predation mortality is significant (ICES-SGMSNS, 2006; ICES WGSAM, 2011; ICES WGSAM, 2014; Cormon *et al.*, 2016; see review in Nielsen, 2016, Section 6; Stock Annex). Especially high abundance of saithe predators and high stock level of northern mackerel as likely predators on smaller Norway pout are likely to significantly affect the Norway pout population dynamics. Biological interactions with respect to intraspecific and interspecific relationships for Norway pout stock dynamics and important predator stock dynamics have been reviewed and further analysed in Nielsen (2016; Section 6) and there is referred to the general conclusions here. Also, interspecific and intraspecific density patterns in Norway pout mortality and maturity have been documented (Nielsen *et al.*, 2012; Lambert *et al.*, 2009; Cormon *et al.*, 2016; see review in Nielsen, 2016, and in Stock Annex). Natural mortality levels by age and season used in the stock assessment do include the predation mortality levels estimated for this stock (ICES WGSAM, 2011; ICES WGSAM, 2014), and in the 2012 inter-benchmark assessment revised values for natural mortality have been used based on the results from Nielsen *et al.* (2012).

Ecosystem impacts of fishery: In order to protect other species (cod, haddock, whiting, saithe and herring as well as mackerel, squids, flatfish, gurnards, *Nephrops*) there is a row of technical management measures in force for the small-meshed fishery in the North Sea such as the closed Norway pout box, bycatch regulations, gear selective devices, minimum mesh size, and minimum landing size. A review of regulations on the Norway pout stock and be found in Nielsen *et al.* (2016). Benthic impacts of the Norway pout fishery have also been evaluated in Bigné *et al.* (2019).

10.1.2 Fisheries

The fishery is nearly exclusively performed by Danish and Norwegian vessels using small mesh trawls in the northwestern North Sea, especially at the Fladen Ground and along the edge of the Norwegian Trench in the northeastern part of the North Sea. Main fishing seasons are the 3rd and 4th quarter of the year with also high catches in the 1st quarter of the year especially before 1999. Recent catches in 1st quarter are relatively low. Some catch also originates from Norwegian fishery in the second quarter. Danish catches are in most recent years highest in 4th quarter. The Norway pout fishery is a mixed commercial, small-meshed fishery directed towards Norway pout as one of the target species together with blue whiting in the Norwegian fishery. Long term trends, as well as detailed changes, in fishing patterns are described in the Stock Annex. Furthermore, the Stock Annex also includes the long-term trends in average exploitation pattern. Recently, the Danish large vessel pelagic fleet fishery has been analysed in Paoletti *et al.* (2021) which also provide yearly, seasonal, and geographical fishing patterns with respect to effort allocation, catches and value of landings for the part of the Norway pout fishery conducted by this fleet for the period 2015–2020. The relative part of the total Danish catch has been increasing for the Danish large vessel pelagic fleet in recent years compared to the Danish demersal otter board fleet.

Landings have been relatively low since 2001 except for 2010 and recently in 2019–2020, and the 2005 landings were the lowest on record (Tables 10.2.1–3; Figure 10.3.5). The directed fishery for Norway pout was closed in 2005, in the first half of 2006, and in 2007 as well as in the first half of 2011 and 2012. In the periods of closures, bycatch quotas were set for Norway pout in the Norwegian mixed blue whiting fishery around 5 kt in some years, as well as in a small experimental fishery in 2007 (1 kt). In the open periods of 2008, 2009, and 2011 the fishing effort and catches have been low. Catches were above 100 kt in 2010 but have in the period 2012–2018 been well below 100 kt, while they increased again in 2019–2021 to be above 100 kt in 2020. The landings in 2021 were 72.0 kt and decreased again in 2022–2023 to around 30 kt and have been very low in 2024. The directed Norway pout fishery has been closed in 2025. The quota has not been taken in those years. The fishery has in these periods mainly been based on the 2008, 2009, 2012, 2014, 2016, 2018–2020 and 2022-year classes being above the long-term average level. The TAC was not taken in 2008–2010 and 2012–2024, while the small TAC in 2011 was taken. The lack of full quota uptake is likely due to targeting of other industrial species like sprat for which fishing costs are lower, but also high fishing (fuel) costs compared to catch rates and also bycatch regulations (mainly in relation to herring and whiting bycatch) have an impact (see also details in Nielsen *et al.*, 2016; Bigné *et al.*, 2019; Paoletti *et al.*, 2021; and the Stock Annex). The bycatch quota regulation of max 20% herring catch on trip basis, the individual quotas for the Danish fishery in general, as well as the implementation of a general herring bycatch quota in the North Sea, play a role in the uptake. Trends in yield are shown in Table 10.3.6 and Figure 10.3.5.

Bycatch of herring, saithe, cod, haddock, whiting, and monkfish at various levels in the small-meshed fishery in the North Sea and Skagerrak directed towards Norway pout has been documented in a row of studies as described in detail in the Stock Annex. Bycatches of these species have been relatively low in the recent decade, and in general, the bycatch levels have decreased in the Norway pout fishery over the years. The declining tendency of bycatch of other species in

the Norway pout fishery also appears from Table 10.2.1. However, here it can also be observed that the bycatches have increased slightly in 2019–2020 but declined again in 2021–2023.

Review of scientific documentation show that gear selective devices and technical measures in the Norway pout fishery, can significantly reduce bycatches of juvenile gadoids, larger gadoids, and other non-target species as well as herring as described in detail in the Stock Annex. Sorting grids are at present used in the Norwegian and Danish fishery (partly implemented as management measures for the larger vessels), but modification of the selective devices and their implementation in management is still ongoing. Existing technical measures such as the closed Norway pout box, minimum mesh size in the fishery, and bycatch regulations to protect other species have been maintained. Norwegian trawlers have not been allowed to fish Norway pout in the British zone (previous EU zone) in 2021–2024. From 2013 to 2020, between 23% and 50% of the annual Norwegian landings of Norway pout were caught within this British zone. The consequences of Brexit on Danish Norway pout fisheries in the British zone from 2025 onwards should also be investigated.

A detailed description of the regulations and their background can be found in Nielsen *et al.*, (2016) and in the Stock Annex.

The quality of the landings statistics in Norway and Denmark is described in the ICES WKPOUT (2016) and associated Annexes (Nielsen *et al.*, 2016; Johnsen *et al.*, 2016) as well as in the Stock Annex. The quality seems to be relatively constant during the last 20 years and of a higher quality than in the years before. From April 2020 onwards, the sampling intensity of the Danish Norway pout fishery has increased where every landing is now sampled, and the number of required samples increase with the landing weight from a minimum of 6 to a maximum of 24 per landing. From 2020, the catch at age and the individual weight estimates of the Norwegian landings of Norway pout have used samples based on a probabilistic survey method for at sea sampling (Otterå *et al.*, 2023). Trends in bycatch levels in the samples from monitoring of the Danish and Norwegian commercial Norway pout fishery should be further analysed in future benchmark assessments as the revised monitoring and sampling systems of catches may give more precise bycatch estimates.

The discard level of Norway pout in the North Sea fisheries is considered to be low (Nielsen *et al.*, 2016).

10.1.3 ICES advice

In September 2024, the advice on North Sea Norway pout was updated. Based on the estimates of SSB in September 2024, ICES classified the stock not to show full reproductive capacity by the end of 2024. The forecast calculated the catch of Norway pout from 1 October 2024 to 31 October 2025 with F scaled such that the fifth percentile of the SSB distribution one year ahead (1 October 2025) equals B_{lim} , i.e. where the probability of SSB being below B_{lim} by 1 October in the forecast year is less than 5%. The forecast resulted in a catch up to 0 kt (40 t) in the directed Norway pout fishery in the period 1 October 2024 to 31 October 2025 which corresponds to a $F_{bar(1-2)}$ of 0.001 and a SSB at 47 kt (46 796 t) by 1 October 2025. It was noted that even with a zero catch it was not possible to obtain a SSB 5th quantile that is at B_{lim} , and accordingly the advice was a zero catch (0 t) and zero fishing mortality ($F_{bar(1-2)}$ of 0.000) resulting in a SSB of 47 kt (46 831 t). This is caused by the very low recruitment in 2023 and 2024. Norway pout is a short-lived species. Recruitment is highly variable and strongly influences the spawning stock and total biomass. The default ICES approach to MSY-based management for short-lived species is an escapement strategy, i.e. to maintain SSB, with 95% probability, above B_{lim} after the fishery has taken place. The forecast is stochastic and uncertainties in the assessment and forecast are directly considered to ensure the SSB stays above B_{lim} with 95% probability according to the ICES MSY and Precautionary

Approach for short-lived species. For the implementation of the escapement strategy, which aims to maintain the SSB above B_{lim} after the fishery has taken place, SSB is calculated for quarter 4 as a proxy for SSB at spawning time (quarter 1). The B_{lim} value was adjusted in the benchmark assessment in 2016 and again in the inter-benchmark and MSE in 2020 due to changed IBTS indices (Brooks and Nielsen, 2020). The B_{lim} estimate in the 4th quarter is lower than the previous value of B_{lim} for the 1st quarter because the 0-group and many of the 1-group fish are not yet included in the estimate of SSB. The yearly catch forecast is for the period 1 October to 30 September. ICES considered that this forecast could be used directly for management purposes for the period 1 November to 31 October. In recent years, the escapement strategy has been practiced in reality in management.

The September 2024 ICES advice was according to the escapement strategy a zero catch from 1 October 2024 to 31 October 2025 as described above due to the very low 2023 and 2024 recruitment. The ICES advice in September 2023 was that with catches up to 20 583 t in the directed Norway pout fishery in the period 1 November 2023 to 31 October 2024 corresponding to a F around 0.12 taking into account a F_{cap} of 0.70 and that the 5th percentile of the spawning-stock biomass in the 4th quarter 2024 will remain above a reference level of B_{lim} (42 573 t). The SSB was expected to decline during 2023 and 2024 due to the below average 2021, 2023 and 2024 recruitment, the growth, and 20% mature as 1-group, and still considering the high natural mortality as well as the short lifespan of the stock.

According to the escapement strategy, the fishery was closed 1 January 2012 because of the well below, nearly historical low, recruitment in 2010 and 2011. A small TAC of 6 kt was set for the second half year 2011 which was taken. Based on the high recruitment in 2012, the fishery was opened again for second half year 2012. Based on the above average recruitment in 2012, 2014, 2016, 2018, 2019, 2020 and 2022, as well as a just below average recruitment in 2015 and 2017, and the low recruitment in 2021 and 2023, the fishery has remained open for all of 2013–2023 and first part of 2024. The quota uptake has been low in recent years (Nielsen *et al.* 2016). The quota uptake in 2019 was below 75%, below 80% in 2020, below 35% in 2021–2022 and below 25% in 2023.

Fishing mortality has generally been lower than the natural mortality for this stock and has decreased in recent years below the long-term average F (0.30) as estimated from the assessment in September 2024.

There is biannual information available to perform real-time monitoring and management of the stock. This can be conducted both with fishery-independent and fishery-dependent information as well as a combination of those. Real time advice (forecast) and management options for 2025 (up to 31 October) was provided for the stock in autumn 2024 as well.

ICES advises that there is a need to ensure that the stock remains high enough to provide food for a variety of predator species. It is advised that bycatches of other species should also be considered in management of the fishery. Also, it is advised that existing measures to protect other species should be maintained.

10.1.4 Management up to 2022

There is no specific management objective set for this stock. With present fishing mortality levels, the status of the stock is more determined by natural processes and less by the fishery. The European Community has decided to apply the MSY approach for short lived species in taking measures to protect and conserve living aquatic resources, to provide for their sustainable exploitation and to minimize the impact of fishing on marine ecosystems.

ICES advised in 2005 real-time management of this stock. In previous years, the advice was produced in relation to a precautionary TAC, which was set to 198 000 t in the EC zone and 50 000 t

in the Norwegian zone. On basis of the real-time management advice from ICES, EU and Norway agreed to close the directed Norway pout fishery in 2005, first part of 2006, all of 2007 and in first part of 2011 and 2012. In 2005 and 2007, the TAC was 0 in the EC zone and 5000 t in the Norwegian zone – the latter to allowed for bycatches of Norway pout in the directed Norwegian blue whiting fishery. The final TAC set for 2008 was 115 kt (EU), 116 kt (EU) for 2009, 163 kt (EU) for 2010, 8 kt for 2011, 96 kt for 2012, 323 kt for 2013, 251 kt for 2014, 328 kt for 2015, 360 kt for 2016, 346 kt for 2017, 173 kt for 2018, 137 kt for 2019, 171 kt for 2020, 255 kt for 2021, 119 kt for 2022, 117kt for 2023, and 20.6 kt for 2024 (distributed with 8.2 kt to EU, 10.3 kt to Norway, and 2.1 kt to UK), however, the TACs were not taken during this period except for the small TAC in 2011. The ICES catch advice for 2025 was according to the escapement strategy a zero catch from 1 October 2024 to 31 October 2025, and the TAC was set to 2.9 kt (distributed with 0.3 kt to EU, 0.1 kt to UK, and 2.5 kt to Norway) where the TAC may only be fished from 1 November 2024 to 31 October 2025 exclusively for by-catches. No direct fisheries for Norway pout are permitted under this quota. By-catch quota may be fished in United Kingdom and Union waters of ICES divisions 2a, 3a and ICES subarea 4 only. Accordingly, directed fishery for Norway pout was closed in this period. Bycatch regulations have sometimes been restrictive (e.g. in 2009 and 2010 mainly in relation to whiting bycatch).

In managing this fishery, bycatches of other species have been considered. Existing technical measures such as the closed Norway pout box, minimum mesh size in the fishery, selective sorting grids in the gears and bycatch regulations to protect other species have been maintained.

Long-term management strategies have been evaluated for this stock as summarized in Section 10.9.1 below and in the Stock Annex. There is here referred to those descriptions. No decision on long-term management plans is currently available for the Norway pout in area 4 and 3.a based on the identified sustainable scenarios. The stock is still managed according to the escapement strategy with a F_{cap} of 0.7 and with no lower TAC_{min} bound or upper TAC_{max} bound set. See also Sections 10.6 and 10.9 below.

An overview of recent relevant management measures and regulations for the Norway pout fishery and the stock can be found in Nielsen *et al.* (2016) and in the Stock Annex.

10.2 Data

10.2.1 Landings/catches

Data for annual nominal landings of Norway pout as officially reported to ICES are shown in Table 10.2.1. The landings equal the catches of Norway pout as discards in this small-meshed fishery is negligible (see also Nielsen *et al.*, 2016). Historical data for annual landings (catches) as provided by ICES (Working Group) are presented in Table 10.2.2, and data for national landings (catches) by quarter, year and by geographical area are given in Table 10.2.3. Total observed and predicted (by the SESAM stochastic assessment model) catches by quarter is given in Table 10.2.3a. Both the Danish and Norwegian landings (catches) of Norway pout were very low in 2005 (1.9 kt), 2007 (5.7 kt) and 2011 (6.5 kt). The landings were moderate in 2008–2009, 2012, 2014 and 2017–2018, higher in 2013 and 2015–2016, and high in 2010 (126 kt), 2019 (98 kt) and 2020 (129 kt). The landings in 2021 were moderate on 72 kt and low in 2022–2023 on respectively 36 and 27 kt, while landings were again very low in 2024 (3.9 kt). The TAC was not fully fished in any of those recent years, except for 2011. The most recent catches have been included in the assessment for the 1st to 3rd quarter 2025, which were also very low (1.1 kt). Catches for the 3rd quarter 2025 include Danish and Norwegian catches up to 15th September 2025. Catches in the last 15 days of 3rd quarter 2025 are assumed to be relatively low and no estimates on that have been included in the assessment.

10.2.2 Age compositions in landings

Age compositions were available from Norway and Denmark (except for Norway in 2007 and 2008). Catch in numbers-at-age by quarter of year is shown in Table 10.2.4. Only very few biological samples were taken from the low Norway pout catches in 2005 and 2011, as well as in first half year 2006, 2007, and 2012, and for 2024 and 2025. The data are in the InterCatch database.

As no age composition data for Norwegian landings have been provided for 2007 and 2008 because of small catches, the catch-at-age numbers from Norwegian fishery are calculated from Norwegian total catch weight divided by mean weight at age from the Danish fishery for those years. As no age composition data for the Danish landings in first half year 2010 have been sampled because of very small catches the catch-at-age numbers from Danish fishery is calculated from Danish total catch weight divided by mean weight at age from the Norwegian fishery in 2010. There were very little samples providing information on age composition from the very low Danish and Norwegian catches in 2024 and 2025. For the periods with very few samples it has been necessary to supplement them with biological information from Danish catches in the same quarters and areas in the previous years and adjacent areas of the same year as well as from survey data in adjacent areas and seasons for the Norwegian fishery. Only fishery targeting Norway pout where more than 70% of the catch is Norway pout is sampled from Danish catches, and the main catches in this period have been by-catch of Norway pout mainly in the herring small meshed fisheries. As the catch is very low in those quarters the impact of this on the output of the assessment is insignificant.

A full-scale Norway pout age reading check and otolith exchange program was made in 2018 with participation of 14 readers from seven countries (Denmark, Norway, Scotland, UK, France, Netherlands, and Germany; ICES WGBIOP, 2018). Different methods were applied for age determination of this species; whole, broken, and sectioned otoliths and images were provided of samples prepared using each method. Samples were collected during the 2016 Q3 IBTS and 2014 Q4 commercial fishing trips from ICES area 27.4.a covering the length range of the fish and considered adequately representative of the stock. Results based on sectioned otoliths were exceptional with an overall percentage agreement based on modal age of 99% and an average CV of 3% (ICES WGBIOP, 2018). For the whole and broken otoliths, the average percentage agreement based on modal age is 82%, with an average CV of 20%. There is a slight tendency for some readers to overestimate the age at modal age 0 and 1 and underestimate compared with modal age 2. The bias that existed between the primary readers from Norway and Denmark in 2016 is still apparent. These results are based only on those readers who provide age data for assessment purposes. In conclusion, there is an overall high level of agreement between readers of the Norway pout - nop.27.3a4 stock. The agreement is higher between the countries who read sectioned otoliths (Germany and UK-England) compared to those who read whole (Denmark) and broken otoliths (Denmark, Norway, and UK-Scotland; ICES WGBIOP, 2018). Further details on the age reading checks and analyses can be found in the Stock Annex, section B.2.

10.2.3 Weight-at-age

Mean weight-at-age in the catch is estimated as a weighted average of Danish and Norwegian data. Mean weight-at-age in the catch is shown in Table 10.2.5 and the historical levels, trends, and seasonal variation in this is shown in Figure 10.2.1. Mean landings weight-at-age from Danish and Norwegian fishery from 2005–2008, 2011, as well as from the Danish and Norwegian fishery in 2024 and 2025, are uncertain because of the few observations and very low catches here. Missing values have been filled in using a combination of sources, covering values from the same quarters in previous year, as well as from adjacent quarters and areas and from other countries within the same year, for the period 2005–2008, 2011, and in first half year 2010. Finally,

mean weight-at-age information from Norway has in 2011 and 2025 involved survey estimates. The assumptions of no changes in weight-at-age in catch in these years and quarters do not affect assessment output significantly because the catches in the same period were very low. Mean weight-at-age data are available from both Danish and Norwegian fishery in 2009, second half 2010, second half 2011, second half 2012, and all of 2013–2023, as well as for quarter 3 2024 for Danish fishery and quarter 4 2024 for Norwegian fishery. Relative low mean weights-at-age have been observed for age groups 1–2 in quarter 1 in 2019–2020. Danish data and age readings have been checked according to this. Very small fish were observed in this period in the Danish catches, so this is not an artefact.

Mean weight-at-age in the stock is given in Table 10.2.6. The inter-benchmark assessment in spring 2012 (ICES IBPNorwayPout, 2012b) introduced revised estimates of mean weight-at-age in the stock used in the Norway pout assessment. The background and rationale behind the revision of mean weight-at-age in the stock is described in the IBPNorwayPout report (ICES, 2012b) and primary literature (e.g. Lambert *et al.*, 2009). On this basis, the same mean weight-at-age by quarter in the stock is used for all years, and mean weight-at-age in the catch is partly used as estimator of weight in the stock. Based on the 2016 benchmark assessment (ICES WKPOUT 2016), slightly revised constant mean weight-at-age estimates in the stock were used both in the benchmark assessment and in the following update assessments for the 1-, 2- and 3-groups by quarter taken as the long-term averages from the commercial mean weight-at-age data. Data for mean weight-at-age by quarter for age 0 were kept constant as estimated in the 2012 inter-benchmark. The revised mean weight at age in the stock was applied in the assessment runs as obtained from long-term averages measured from the commercial fishery catch.

In the Stock Annex and in the ICES WKPOUT (2016) including Nielsen (2016), the benchmark revisions in 2012 and 2016 of the population dynamic parameters in the assessment are described. The estimation of mean weights-at-age in the catches and the used mean weights in the stock in the assessment is described in ICES WKPOUT (2016), Nielsen (2016) and in the Stock Annex. The data are in the InterCatch database.

10.2.4 Maturity and natural mortality

The inter-benchmark assessment in spring 2012 (ICES IBPNorwayPout, 2012b) introduced revised estimates of maturity and natural mortality-at-age used in the Norway pout stock assessment. These values were confirmed in the ICES 2016 benchmark assessment (ICES, 2016). The background and rationale behind the revision of the natural mortality and maturity parameters is described in the IBPNorwayPout report (ICES, 2012b) and primary literature (e.g. Nielsen *et al.*, 2012; Lambert *et al.*, 2009; ICES WGSAM, 2011; ICES WGSAM, 2014). In Nielsen (2016) and in the Stock Annex a summary is given of the inter-benchmark revisions of the population dynamic parameters used in the assessment where maturity and natural mortality used in the assessment is described. Proportion mature and natural mortality by age and quarter used in the assessment is given in Table 10.2.6.

The same proportion mature, and natural mortality are used for all years in the assessment. The proportion mature used is 0% for the 0-group, 20% of the 1-group and 100% of the 2+-group independent of sex. The revisions of the maturity ogive which have been implemented in the 2012 inter-benchmark assessment as well as in the present assessment is based on results from a paper by Lambert *et al.* (2009) indicating that the maturity rate for the 1-group is close to 20% in average (varying between years and sex) with an increasing tendency over the last 20 years. Furthermore, the average maturity rate for 2- and 3-groups in 1st quarter of the year was observed to be only around 95% as compared to 100% used in the assessment.

Instead of using a constant natural mortality set to 0.4 for all age groups in all seasons as used in the previous assessments, then variable natural mortality between ages have been introduced in the 2012 ICES IBPNorwayPout inter-benchmark assessment (ICES, 2012b; Stock Annex) and the present assessment. The revision of the natural mortality parameters is based on results in Nielsen *et al.* (2012) and the ICES WGSAM (2011) and ICES WGSAM (2014) multispecies assessment reports. The revised values are shown in Table 10.2.6.

10.2.5 Summary of inter-benchmark assessment on population dynamic parameters

A summary of the ICES 2012 and 2016 benchmark assessments with revised weight, maturity and natural mortality parameters at age included in the assessment is given in Nielsen (2016) and in the Stock Annex as well as in the ICES IBPNorwayPout inter-benchmark assessment report (ICES, 2012b) and the ICES WKPOUT (2016) report.

10.2.6 Catch, effort and research vessel data

Description of catch, effort and research vessel data used in the assessment is given in the ICES WKPOUT 2016 Benchmark Report (ICES, 2016) and its Annexes, in Section 10.2.6 below, as well as in the Stock Annex (see also Table 10.3.1).

10.2.6.1 Commercial fishery data

Catch information for 1984–2025 is included in this assessment as presented in tables 10.2.1–10.2.5 and Figure 10.2.1. Catches in all of 2005, 1st quarter 2009, first half year 2011 and 2012, and first quarter 2013 were nearly 0 and only very limited information exists about this catch. Consequently, there has been assumed and used low catches of 0.1 million individuals per age (for age groups 1–3) per quarter in the assessment for all of 2005. The fishing effort and catch efficiency (catch per unit effort) and of the Danish and Norwegian commercial fishery according to year and quarter of year are shown in tables 10.2.7 and 10.2.8, respectively, and according to year and fishing vessel engine horsepower category in Tables 10.2.9 and 10.2.10, respectively. Furthermore, trends herein are shown in Nielsen *et al.* (2016), in Johnsen *et al.* (2016), in Bigné *et al.* (2019), and in Paoletti *et al.* (2021).

No commercial fishery tuning fleet is included in the assessment from 2006 onwards based on the decisions made in the Norway pout benchmark assessment in September 2016 (ICES WKPOUT, 2016).

10.2.6.2 Research vessel data

Fishery-independent survey data used as tuning fleets in the present assessment is given in Table 10.2.11 and Figure 10.2.2 (see also Table 10.3.1).

Survey indices series of abundance of Norway pout by age and quarter are for the assessment period available from the IBTS (International Bottom Trawl Survey 1st and 3rd quarter) and the EGFS (English Ground Fish Survey, 3rd quarter) and SGFS (Scottish Ground Fish Survey, 3rd quarter), Table 10.2.11. The new survey data from the 1st quarter 2025 IBTS and the 3rd quarter 2024 IBTS research surveys have been included in this September 2025 assessment as well as the 3rd quarter 2025 EGFS and SGFS research survey information. The survey data time-series including the new information is presented in Table 10.2.11, as well as trends in survey indices in Figure 10.2.2. Surveys covering the Norway pout stock are described in detail in ICES WKPOUT (2016), Nielsen (2016) and in Johnsen and Søvik (2016) as well as in the Stock Annex. Survey data time-series used in tuning of the Norway pout stock assessment are described below.

From 2009 and onwards, the SGFS changed its survey area slightly with a few more hauls in the northern North Sea and a few less hauls in the German Bight. This is not evaluated to influence the indices significantly as the indices are based on weighted subarea averages.

In 3rd quarter 2015–2016 test trials were conducted in the international third quarter IBTS with 15 min duration hauls compared to 30 min duration hauls. The new 15 min test hauls have been included in the index calculation for 3rd quarter 2015–2016, and will potentially affect the Norway pout indices for the SGFS and the combined IBTS Q3 index. It has been necessary to include the 15 min hauls in the SGFS 2015–2016 data as extensive areas (of the total SGFS survey area) are only covered with this type of hauls. Only one 15 min test haul was included in the EGFS 2015 and none in 2016. There has been no continuation of the tow duration experiment in the Q3 surveys in 2017–2025 and, accordingly, no new 15 min hauls have been conducted and included in the Q3 2017–2025 SGFS and EGFS survey indices (and consequently in the combined Q3 IBTS survey index). Analyses of this are still ongoing and nothing conclusive is available at present concerning potential significant impacts of this on the indices. Preliminary analyses indicate no significant differences in catch rates of Norway pout between the 15 min hauls and the 30 min hauls in the SGFS, however, the variability is very high and there are only very few observations available. Long time-series and many observations are necessary to make statistically robust evaluations of and conclusions on potential differences.

In September 2015, the EGFS survey indices were revised as to incorporate the relevant primes within the Norway pout area following the IBTS Manual (2015), i.e. in the selection of the prime stations to be included in the Norway pout index calculation. The revision is described in detail in an ICES working document to ICES WGNSSK 2015 (Silva, 2015). This has changed the EGFS indices for Norway pout for all years and ages since 1992. Especially, the indices for the 0-group have changed significantly without any obvious trends over time. However, the perception of the dynamics in the stocks (e.g. strong year classes as 0-group and also as older ages in the cohorts) seems not to have changed in relative terms for this survey. Consequently, there is consistency in this to the previous EGFS indices and in relation to the other survey indices also for Norway pout. In the EGFS Q3 2017–2025, an additional haul has been taken (prime 77 – DATRAS haul number 147) fished on behalf of the Scottish (SGFS) that falls inside ICES rectangle 40E8 and, therefore, inside the Norway pout index area according to the IBTS manual. This prime is expected to be fished from now on by the English (EGFS) so it will fall inside the English survey index instead of the Scottish survey index. To make the EGFS time-series consistent over time it has been decided to exclude the Prime 77 haul in the 2017–2025 indices used in the assessment. By comparison, it appears that the survey trends seem similar without prime 77 in the EGFS for 2017–2025. In the 2020, 2021 and 2023, EGFS survey, all 77 prime stations were successfully fished aimed at 30 minutes tows, though with some reduced to at least 20 minute tows for operational reasons. Due to technical issues with the research vessel in the 2022 EGFS survey, not all the tows were completed as intended, completing 66 or the 77 planned tows with an additional 5 tows completed on behalf of our Scottish colleagues (SGFS survey). Of the 45 planned tows that contribute to the Norway Pout indices, 5 rectangles were missed (44F4, 45F3, 51F0, 51F1 and 51F2) without significant geographical trends. However, these tows may have been fished by other nations on the EGFS behalf, with 51F0–51F2 likely to have been picked up by UK(Scotland). The five tows completed on behalf of UK(Scotland) were fished in 40E8, 43F4, 46E9, 46E8 and 46E7. None of these have been included in the EGFS Norway pout indices calculation in order to be consistent. Additionally, of note, the station in 45E8 was shot in the incorrect rectangle (44E8) but was hauled in the correct rectangle. This too has been set as 45E8 in the data so is correctly accounted for within these indices. All 66 Prime stations and the additional 5 fished for Scotland were successfully fished, were aimed at being 30 minutes tows, though some were reduced to be either 15 or 20 minutes tows for operational reasons. In 2024, 76 of the planned tows in the EGFS were completed as intended with one station missed being an agreed “swap” between the

English and Norwegian IBTS Q3 survey. In 2025, 73 of the 78 planned tows were completed with 5 stations missed located on the northern areas of the survey grid in 49E9, 51E8, 51E9, 51F0, and 51F1. The above modifications in the EGFS are not expected to change the perception of the trends in this survey index and not expected to have significant effect on the assessment results.

With respect to the SGFS 2017 Q3 index, around 5 survey days was lost in 2017 due to vessel issues. Hence, there were only 76 hauls in 2017 compared to 99 hauls in 2016. In 2016, there was almost a 50/50 split by ICES Subarea with 50 hauls undertaken in 4A and 49 in 4B in the SGFS. In 2017, this was slightly more unbalanced with 43 hauls taking place in 4A and 33 in 4B. In 2019, there has been a slight revision of the SGFS indices from 2013–2018 because of additional data check and removal of invalid hauls. This have resulted in very slight changes. As expected, the divergence was very small and typically around 1–3% increase and obviously were dependent on how many invalid hauls were recorded during each survey year. This does not at all change the perception of the trends in this survey index and does not have significant effect on the assessment results. Also, a few invalid hauls during the 2019 survey were encountered with the result that in order to ensure that there would be no loss to the overall survey Norway covered 6 of the stations normally completed by Scotland within the most North-Easterly 2 legs of the SGFS survey. These were stations 50F0, 50F1, 50F2, 48F1, 48F2 and 48F3. In 2018, these stations accounted for around 2% of the overall Norway Pout abundance for the survey so it is expected that although not an ideal situation from the perspective of providing consistent coverage the impact of this change will be minimal. In the SGFS 2020 survey, there was only one invalid haul, and the SGFS 2021 and 2023 surveys were conducted as planned. For the 2022 SGFS survey there was a delay in MS Scotia commencing this survey (2 weeks late). Accordingly, the area covered by Scotland was not quite as originally intended with some stations being covered by other nations whereas MS Scotia was out of action. In addition, MS Scotia was not able to secure access to Norwegian or Danish waters thereby restricting the eastern extent of the survey. Despite this, the SGFS survey coverage overall in 2022 was broadly in line with previous years with regard to effort with the number of stations completed within each of the old SGFS demersal areas being broadly in line (as a proportion of total) with 2021 aside from the numbers of stations completed around the Shetland area where there was an increase in effort largely due to other nations prioritizing other stations to complete at the point when the SGFS survey was under threat of cancellation due to vessel breakdown. However, based on the SGFS survey specialists have made comparative plots of the two SGFS survey years 2021 and 2022 with respect to geographical coverage, catch rates and abundance indices and compared the two plots for both years, they conclude that the index provided in 2022 is robust despite the necessary changes effected during 2022 quarter 3 SGFS survey programme. All hauls were completed as planned in the SGFS 2023 survey, but for the SGFS 2024 survey the spatial coverage has been impacted by the fact that the German IBTS survey was cancelled at the last minute so all the remaining survey participants had to try and move things around a little in order to minimise the impact felt by the loss of one of the surveys. For Scotland this meant that they ended up completing a second repeat station on their Southern boundary at 37F0, 37F1 and 37F2 and similarly they also had an additional repeat station completed at 51E8 right on their Northern boundary. Consequently, they did not complete their 5 stations on the transect 48E9 – 48F3 as that was covered for Scotland by Norway which allowed the SGFS the time to provide much needed survey cover further South. In the 2025 SGFS survey the stations were covered as planned and most squares were as usual covered by one GOV station. However, what was non-standard was that for the 16 ICES statistical squares where Scotland was the sole surveying nation (and that requires Scotland to deploy the trawl twice) they completed one of the stations by square with the typical GOV as per the norm but for the second station it was undertaken using the new GOV replacement trawl net (JTS/BT238). Accordingly, 16 stations out of 86 stations mainly covering the North-Western part of the survey area covering 41E8, 41E9, 44E8, 46E7, 46E8, 46E9, 47E6, 48E6, 48E7, 48E8, 49E6, 49E7, 49E8, 50E7, 50E8, and 50E9. The above modifications are not expected to change the

perception of the trends in this survey index and accordingly not to have significant effect on the assessment results.

Additionally, it should be noted that in the 2014 IBTS Q1 survey, less hauls were conducted in the northern part of the North Sea than usual. This did not result in change in the perception of the stock dynamics.

From 3rd quarter 2018, the depth range of the IBTS survey has been extended to 250 m (previously 200 m). The tows deeper than 200 m are extra stations. These stations have not been included in the NP survey indices. Obviously, those additional hauls cannot be included into the standard indices before the effects are statistically robustly evaluated and before reasonable time-series and adequate number of observations are available to analyse the potential effects of inclusion of the deeper tows in the indices.

In 2020, the IBTS quarter 1 (Q1) and quarter 3 (Q3) indices have been substantially revised (https://github.com/ices-tools-prod/DATRAS/tree/master/ALK_substitution) also covering the full Norway pout index time-series for all age groups. The changes in the survey indices and their influence on assessment results as well as sustainability reference points are shown, described, and evaluated in Brooks and Nielsen (2020). See also further details in Section 10.1.4 above and Sections 10.6 and 10.9 below. Again in 2023–2024, there seem to have been extensive updates of the IBTS Q1 and Q3 survey indices for Norway pout on DATRAS, and a few have changed substantially. However, for the most recent years this is not extensive. For the above reasons, it was in the 2020 inter-benchmark assessment decided to keep the survey indices calculated with the new method constant in the period 1984–2019 as estimated in 2020 for future assessments, and only update the indices from 2020 onwards from the DATRAS IBTS Q1 and IBTS Q3 data and indices.

The survey data time-series including the new information are presented in Table 10.2.11.

10.2.6.3 Revision of assessment tuning fleets

The revision of the tuning fleets used in the benchmark 2004 assessment - as used in the 2005–2006 and 2007–2015 assessments – and the additional revisions of the tuning fleets in the benchmark 2016 assessment – as used in the September 2016 and future assessments - is summarized in Table 10.3.1. Details of the revision are described in the Stock Annex and in the ICES WKPOUT 2016 Report (ICES, 2016) and its Annexes.

The overall assessment period has been changed by cutting off the first assessment year (1983), so the assessment period is from 1984–2025, and the assessment tuning fleets have been changed by removing the quarter 1, 3, and 4 commercial tuning fleets and keeping the same survey fleets. The assessment biological parameter settings are the same according to the inter-benchmark assessment in spring 2012 (ICES IBPNorwayPout, 2012b) and in the benchmark assessment in 2016 (ICES WKPOUT 2016) with respect to the population dynamic parameter settings in the assessment for natural mortality, maturity-at-age, and mean weight at age in the stock (see also Table 10.3.1).

10.3 Assessment: Catch-at-age data analyses

10.3.1 Review of assessment

The September 2024 assessment was accepted, and similar for the 2023 assessment no major recommendations for changes and comments were given for the assessment and forecast either in the audit or the ADGPOUT 2024. However, the ADGPOUT 2024 recommended that if recruitment in 2025 is also low, similar to recruitment in the two previous years (2023–2024), then it is

recommended to only use more recent years for the forecasted recruitment values. First of all, there is nothing indicating regime shifts in recruitment to support this. Second, even with a zero catch in 2025-2026 the fifth percentile of the SSB distribution one year ahead will not reach B_{lim} , i.e. be significantly below B_{lim} , according to the forecast drawing among the full set of historical recruitments. Accordingly, the advice of a zero catch will not change by drawing from a more recent set of recruitments. Furthermore, it is necessary to have a certain time series available to estimate the 5th percentiles adequately precisely. The minutes from the ADGPOUT 2022 pointed out that the residuals of catches are larger than residuals from surveys. Also, the ADG commented in September 2022 on discrepancies between observed and model predicted catches in 2017 and 2019. This was responded to by the assessor in the draft advice in 2023: “The catch residuals are higher than the survey residuals in the assessment, but show no trends in the recent period from 2000 onwards. The observed and assessment model predicted catches and their seasonal patterns are also in good agreement in the same period, and the assessment model capture the seasonal fishing patterns well. The assessment is sensitive to the abrupt changes in the seasonal fishing patterns.” There is also good agreement in the most recent years despite the tendency for 4th quarter catches to be more dominating relative to 3rd quarter catches for the most recent years. The ADGPOUT 2023 decided to remove this from the final advice as they concluded that “it is really not a problem with the assessment.” Potential retrospective patterns in SSB and R were discussed at the ICES WGNSSK meeting in May 2018 as well in the following meetings, but no major issues and problems were pointed at, and it was concluded that the assessment has been performed correctly and performs relatively well. It should be noted that the retrospective patterns are within the uncertainty bounds of the assessment estimates for spawning-stock biomass, fishing mortality and recruitment. In the 2014 assessment review, it was only noted that potential area specific assessment should be considered in relation to a benchmark assessment.

10.3.2 Final assessment

A seasonal extension to the State-space Assessment Model (SAM) was used during this September 2025 assessment (SESAM), and in the benchmark 2016 Norway pout assessments reported in ICES WKPOUT (2016). In the latter, the SESAM assessment model was evaluated and compared with the assessment model previously used (Seasonal extended survivors analysis SXSA). It was found that the SESAM model estimates very similar trends in SSB and fishing mortality compared to SXSA. The SESAM model was preferred by the ICES WKPOUT (2016) benchmark assessment group due to its ability to incorporate process and observation error and estimate uncertainties in all quantities, including the forecast.

The method is described in detail in Nielsen and Berg (2016; WD6 of the ICES WKPOUT 2016), and the source code, input data and output is available online at www.stockassessment.org under “NorPoutBench2016”, and for the current September 2025 assessment under “NP_Sep2025_v1” at the same website. A summary description of the model and its settings are given in the Stock Annex similar to the previously used assessment models used for the Norway pout stock.

In brief, the model is the same as the SAM model, except that the time-step used is one quarter of a year rather than a full year. Recruitment is assumed to occur in quarter 3 only. The logarithm of the fishing mortality-at-age and quarter is assumed to follow a multivariate random walk with lag 4 and correlated increments, i.e. the log F-at-age in a given quarter is given by the log F-vector in the same quarter one year earlier plus a correlated noise term with mean zero.

The observation equations in SESAM are also extended to deal with zero observations (both surveys and catches), which are usually treated as missing values in SAM. This is done by introducing a detection limit for each fleet, and defining the likelihood of a zero observation to be the

probability of obtaining a value less than the detection limit. The detection limit is set to 0.5 times the smallest positive observation by fleet.

A special option is included to down-weight the influence of large jumps in $\log F$ on the estimated random walk variance due to periods where the fishery was closed. This option reduces the estimated $\log F$ process variance considerably.

In the ICES WKPOUT (2016) benchmark, a number of variants of the SESAM model were investigated and compared to the previous assessment model, SXSA. These variants included the use (or not) of commercial CPUE data, omission of the earliest years of data from the assessment, alternative settings for the detection threshold used to manage zero-valued data, and omitting the years of fishery closure when estimating the random walk variance on fishing mortality.

The final SESAM model also used in this September 2025 assessment excludes commercial CPUE data, omits 1983 data from the assessment, use age 3+-group, and omits the years of fishery closure from the random walk variance calculation. In relation to evaluation of stock sustainability and forecast, B_{lim} is set equal to B_{loss} based on quarter 4 SSB values to align with the new fishing season (1 November to 31 October). The short-term forecast is stochastic, which allows the probability of SSB being below B_{lim} to be evaluated immediately following the fishing season.

Stock indices and assessment settings used in the assessment are presented in Tables 10.3.1–10.3.2.

Results of the SESAM analysis are presented in Tables 10.3.1–10.3.2 (assessment model parameters, settings, and options), Table 10.3.3 (population numbers-at-age; recruitment), Table 10.3.4 (fishing mortalities by year and quarter), Table 10.3.5 (diagnostics), and Table 10.3.6 (stock summary). The summary of the results of the assessment are shown in Table 10.3.6 and Figures 10.3.1 (spawning-stock biomass, SSB), 10.3.2 (total-stock biomass, TSB), 10.3.3 (fishing mortality, F_{bar}), 10.3.4 (recruitment), 10.3.5 (yield, observed and predicted catches on yearly and quarterly basis), and 10.3.6–10.3.7 (stock–recruitment plots for quarter 1 and quarter 3, respectively). The retrospective patterns and the residuals from the SESAM September 2025 assessment are given in Figure 10.3.8 and Figures 10.3.9–10.3.11, respectively.

Fishing mortality has generally been lower than natural mortality and has decreased in the recent 20 years below the long-term yearly average (0.26, Tables 10.3.4 and 10.3.6). Fishing mortality for the 1st and 2nd quarter has in general decreased in recent years, while fishing mortality for 3rd and 4th quarter, that historically constitutes the main part of the annual F – and here especially 4th quarter in most recent years, has also decreased moderately during the last 20 years. Fishing mortality in 2005, first part of 2006, 2007, 2008, 2011, in first part of 2012 and in 2025 was close to zero due to the closure of the Norway pout fishery in those periods. Fishing mortality was moderate in 2009 and 2010 as well as in second half 2012 and in 2013–2024, and the TACs have not been fished up in any of these recent years. In recent years, the quota uptake has been below 35% (see Nielsen *et al.*, 2016), and in 2019 the quota uptake was below 75%, below 80% in 2020, below 35% in 2021–2022, and below 25% in 2023 and 2024. The low TAC of 6 kt in 2011 was taken in second half year resulting in a very low F in 2011, and only around 50% of the very low TAC of 2.9 kt in 2025 set for by-catch of Norway pout has so far been taken.

Spawning-stock biomass (SSB) has since 2001 decreased continuously until 2005 but has in later years increased due to the strong 2008, 2009, 2012, 2014, 2016, 2018, 2019 and 2020 year classes as well as the about long-term average recruitment in 2022, and the lowered fishing mortality. The stock biomass fell to B_{lim} in 2005 which until 2025 was the lowest level ever recorded. By 1 January 2007 and 2008 the stock was at B_{pa} (= $MSY B_{escapement}$; i.e. at increased risk of suffering reduced reproductive capacity), while the stock by 1 January 2009–2012 and by 1 January 2014–2024 was above B_{pa} (i.e. the stock show full reproductive capacity). The stock was well above B_{pa} by end of

2024 but has drastically decreased to be below B_{lim} by end of 2025 being the lowest SSB ever recorded because of the very low recruitment in 2023, 2024 and 2025.

The recruitment in 2010 was very low and at the same level as the low 2003- and 2004-year classes where these three-year classes were the lowest on record since the mid-1980s. The recruitment in 2008, 2009, 2012, 2014, 2016, 2018, 2019 and 2020 was high. Recruitment in 2011 and 2013 was also very low, and the recruitment in 2015, 2017, 2021 and 2023 was below long-term average (51 billion), but because of the strong 2012, 2014, 2016, 2018, 2019- and 2020-year classes and the about average 2022 year class the SSB has been well above B_{pa} (= $MSY B_{escapement}$) by 1 January 2014–2025 even with a high yearly TAC in 2014–2022 (up to 3rd quarter) considering growth, high natural mortality, and 20% maturation at age 1. The 2023-, 2024- and 2025-year classes have been very weak with recruitment at similarly low levels as observed in 2003–2004 and 2010 below 10 billion, reducing the spawning stock biomass to be below B_{lim} by the end of 2025. The 2021 recruitment was about half of the long-term average (51 billion), while the strong 2020 and average 2022 recruitment kept the stock well above B_{pa} up to the end of 2024.

10.3.3 Comparison with 2015–2024 assessments

The final, accepted September 2015 SXSA assessment run was compared to the inter-benchmark May 2012 and the update September 2014 and May 2014 Scenario 2 SXSA assessments. The result of the comparative runs between the September 2015 and the September 2014 and May 2014 assessments are shown in the ICES WGNSSK 2015 Report. The resulting outputs of these assessments were shown to be identical giving similar perception of stock status and dynamics.

The WKPOUT 2016 benchmarking comparison of the SESAM and SXSA May 2014 assessments are presented in the ICES WKPOUT 2016 Report (ICES, 2017). The overall conclusions were that the two assessments give the same perception of stock dynamics with respect to abundance (SSB) and recruitment over time. There was some variability of the estimates of fishing mortality especially in the middle of the assessment period, however, the SXSA estimates lies within the confidence intervals of the SESAM estimates of fishing mortality.

The retrospective analysis based on the SESAM September 2025 assessment is shown in Figure 10.3.8. There is a tendency towards the retrospective analyses do not fully converge although being at the same level and showing the same perceptions of the stock dynamics. For the latest years it converges for SSB, but for a few previous years to this the convergence is not as high. No strong retrospective patterns are observed for SSB, and the Mohns rho values are below the threshold of ± 0.3 for short lived species. For the most recent years the convergence is high for SSB (Mohn's rho -12%) and for all years the retrospective patterns are within the confidence limits of the estimates for both SSB, F and R, except for F in 2024 (Figure 10.3.8). There is a tendency that the assessment overestimate terminal Fs (Mohn's rho 43%), and accordingly the Fs for previous years are downscaled. The catches in 2022–2025 have been low, and the assessment perceives this in the fishing mortalities. The SESAM model is sensitive to changes in seasonal fishing patterns. Usually, the 4th quarter catches are higher than the 1st, 2nd and 3rd quarter catches, respectively, but the catches in 4th quarter 2023 and 4th quarter 2024 were very low compared to the catches in 3rd quarter 2023 and 3rd quarter 2024, respectively. There is a strong positive retrospective pattern for recruitment with a tendency to overestimate recruitment in the terminal assessment year (Mohn's rho 74%). This is partly due to not full consistency between in-year-0-group indices and 1-group-indices the following year for the EGFS and SGFS Q3 surveys. The current assessments have downscaled the strong 2022-year class, and the recruitment in 2023, 2024 and 2025 is very low. It should be noted that there is some difference between estimates of the B_{loss} level in the start of Q4 in 2005 between assessments.

10.3.4 Historical stock trends

The assessment and historical stock performance is consistent with previous years' assessments, i.e. the perception of stock dynamics of the SSB, the fishing mortality for ages 1 and 2 and recruitment over time are consistent. However, there is in general a tendency to overestimate recruitment in the terminal assessment year. According to the benchmark assessments, the SXSA estimates of fishing mortality is within the confidence limits of the SESAM estimates of fishing mortality, which is generally also the case for the September 2025 assessment, except for the 2023 F. In the latest assessment, the Mohn's rho on F is 43% (see further comments on this in Section 10.3.3 above and further below in Section 10.3.4).

There has been a smaller consistent decrease in $F_{\text{bar}(1-2)}$, because of the introduction of the revised IBTS Q1 and Q3 index time-series for Norway pout of all age groups in 2020 (Brooks and Nielsen, 2020). The changes are not affecting TSB (Total-stock biomass) and recruitment very much. This is because the changes have been relatively higher for the indices of the older mature age groups in the population.

Recruitment estimates

The long-term average recruitment (age 0, second quarter) is 51 billion (arithmetic mean) for the period 1984–2025 (Table 10.3.6). Recruitment is highly variable and influences SSB and TSB rapidly due to the short lifespan of the species and because 20% reach maturity as 1-group. The recruitment reached historical minima in 2003–2004, in 2010 and again in 2023, 2024 and 2025. The recruitment in 2008, 2009, 2012, 2014, 2016, 2018, 2019 and 2020 was high and about long-term average in 2022. Recruitment in 2011 and 2013 was very low, and the recruitment in 2015, 2017, and 2021 has been below long-term average (51 billion).

10.4 Short-term forecast

The short-term forecast is stochastic based on the SESAM September 2025 assessment, which allows the probability of SSB being below B_{lim} to be evaluated immediately following the fishing season. The SESAM is, like the SXSA, a quarterly based model estimating biomass at the start of each quarter of the year.

Short-term projections are conducted as follows.

1. Assume values for M , weight-at-age in the catches and in the stock, and maturity-at-age for the projection period. Since all of those quantities except weight-at-age in the catches are assumed constant over time, only weight-at-age requires special treatment. A procedure for forecasting catch weights is described in ICES WKPOUT (2016, WD6, Nielsen and Berg, 2016) but see also below and Stock Annex.
2. Draw K samples from the joint posterior distribution of the states ($\log N$ and $\log F$) in the last year with data, and the recruitment in all years.
3. Assume that $\log F_t = \log F_{t-4} + \log G_t$, for all future values of t where G_t is some chosen vector of multipliers of the F -process. If $G_t = 1$ for all t this corresponds to assuming the same level and quarterly pattern in F for all future time-steps as in the last data year.
4. Create K forecasting trajectories starting from the samples of joint posterior distribution of the states. This is done by sampling K recruitments from the vector of historic recruitments obtained in step 2, and then projecting the states forward in time using the stock equation with randomly sampled process errors from their estimated distribution.

Each forecast is based on 4000 simulations where there is drawn by random from the vector of historic recruitments. In the forecast procedure there is made a forecast for each resampled recruitment by each simulation, and the median of the outputs are used. The median of the resampled recruitment values is approximately the same as the geometric mean of the vector of

historic recruitment and represent the input recruitment in the forecast year. It should be noted that the short-term forecast only uses the observed 2025 recruitment (Q3 2025) in the SSB estimate by 4th quarter 2025. The recruits in 2026 do not become a part of SSB by 4th quarter (1 October) 2026 because they have not reached maturity yet by 4th quarter 2026 but will do that by 1 January 2027 (20% mature as 1-group here). However, the forecast is just run up to 4th quarter 2026, and the recruits in 2026 is accordingly not used (and shall not be that) in the forecast SSB estimate in Q4 2026.

5. Find G_t such that the 5th (or any other) percentile of the catches (total mass) in the projections equal some desired level such as B_{lim} (optional).

Forecasting weight-at-age in the catches

There is substantial variation in weight-at-age in the commercial catches from year-to-year, which means that usual methods of using running averages will be quite sensitive to the bandwidth of the running average. This is important since TAC estimates calculated in step 5 above depend directly on the catch weight-at-age. See also Stock Annex.

The following model is used:

$$E(\sqrt{CW_{a,q,t}}) = \mu_{a,q} + s(\text{cohort}, a) + U_t$$

where $\mu_{a,q}$ is a mean for each combination of quarter and age, $s(\cdot)$ is tensor product smoothing spline, and U_t are normal distributed random effects. The square root transform is used to achieve variance homogeneity in the residuals. See Figure 1 in ICES WKPOUT (2016, WD6, Nielsen and Berg, 2016).

The projected mean weight at ages in the catch used in the forecast are shown in Table 10.6.1.

Forecasts

The first forecast provides a TAC advice according to a calculated yield in the forecast year where the probability of SSB being below B_{lim} by 1 October in the forecast year is less than 5%, i.e. the forecast estimates the yield according to SSB that meets the 5% criterion at the B_{lim} date which is 1 October as explained in Sections 10.1.4, 10.6 and 10.9. The purpose of the first forecast is to calculate the catch of Norway pout from 1 October 2025 to 31 October 2026 with F scaled such that the fifth percentile of the SSB distribution one year a head (1 October 2026) equals B_{lim} , i.e. where the probability of SSB being below B_{lim} by 1 October in the forecast year is less than 5%. The results of the forecast are presented in Table 10.6.2 and Figure 10.6.1, and this results in a catch up to 0 kt (15 t) in the directed Norway pout fishery in the period 1 October 2025 to 31 October 2026 which corresponds to a $F_{bar(1-2)}$ of 0.0002 and a SSB at 42 kt (42 162 t) by 1 October 2026. It should be noted that even with a zero catch it is not possible to obtain a SSB 5th quantile that is at B_{lim} , and accordingly the advice will be a zero catch (0 t) and zero fishing mortality ($F_{bar(1-2)}$ of 0.0000) resulting in a SSB of 42 kt (42 171 t) by 1 October 2026.

The purpose of the second forecast is to calculate the catch of Norway pout from 1 October 2025 to 31 October 2026 with F scaled to zero. The results of the forecast are presented in Table 10.6.3 and Figure 10.6.2 resulting in no catch in the directed Norway pout fishery in the period 1 October 2025 to 31 October 2026 which corresponds to a $F_{bar(1-2)}$ of 0.0000 and a SSB at 42 kt (42 171 t) by 1 October 2026.

The purpose of the third forecast is to calculate the catch of Norway pout from 1 October 2025 to 31 October 2026 with F scaled to F status quo for previous year up to 1 October 2025. The results of the forecast are presented in Table 10.6.4 and Figure 10.6.3 where catches up to 2 kt (2 239 t)

can be taken in the directed Norway pout fishery in the period 1 October 2025 to 31 October 2026 which corresponds to a $F_{\text{bar}(1-2)}$ of 0.034 and a SSB at 41 kt (40 801 t) by 1 October 2026.

The purpose of the fourth forecast is to calculate the catch of Norway pout from 1 October 2025 to 31 October 2026 with F scaled such that the median of the SSB distribution one year a head (1 October 2026) equals B_{lim} . The results of the forecast are presented in Table 10.6.5 and Figure 10.6.4 where catches up to 0 kt (15 t) can be taken in the directed Norway pout fishery in the period 1 October 2025 to 31 October 2026 which corresponds to a $F_{\text{bar}(1-2)}$ of 0.0002 and a SSB of 42 kt (42 162 t) by 1 October 2026.

The purpose of the fifth forecast is to calculate the catch of Norway pout from 1 October 2025 to 31 October 2026 with F scaled such that the median of the SSB distribution one year a head (1 October 2026) equals B_{pa} . The results of the forecast are presented in Table 10.6.6 and Figure 10.6.5 where catches up to 0 kt (15 t) can be taken in the directed Norway pout fishery in the period 1 October 2025 to 31 October 2026 which corresponds to a $F_{\text{bar}(1-2)}$ of 0.0002 and a SSB of 42 kt (42 162 t = B_{pa}) by 1 October 2026. It should be noted that even with a zero catch it is not possible to obtain a SSB median that is at B_{pa} .

The purpose of the sixth forecast is to calculate the catch of Norway pout from 1 October 2025 to 31 October 2026 with F scaled to 0.2, i.e. with a $F_{\text{cap}} = 0.2$. The results of the forecast are presented in Table 10.6.7 and Figure 10.6.6 where catches up to 12 kt (12 358 t) can be taken in the directed Norway pout fishery in the period 1 October 2025 to 31 October 2026 which corresponds to a $F_{\text{bar}(1-2)}$ of 0.201 and a SSB of 36 kt (35 721 t) by 1 October 2026.

The purpose of the seventh forecast is to calculate the catch of Norway pout from 1 October 2025 to 31 October 2026 with F scaled to 0.3, i.e. with a $F_{\text{cap}} = 0.3$. The results of the forecast are presented in Table 10.6.8 and Figure 10.6.7 where catches up to 18 kt (17 846 t) can be taken in the directed Norway pout fishery in the period 1 October 2025 to 31 October 2026 which corresponds to a $F_{\text{bar}(1-2)}$ of 0.302 and a SSB of 33 kt (33 012 t) by 1 October 2026.

The purpose of the eighth forecast is to calculate the catch of Norway pout from 1 October 2025 to 31 October 2026 with F scaled to 0.4, i.e. with a $F_{\text{cap}} = 0.4$. The results of the forecast are presented in Table 10.6.9 and Figure 10.6.8 where catches up to 23 kt (22 957 t) can be taken in the directed Norway pout fishery in the period 1 October 2025 to 31 October 2026 which corresponds to a $F_{\text{bar}(1-2)}$ of 0.403 and a SSB of 31 kt (30 713 t) by 1 October 2026.

The purpose of the ninth forecast is to calculate the catch of Norway pout from 1 October 2025 to 31 October 2026 with F scaled to 0.5, i.e. with a $F_{\text{cap}} = 0.5$. The results of the forecast are presented in Table 10.6.10 and Figure 10.6.9 where catches up to 28 kt (27 704 t) can be taken in the directed Norway pout fishery in the period 1 October 2025 to 31 October 2026 which corresponds to a $F_{\text{bar}(1-2)}$ of 0.503 and a SSB of 29 kt (28 793 t) by 1 October 2026.

The purpose of the tenth forecast is to calculate the catch of Norway pout from 1 October 2025 to 31 October 2026 with F scaled to 0.6, i.e. with a $F_{\text{cap}} = 0.6$. The results of the forecast are presented in Table 10.6.11 and Figure 10.6.10 where catches up to 32 kt (32 135 t) can be taken in the directed Norway pout fishery in the period 1 October 2025 to 31 October 2026 which corresponds to a $F_{\text{bar}(1-2)}$ of 0.604 and a SSB of 27 kt (26 958 t) by 1 October 2026.

The purpose of the eleventh forecast is to calculate the catch of Norway pout from 1 October 2025 to 31 October 2026 with F scaled to 0.7, i.e. with a $F_{\text{cap}} = 0.7$. The results of the forecast are presented in Table 10.6.12 and Figure 10.6.11 where catches up to 36 kt (36 244 t) can be taken in the directed Norway pout fishery in the period 1 October 2025 to 31 October 2026 which corresponds to a $F_{\text{bar}(1-2)}$ of 0.705 and a SSB of 25 kt (25 272 t) by 1 October 2026.

According to the long-term management strategy evaluation based on the joint EU-Norway request from November 2017 and the resulting released advice by ICES in May 2018 evaluating

long-term management strategies for Norway pout in area 4 and 3.a (ICES WKNPOUT, 2018) it was shown that the current procedure for providing TAC advice for Norway pout, based on an escapement strategy where the probability of SSB being below B_{lim} by 1 October in the forecast year is less than 5% is only precautionary with the addition of an F_{cap} at 0.7. See also Sections 10.1.4, 10.6, and 10.9.

10.5 Medium-term projections

No medium-term projections are performed for this stock. The stock contains only a few age groups and is highly influenced by recruitment.

10.6 Biological reference points

As explained in the ICES WKPOUT 2016 Report (ICES, 2017), Section 3.8, the benchmark has recommended that the $B_{lim} = B_{loss}$ should be the lowest SSB estimated in quarter 4, because this is closest to the beginning of the fishing season (1 November), and would be the most appropriate to use as a B_{lim} reference point, because the probability of SSB being below B_{lim} can then be evaluated immediately after the fishing season for which a TAC is being calculated. It was argued that the quarter 4 SSB (an existing output of the SESAM model) was adequate for this purpose because any attempt to calculate an SSB corresponding to 1 November would require further assumptions and would effectively only be an interpolation between the quarter 4 and subsequent quarter 1 SSBs, thus unnecessarily complicating the calculation of the SSB. The forecast provides a TAC advice according to a calculated yield in the forecast year where the probability of SSB being below B_{lim} by 1 October in the forecast year is less than 5%, i.e. the forecast estimates the yield according to SSB that meets the 5% criterion at the B_{lim} date which is 1 October. Accordingly, it is recommended that this TAC is used for the management year 1 November–31 October. This is an approximation and will be sustainable unless radical changes occur in the seasonal fishing pattern used in the forecast. In the period between 1 October and 1 November in the forecast year there will be provided a new assessment.

In Table 10.6.13, quarterly minima of the estimated SSB time-series (1984–2016) are shown from the SESAM Benchmark Assessment Baseline Run from the Norway pout benchmark assessment in ICES WKPOUT (2016). The estimates are quarterly minima estimated at the beginning of the season. The lowest observed biomasses in the assessment period are in 2005. The estimates are B_{loss} estimates which equals B_{lim} according to the ICES WKPOUT 2016 benchmark assessment which by 1 October is $B_{lim} = 39\,450$ t (ICES, 2017). In Table 10.6.14, the same minima for the same period are shown using the new IBTS Q1 and Q3 survey indices introduced in the assessment from 2020 onwards. See also Sections 10.1.4 and 10.9. This revision provides a $B_{lim} = 42\,573$ t by 1st October (4th quarter).

The B_{lim} SSB estimate in Q4 is low because of the 0-group and many of the 1-group fish are not in the SSB yet at that time. However, in the forecast there is a change in maturity and an age-class shift by 1 January, i.e. the 0-group becomes 1-group and 20% of those become mature, and the 1-group becomes 2-group and 100% of those become mature. This is in the forecast calculated into the SSB available for spawning in 1 quarter of the forecast year.

The fishing pattern has not changed significantly in the most recent years with most important catches in the second half of the year. Accordingly, the use of B_{lim} by Q4 should be sustainable.

It should be noted that there is a tendency towards the retrospective analyses for SSB do not fully converge although being at the same level (see also Sections 10.3.3 and 10.3.4 above). It should also be noted that there is quite some difference between estimates of the B_{loss} level in the start of Q4 in 2005 between assessments.

Framework	Reference point	Value	Technical basis	Source
MSY approach	MSY $B_{\text{escapement}}$	Not defined*		
	F_{MSY}	Not defined		
	F_{cap}	0.70	A long-term management strategy evaluation, indicating that an escapement strategy for Norway pout is only pre-cautionary with the addition of an upper limit on fishing mortality = F_{cap} ($F_{\text{bar}[1-2]}$) at 0.7	Brooks and Nielsen (2020)
Precautionary approach	B_{lim}	42 573 tonnes (4 th quarter)	$B_{\text{lim}} = B_{\text{loss}}$, the lowest observed biomass in 2005 (as estimated in the updated benchmark assessment)	Brooks and Nielsen (2020)
	B_{pa}	69 736 tonnes (4 th quarter)	$B_{\text{pa}} = B_{\text{lim}} e^{0.3 \times 1.645}$	Brooks and Nielsen (2020)
	F_{pa}	Not defined		
Management plan	SSB_{MGT}	Not applicable		
	F_{MGT}	Not applicable		

* MSY $B_{\text{escapement}}$ has not been defined, as the escapement strategy uses directly the 95% probability of SSB being above B_{lim} .

No F-based reference points are advised for this stock except for an F_{cap} (see below and Sections 10.1.4 and 10.9).

Norway pout is a short-lived species and most likely a one-time spawner. The population dynamics of Norway pout in the North Sea and Skagerrak are very dependent on changes caused by recruitment variation and variation in predation (or other natural) mortality, and less by the fishery. Recruitment is highly variable and influences SSB and TSB rapidly due to the short lifespan of the species. (Basis: Nielsen *et al.*, 2012; Sparholt *et al.*, 2002a,b; Lambert *et al.*, 2009). Furthermore, 20% of age 1 is considered mature and is included in SSB (Lambert *et al.*, 2009). Therefore, the recruitment in the year after the assessment year does influence the SSB in the following year. Also, Norway pout is to limited extent exploited already from age 0. Overall, the stock is very dependent of yearly dynamics and should be managed as a short-lived species.

On this basis, advice on yield in the forecast year where the probability of SSB being below B_{lim} by 1 October in the forecast year is less than 5% is considered sustainable. That is where F is scaled such that the fifth percentile of the SSB distribution one year ahead (1 October in forecast year) equals B_{lim} . According to the long-term management strategy evaluation based on the joint EU-Norway request from November 2017 and the resulting released advice by ICES in May 2018 evaluating long-term management strategies for Norway pout in area 4 and 3.a (ICES WKNPOUT, 2018) it was shown that the current procedure for providing TAC advice for Norway pout, based on an escapement strategy where the probability of SSB being below B_{lim} by 1 October in the forecast year is less than 5% is only precautionary with the addition of an F_{cap} at 0.7 (see also Sections 10.1.4 and 10.9).

B_{pa} has been calculated from

$$B_{\text{pa}} = B_{\text{lim}} e^{0.3 \times 1.645} (\text{SD}).$$

A SD estimate around 0.3–0.4 is considered to reflect the real uncertainty in the assessment. This SD-level also corresponds to the level for SD around 0.2–0.3 recommended to use in the manual for the Lowestoft PA Software (Cefas, 1999). The relationship between the B_{lim} and B_{pa} (42 573 and 69 736 t) is 0.6.

It is obvious that the Norway pout, being a short-lived species, has no well-defined breakpoint (inflection) in the SSB-R relationship (ICES IBPNorwayPout 2012b; ICES WKPOUT, 2016) and therefore there is no clear point at which impaired recruitment can be considered to commence (i.e. SSB does not impact R negatively, and that there is a relatively high recruitment observed at B_{loss} as well as more observations above than below the inflection point).

The $B_{lim} = B_{loss} = 42\,573$ t (quarter 4) is based on the lowest observed SSB in 2005 providing good recruitment.

Revision of Reference points in 2020

Due to introduction of revised IBTS (International Bottom Trawl Survey) quarter 1 (Q1) and quarter 3 (Q3) indices for the full survey time-series for all age groups of Norway pout by ICES in 2020 (https://github.com/ices-tools-prod/DATRAS/tree/master/ALK_substitution) the long-term sustainability of the B_{lim} and $F_{cap} = 0.7$ reference points were during summer 2020 evaluated during and inter-benchmark and presented in Brooks and Nielsen (2020).

The analyses showed a slight change in B_{lim} of less than 10% from $B_{lim} = 39\,447$ t (Benchmark ICES WKPOUT, 2016 estimate) to $B_{lim} = 42\,573$ t by running the benchmark assessment with the new IBTS indices (Brooks and Nielsen, 2020).

Furthermore, Brooks and Nielsen (2020) evaluated harvest control rules (HCRs) within the escapement strategy currently practiced (aimed at retaining a minimum stock size in the sea every year after fishing) that are based on the new B_{lim} value and simulated to be restricted by a combination of an upper limit on F values (F_{cap}), different F_{max} values (between the historical observed F_{max} of 0.67, i.e. the $F_{historical}$ for the assessment using the revised IBTS data, and up to a F_{max} value of 2) as well as different TAC upper bounds (TAC_{max}) for setting the TAC. The TAC_{max} values evaluated was from 200 kt up to infinite (i.e., with no upper TAC bound). The sustainability of the current $F_{cap} = 0.7$ was evaluated through long-term management strategy evaluation simulations with the new B_{lim} reference point and according to the different F_{max} and TAC_{max} values applied as described above and detailed in Brooks and Nielsen (2020).

These evaluations showed that the currently implemented F_{cap} of 0.7 is also precautionary and sustainable with the slightly revised B_{lim} reference point (Brooks and Nielsen, 2020).

This is the case also in extremely unrealistic scenarios of an infinite TAC_{max} and with F_{max} values between 0.67 and up to 2 (Brooks and Nielsen, 2020). All scenarios for $F_{max} = 0.67$ and for a very unrealistic high $F_{max} = 1$ with infinite TAC_{max} are sustainable. Even with the unrealistically high maximum implementable F of 2 then the risk only goes above 0.05 with an $F_{cap} = 0.7$ (when rounded to the nearest 0.01 units) for the risk3.long.Q4. All other scenarios for $F_{max} = 2$ values are sustainable (Brooks and Nielsen, 2020). This means that if there were an unrealistic high F_{max} of around 1.6 which is similar to the natural mortality level for the stock then all scenarios of $F_{cap} = 0.7$ would obviously be sustainable.

The WGNSSK working group has on this basis decided to switch to the new B_{lim} reference point, and on this basis to calculate a new B_{pa} reference point, and continue with the currently implemented F_{cap} of 0.7. It should again be noted that no lower bound TAC_{min} or upper bound TAC_{max} boundaries have been implemented in the management (see also Sections 10.1.4 and 10.9).

In Table 10.6.13 quarterly minima of the estimated SSB time-series (1984–2016) according to recruitment are shown from the SESAM updated Benchmark Assessment Run (Run: NP_Sep17_fixC_Benchmark2016Data_NewIBTS, www.stockassessment.org) and with new IBTS

Q1 and Q3 survey indices for Norway pout made available in 2020 (Brooks and Nielsen, 2020) (Table 10.6.14). The estimates are quarterly minima estimated at the beginning of the season. The lowest observed biomasses providing a good year class in the assessment period are still in 2005. The estimates are B_{loss} estimates which equals B_{lim} which by 1 October is $B_{lim} = 42\,573$ t, i.e. based on the lowest observed SSBs in 2005 with high recruitment.

10.7 Quality of the assessment

The estimates of the SSB, recruitment and the average fishing mortality of the 1- and 2-group are consistent with the estimates of previous year's assessment, except that SSB has further decreased because of introduction of the survey Q1 and Q3 indices in 2025 (see Sections 10.1.4, 10.6 and 10.9). The overall perception of stock dynamics with respect to abundance (SSB), fishing mortality and recruitment over time is the same. The estimates of Mohn's Rho in the retrospective analyses are of the baseline SESAM assessment September 2025, with terminal assessment year ranging from 2018–2025, is -12% for SSB, 43% for F_{bar} , and 74% for R shown in Figure 10.3.8. However, there is a tendency that the assessment overestimate terminal F_s (Mohn's rho 43%), and accordingly the F_s for previous years are downscaled. The catches in 2022–2025 have been low, and the assessment perceives this in the fishing mortalities. The SESAM model is sensitive to changes in seasonal fishing patterns. Usually, the 4th quarter catches are higher than the 1st, 2nd and 3rd quarter catches, respectively, but the catches in 4th quarter 2023 were very low only about one third of the catches in 3rd quarter 2023, while the catches in 4th quarter 2024 was at the same level (slightly lower) than the catches in 3rd quarter 2024. Despite these tendencies, then the terminal year estimates lie within the confidence limits of the model estimates except for F in 2023 which appear from Figure 10.3.8 (see also Sections 10.3.3 and 10.3.4 above).

The assessment is considered appropriate to indicate trends in the stock and immediate changes in the stock because of the assessment considering the seasonality in fishery, use of seasonal based fishery-independent information, and using most recent information about recruitment. The assessment provides stock status and year-class strengths of all year classes in the stock up to the end of third quarter of the assessment year. The assessment method gives a good indication of the stock status the 1 October the following year based on projection of existing recruitment information in 3rd quarter of the assessment year.

10.8 Status of the stock

Based on the estimates of SSB in September 2025, ICES classifies the stock not having full reproductive capacity by end of 2025.

With F scaled such that the fifth percentile of the SSB distribution one year ahead (1 October 2026) equals B_{lim} , i.e. where the probability of SSB being below B_{lim} by 1 October in the forecast year is less than 5% catches up to 0 kt (15 t) can be taken in the directed Norway pout fishery in the period 1 October 2025 to 31 October 2026 which corresponds to a $F_{bar(1-2)}$ of 0.0002 and a SSB at 42 162 t (42 kt) by 1 October 2026. It should be noted that even with a zero catch it is not possible to obtain a SSB 5th quantile that is at B_{lim} , and accordingly the advice will be a zero catch (0 t) and zero fishing mortality ($F_{bar(1-2)}$ of 0.0000) resulting in a SSB of 42 kt (42 171 t) by 1st October 2026. This is due to the strong 2020 recruitment being above the long-term average recruitment (51 billion), a 2021 recruitment around half (25 billion), and a 2022 recruitment around average (48 billion) of the long-term average (51 billion), growth of the stock, and still taking into consideration the high natural mortality as well as the short lifespan of the stock.

Fishing mortality has generally been lower than the natural mortality for this stock and has decreased in recent years below the long-term average F (0.26). If the advised catch is taken (zero

catch), the F is expected stay well below this average for the period 1 October 2025 to 31 October 2026. Targeted fishery for Norway pout was closed in 2005, first half year 2006, in all of 2007, in first half of 2011 and 2012, and again in the last part of 2024 and in 2025, and fishing mortality and effort has accordingly reached historical minima in these periods (Table 10.3.6). The fishery was open for the second half 2006, 2011 and 2012 as well as in all of the years 2008–2010 and 2013–2023, and in first part of 2024. Here, the fishing mortality was low in 2008 and 2011 and moderate in 2009 and 2010 as well as in second half 2012 and in 2013–2023, but still below the long-term average. The TACs have not been fished up in any of these recent years. Less than 75% of the quota was taken in 2019, less than 80% in 2020, less than 35% in 2021–2022 and less than 25% in 2023 and 2024, while only around 50% of the very low TAC of 2.9 kt in 2025 set for by-catch of Norway pout has so far been taken.

The recruitment reached historical minima in 2003–2004 and again in 2023–2025, and the 1987, 2002, 2006, and 2010 year classes were weak. The recruitment in 2008, 2009, 2012, 2014, 2016, 2018, 2019 and 2020 was high well above the long-term average (51 billion). Recruitment in 2011, and 2013 was also very low, while the recruitment in 2022 was about long-term average, and the 2015, 2017, and 2021 recruitment has been below the long-term average (Table 10.3.6).

10.9 Management considerations

There are no management objectives for this stock.

From the results of the forecast presented here with a F scaled such that the fifth percentile of the SSB distribution one year ahead (1 October 2026) equals B_{lim} , i.e. where the probability of SSB being below B_{lim} by 1 October in the forecast year is less than 5% catches up to 0 kt (15 t) can be taken in the directed Norway pout fishery in the period 1 October 2025 to 31 October 2026 which corresponds to a $F_{bar(1-2)}$ of 0.0002 and a SSB at 42 kt (42 162 t) by 1 October 2026. It should be noted that even with a zero catch it is not possible to obtain a SSB 5th quantile that is at B_{lim} , and accordingly the advice will be a zero catch (0 t) and zero fishing mortality ($F_{bar(1-2)}$ of 0.0000) resulting in a SSB of 42 kt (42 171 t). This is due to the very low recruitment in 2023, 2024 and 2025 and a 2022 recruitment around average (48 billion) of the long-term average (51 billion), growth of the stock and still taking into consideration the high natural mortality as well as the short lifespan of the stock.

Norway pout is a short-lived species and most likely a one-time spawner. The population dynamics of Norway pout in the North Sea and Skagerrak are very dependent on changes caused by recruitment variation and variation in predation (or other natural) mortality, and less by the fishery. Recruitment is highly variable and influences SSB and TSB rapidly due to the short lifespan of the species. (Basis: Nielsen *et al.*, 2012; Sparholt *et al.*, 2002a,b; Lambert *et al.*, 2009). Furthermore, 20% of age 1 is considered mature and is included in SSB (Lambert *et al.*, 2009). Therefore, the recruitment in the year after the assessment year does influence the SSB in the following year. Also, Norway pout is to limited extent exploited already from age 0. Overall, the stock is very dependent of yearly dynamics and should be managed as a short-lived species.

There is a need to ensure that the stock remains high enough to provide food for a variety of predator species. Natural mortality levels by age and season used in the stock assessment reflect the predation mortality levels estimated for this stock from the recent multispecies stock assessment performed by ICES (ICES WGSAM, 2014; 2011; ICES SGMSNS, 2006). Biological interactions with respect to intraspecific and interspecific relationships for Norway pout stock dynamics and important predator stock dynamics have been reviewed and further analysed in Nielsen (2016; Section 6) and there is referred to the general conclusions here.

Existing technical measures such as the closed Norway pout box, minimum mesh size in the fishery, and bycatch regulations to protect other species have been maintained. An overview of

recent relevant management measures and regulations for the Norway pout fishery and the stock can be found in Nielsen *et al.* (2016) and in the Stock Annex.

Historically, the fishery includes bycatches especially of haddock, whiting, saithe, and herring. Existing technical measures to protect these bycatch species should be maintained or improved. Bycatches of these species have been relatively low in the recent decade, and in general, the bycatch levels of these gadoids have decreased in the Norway pout fishery over the years. The declining tendency of bycatch of other species in the Norway pout fishery also appears from Table 10.2.1. However, here it can also be observed, that the bycatches have increased slightly in 2019 and 2020 compared to previous years and to 2021–2024 where bycatches were again low. From April 2020 onwards, the sampling intensity of the Danish Norway pout fishery has increased where every landing is now sampled, and the number of required samples increase with the landing weight from a minimum of 6 to a maximum of 24 per landing. This new sampling system may give more precise estimates of bycatches which should be evaluated in future benchmark assessments.

Sorting grids in combination with square mesh panels have been shown to reduce bycatches of whiting and haddock by 57% and 37%, respectively (Eigaard and Holst, 2004; Nielsen and Madsen, 2006; Eigaard and Nielsen, 2009; Eigaard *et al.*, 2012). Sorting grids are at present used in the Norwegian and Danish fishery (partly implemented as management measures for the larger vessels), but modification of the selective devices and their implementation in management is still ongoing. In Eigaard *et al.* (2021) a new bycatch reduction device, termed “Excluder”, is presented as an alternative to a traditional rigid sorting grid, mandatory in the small-meshed Norway Pout trawl fishery in the North Sea. For all bycatch species analysed in Eigaard *et al.* (2021), the Excluder had significantly lower catches relative to the grid: herring (21%), whiting (6%), mackerel (5%), American plaice (70%), witch flounder (15%), and lesser silver smelt (71%). For Norway Pout there was a significant increase in the overall catch efficiency of 32%. These results are explained by a 10 cm smaller L50 (the length of fish with 50% probability of being rejected by the sorting system) of the Excluder and a 15 times larger sorting area, which reduces the risk of clogging and loss of function. The excluder has been implemented in the EU fishing zone from January 2022, so it is optional to choose between sorting grid or excluder. ICES suggests that these devices (or modified forms of those) are fully implemented and brought into use in the fishery. The implementation of these technical measures shall be followed up by adequate control measures of landings or catches at sea to ensure effective implementation of the existing bycatch measures. An overview of recent relevant management measures and regulations for the Norway pout fishery and the stock can be found in Nielsen *et al.* (2016) and in the Stock Annex.

10.9.1 Long-term management strategies

Long-term management strategies have been evaluated for this stock based on joint EU-Norway requests. The evaluations are described and summarised in the Stock Annex. ICES has evaluated and commented on three management strategies in 2007, although these have not been decided on. Long-term management strategies have been evaluated again in September 2012 and June 2013 based on new joint EU-Norway requests (ICES, 2012a) in spring 2012 and spring 2013 to be available for the September 2012 and September 2013 ICES advice, respectively. These MSEs have been presented in special ICES reports (Vinther and Nielsen, 2012; 2013). No long-term management strategies have been decided upon.

ICES has evaluated and commented on three management strategies in 2007, following requests from managers – fixed fishing mortality ($F = 0.35$), Fixed TAC (50 000 t), and a variable TAC escapement strategy. The 2007 evaluation showed that all three management strategies are capable of generating stock trends that stay at or above $B_{pa} = MSY B_{escapement}$, i.e. away from B_{lim} with a

high probability in the long-term and are, therefore, considered to be in accordance with the MSY and precautionary approach. ICES does not recommend any particular one of the strategies.

The choice between different strategies depends on the requirements that fisheries managers and stakeholders have regarding stability in catches or the overall level of the catches. The variable TAC escapement strategy as evaluated in 2007 has higher long-term yield compared to the fixed fishing mortality strategy (and the fixed TAC strategy), but at the cost of a substantially higher probability of having closures in the fishery. If the continuity of the fishery is an important property, the fixed F (equivalent to fixed effort) strategy will perform better.

There should be no shift in management strategies between years. In recent years, the escapement strategy has been practiced.

A detailed description of these long-term management strategies and management plan evaluations can be found in the Stock Annex and in the ICES AGNOP 2007 (ICES CM 2007/ACFM:39), ICES WGNSSK 2007 (ICES CM 2007/ACFM:30, Section 5.3) and the ICES AGSANNOP (ICES CM 2007/ACFM:40) reports as well as in Vinther and Nielsen (2012, 2013).

ICES has again in September–October 2012 and April–May 2013 (Vinther and Nielsen, 2012; 2013) evaluated and commented on long-term management strategies for the stock using updated stock information. In September 2012, ICES evaluated 3 additional management strategies within the escapement strategy (Vinther and Nielsen, 2012): 1) A long-term minimum TAC > 0 together with a maximum TAC (only with one yearly assessment in September) with the result that a minimum TAC up to 27 kt (revised to 20 kt in the 2013 evaluation) and a maximum TAC of 100–250 kt will be long-term sustainable; 2) A long-term fixed initial TAC the first 6 months of the year followed by an date where the TAC for the whole year is set based on a fixed F (only with one yearly September assessment) with the result that an initial TAC between 25–50 kt and a fixed $F = 0.35$ (corresponding to median catch of 60 kt) is long-term sustainable; 3) Similar to 2, but here with a within year update assessment and advice based on the escapement strategy, and the result here is that an initial TAC of up to 50 kt is sustainable when having a within year update assessment. The difference between the MSE 1 and 2–3 is that the initial fixed TAC is assumed to be taken (or possibly lost) within the first six months of the year (MSE 2–3), while the minimum TAC in MSE 1 can be applied all year. As a follow up on this, ICES evaluated in April 2013 one additional management strategy within the escapement strategy (Vinther and Nielsen, 2013): 4) A long-term minimum TAC > 0 and a maximum TAC, but where the TAC year is from 1 November–31 October rather than from 1 January to 31 December, and one annual advice from the September assessment, with the result that a minimum TAC up to 20 kt with maximum TAC of 100 kt ($F_{\max/\text{cap}} = 0.8$) or with maximum TAC of 200 kt ($F_{\max/\text{cap}} = 0.6$) will be long-term sustainable with some level of F control according to those F_{cap} levels.

With the changes introduced by the August 2016 Norway pout benchmark assessment (ICES WKPOUT, 2016 and Annexes) involving change of assessment model, change of assessment year, change of assessment period, removal of the commercial fishery tuning fleet in the assessment, change of the plus-group in the assessment from 4+ to 3+ and change of stock MSY reference level these above previous MSEs cannot be used anymore for long-term management plans of the stock (including the F_{cap} estimates made there).

Long-term management strategy evaluation according to the new assessment and the revised reference levels as established from the benchmark assessment in August 2016, have been requested in a joint EU-Norway request from November 2017. Based on this EU / Norway request ICES on 29 May 2018 released its advice evaluating long-term management strategies for Norway pout in area 4 and 3.a (ICES WKNPOUT, 2018) which is based on the work from the ICES WKNPOUT (Report of the Workshop for Management Strategy Evaluation for Norway Pout, ICES, Copenhagen 26–28 February 2018, ICES CM2018/ACOM:38 Ref WGNSSK, 96 pp) as

presented to the ICES WGNSSK and approved by ICES ACOM in May 2018. This is summarized below and in the Stock Annex.

ICES has evaluated sustainability of a range of harvest control rules (HCRs) within the escapement strategy currently used for Norway pout, with additional lower (TAC_{min}) and upper (TAC_{max}) bounds on TAC and optional use of upper fishing mortality values (F_{cap}) (ICES WKNPOUT, 2018). Several HCRs were identified that combined TAC_{min} in the range of 20 000–40 000 t and TAC_{max} less than or equal to 200 000 t (150 000 t or 200 000 t) and F_{cap} values of 0.3 and 0.4, resulting in no more than a 5% probability of the spawning-stock biomass falling below B_{lim} .

ICES has evaluated harvest control rules (HCRs) within the escapement strategy currently practiced (aimed at retaining a minimum stock size in the sea every year after fishing) that are furthermore simulated to be restricted by a combination of TAC lower bounds (TAC_{min}) and upper bounds (TAC_{max}). For some HCRs, an upper limit on F (F_{cap}) is also used for setting the TAC.

Because of uncertainties in the estimate of the incoming year class, escapement strategies for short-lived species, where catch opportunities are very dependent on the strength of the incoming year class, may lead to a TAC where a too high portion of the stock is caught. ICES evaluations were conditioned by a maximum realized level of fishing mortality the fishery can exert (assumed at 0.89; $F_{historical}$), which means that the full TAC will not be taken if the required F to catch the TAC exceeds this value.

The identified combinations of TAC_{min} lower bound, TAC_{max} upper bound and F_{cap} give a less variable TAC and F from one year to the next, but also a lower long-term yield than the default escapement strategy. ICES is not in position to advise on this trade-off between higher yield and stability.

The results are sensitive to the assumption that the fishery stops catching Norway pout when F exceeds $F_{historical}$. Therefore, the HCR should be re-evaluated if future F exceeds $F_{historical}$ (0.89).

The evaluation showed that the current procedure for providing TAC advice for Norway pout, based on an escapement strategy is only precautionary with the addition of an F_{cap} at 0.7.

In consultations between EU and Norway, held on 5-6 September 2018, the advice was presented by ICES and in the following discussions, certain limited additional elements, to be reviewed by ICES, came up. This resulted in an additional EU / Norway request from September 2018 on evaluation of additional elements concerning the ICES advice evaluating long-term management strategies for Norway pout in area 4 and 3.a. Here ICES is requested to assess, following MSY $B_{escapement}$:

- which scenarios of TAC_{min} and TAC_{max} would be precautionary, if the F_{cap} is set at 0.7 (building on request part 2 and 3, pages 3 and 4 of the advice).
- which scenarios of TAC_{min} and TAC_{max} would be precautionary, if an interannual flexibility of +/-10% (both banking and borrowing) was introduced for Norway pout (building on request part 2 and 3, pages 3 and 4 of the advice, plus including precautionary scenarios with an F_{cap} of 0.7 – following from paragraph 1 of this request).

On this basis, ICES has evaluated additional harvest control rules (HCRs) within the escapement strategy currently used for Norway pout, with additional lower (TAC_{min}) and upper (TAC_{max}) bounds on TAC and use of an upper fishing mortality (F_{cap}) at 0.7. As for the scenario made for ICES May 2018 advice (ICES WKNPOUT, 2018), ICES evaluations were conditioned by a maximum realized level of fishing mortality the fishery can exert (assumed at 0.89; $F_{historical}$), which means that the full TAC will not be taken if the required F to catch the TAC exceeds this value.

This is presented in the ICES advice in ICES WKNPOUT (2018). Several HCRs were identified that combined TAC_{min} in the range of 20 000–40 000 t and TAC_{max} less than or equal to 200 000 t, resulting in no more than a 5% probability of the spawning-stock biomass falling below B_{lim} .

Increasing the F_{cap} from 0.4 (which was previously evaluated) to 0.7 results in a higher median and mean TAC, but also in a higher long-term probability of SSB falling below B_{lim} . It also results in a higher probability of being constrained by the TAC_{max} .

The evaluations and ACOM approval of this led to identification of an expanded set of sustainable scenarios with a F_{cap} of 0.7. Tables 1 and 2 in ICES WKNPOUT (2018) summarize the long-term (2023–2037) performance metrics for the (precautionary) combinations that result in no more than 5% probability of SSB falling below B_{lim} in the period 2023–2037. More detailed statistics for both precautionary and non-precautionary HCRs are shown in the Table 3 of this advice.

Given that Norway pout is short-lived and that the HCR scenarios are based on the escapement strategy, the application of an additional interannual quota flexibility of $\pm 10\%$ is not considered precautionary.

No decision on long-term management plans is currently available for the Norway pout in area 4 and 3.a based on the identified sustainable scenarios.

Due to introduction of revised IBTS (International Bottom Trawl Survey) quarter 1 (Q1) and quarter 3 (Q3) indices for the full survey time-series for all age groups of Norway pout by ICES in 2020 (https://github.com/ices-tools-prod/DATRAS/tree/master/ALK_substitution) the long-term sustainability of the B_{lim} and $F_{cap} = 0.7$ reference points were during summer 2020 evaluated and presented in Brooks and Nielsen (2020).

The changes by ICES of the historical IBTS Q1 and IBTS Q3 DATRAS indices for demersal species in the North Sea including the Norway pout indices based on introduction of a new calculation method for the indices resulted in a slight change in B_{lim} of less than 10% from $B_{lim} = 39\,447\text{ t}$ (Benchmark ICES WKPOUT, 2016 estimate) to $B_{lim} = 42\,573\text{ t}$. This change was estimated by running the benchmark assessment with the new IBTS indices (Brooks and Nielsen, 2020).

Brooks and Nielsen (2020) evaluated potential change in the MSY and PA sustainability reference points of using the revised IBTS survey indices in DATRAS compared to the previously used indices and presents output from exploratory Management Strategy Evaluation (MSE) with consequences for the precautionary F_{cap} of changed biomass reference points. That is, whether the F_{cap} of 0.7 is still sustainable with the changed biomass reference points for the stock resulting from this revision of survey data. Furthermore, the working document evaluated harvest control rules (HCRs) within the escapement strategy currently practiced (aimed at retaining a minimum stock size in the sea every year after fishing) that are based on the new B_{lim} value and simulated to be restricted by a combination of an upper limit on F values (F_{cap}), different F_{max} values (between the historical observed F_{max} of 0.67, i.e. the $F_{historical}$ for the assessment using the revised IBTS data, and up to a F_{max} value of 2) as well as different TAC upper bounds (TAC_{max}) for setting the TAC. The TAC_{max} values evaluated was from 200 kt up to infinite (i.e. with no upper TAC bound). The sustainability of the current $F_{cap} = 0.7$ was through long-term management strategy evaluation simulations evaluated with the new B_{lim} reference point and according to the different F_{max} and TAC_{max} values applied as described above and detailed in Brooks and Nielsen (2020).

These evaluations showed that the currently implemented F_{cap} of 0.7 is also precautionary and sustainable with the slightly revised B_{lim} reference point (Brooks and Nielsen, 2020). This is the case also in extremely unrealistic scenarios of an infinite TAC_{max} and with F_{max} values between 0.67 and up to 2 (Brooks and Nielsen, 2020). All scenarios for $F_{max} = 0.67$ and for a very unrealistic high $F_{max} = 1$ with infinite TAC_{max} are sustainable.

The conclusion in Brooks and Nielsen (2020) is that with no limits on TAC, then the assumption of a maximum implementable F has a stronger effect on the simulated stock dynamics. When the maximum implementable F is near F_{cap} , then F_{cap} has very little effect on the stock dynamics. If we assume that the maximum implementable F is extremely large (2.0 which is more than double the maximum estimated value), then the effect of F_{cap} can be seen again. With maximum

implementable F at either its maximum historical estimate or at 1.0, then all risk statistics still show $F_{\text{cap}} = 0.7$ to be precautionary. Furthermore, even with the unrealistically high maximum implementable F , then the only risk that goes above 0.05 (when rounded to the nearest 0.01 units) is risk3.long.Q4 for a $F_{\text{cap}} = 0.7$. The type 3 risk statistics may require more replicates to converge to the true value expected from infinite replicates; if needed, this could be investigated in a benchmark. However, the overall result is that risk 1 statistics all indicate precautionarity even under extreme assumptions for high fishing effort. This means that if there were an unrealistic high F_{max} of around 1.6 which is similar to the natural mortality level for the stock then all scenarios of $F_{\text{cap}} = 0.7$ would obviously be sustainable.

The WGNSSK working group has on this basis decided to switch to the new B_{lim} reference point, and on this basis to calculate a new B_{pa} reference point, and continue with the currently implemented F_{cap} of 0.7. It should again be noted that no lower bound TAC_{min} or upper bound TAC_{max} boundaries have been implemented in the management (see also Sections 10.1.4 and 10.6).

10.10 Benchmark issues

Recommendations for future assessments:

1. Age reading check and otolith exchange program

This has been completed and is summarised in section 10.2.2 and in the Stock Annex Section B.2.

2. Data and stock ID

There are no major data deficiencies identified for this stock, whose assessment is usually of high quality.

Investigate size-at-age and derived weight-at-age in how it affects SESAM model estimation in terms of sampling accuracy and precision achieved under the current design and the most statistically rigorous way to impute values for years where these data are missing or in question.

There are currently two recruit indices (age 0 from SGFS and EGFS) being used in model parameter estimation. To avoid duplicative information being introduced into the assessment, a method should be developed that combines the Scottish and the English indices into a single robust index. In general GAMM analyses should be conducted to explore further integration of survey time series. Also, it should be investigated further, now there is starting to be a time series available to do this, whether there is a tendency since 2014 in the SGFS survey to estimate 0-group and 1-group at a higher level since change of vessel in 2012-2013 and onwards. Potential usefulness of Norwegian shrimp survey or HERAS (2016-2024/25) indices could be investigated as an index for SSB or TSB. Should compare with age specific IBTS indices. (There may be a bias in HERAS indices and tendencies herein for small demersals in relation to large indices of the targeted clupeoids in HERAS which should be investigated as well).

Investigate error variances of the data that concerns sampling mechanics, sampling theoretics and sampling designs for both fishery-independent data, and for those obtained from the fishery. That can involve trends in by-catch levels in the samples from monitoring of the Danish and Norwegian commercial Norway pout fishery and/or by-catch of Norway pout in other fisheries (for example shrimp consume fisheries or other industrial fisheries). Further, it can involve production of updated fishery effort and cpue plots for the Norway pout fishery fleets (e.g. the large scale Danish pelagic fleet fishing on Norway pout, besides what is currently done for the demersal trawl fleet, as there now is a time series available since 2019-2020 to evaluate trends for this fleet as well).

Investigate possibilities for producing RDBES data for the stock from Danish and Norwegian fishery data and samples (and possible estimation in RDBES) for as long a time series as

possible/realistic back in time. This will most likely need a data call which can be similar to the one recently made for sandeel in the North Sea.

3. Assessment

Investigate retrospective patterns in the SESAM assessment among other in relation to the Mohn's Rho values for recruitment, SSB and F. The catch residuals are higher than the survey residuals in the assessment, but show no trends in the recent period from 2000 onwards. This should be explored further in future benchmarks. Among other in relation to this sensitivity to smaller changes in seasonal fishing patterns and 0-observations in catch and surveys.

Introduce procedure in SESAM to make one-out-standard analyses of tuning time series.

Change and develop additional SESAM standard diagnostics, which can for example be: (i) a better format for displaying and interpreting standardized model residuals over time (the bubble plots are horizontally compressed and very difficult to read and interpret); (ii) performance statistics based on prediction skill (e.g., how well does the model predict when a data point is removed; residuals one-observation-ahead, one-year-ahead?); (iii) likelihood profiles (if there is tension in the model, where does it occur?); (iv) some depictions of any gradient problems that may exist; (v) summary tables with AIC/BIC values for models using the same data (i.e., documentation of all intermediate models tested before arriving at the final choice of parameter coupling); (vi) statistics for model goodness-of-fit; (vii) curves for fits of catch-at-age and survey indices as well as selection curves and proportion-biomass-at-age plots. (From WKPOUT 2016: The SESAM model appears to be very flexible, but currently estimating relatively few parameters. The Norway pout assessment exhibits coherent indices and generally behaves well. The reviewers were left with the impression that some additional parameters might be estimable. We suggest that the assessment working group spend some time considering what might be worth investigating in future assessments. For example, a clear linkage between states of maturity and natural mortality).

Investigate possibilities for using a stock production assessment model. If length distributions are needed from the commercial catch data there should be made a data call for this.

4. Forecast and reference points

Sensitivity runs with SESAM on the assumptions of time invariant growth, maturity and natural mortality may need to be considered. For the short term, projections that include different ways to handle mean weight-at-age, including projecting forward with specified uncertainty, should be more fully explored (smoothed historic time series, average over some recent time period, etc.).

The catch residuals are higher than the survey residuals in the assessment, but show no trends in the recent period from 2000 onwards. This could be explored further in future benchmarks. Among other in relation to this sensitivity to smaller changes in seasonal fishing patterns and 0-observations in catch and surveys. (It should be noted here, that in the 2016 benchmark, the observation equations in SESAM are also extended to deal with zero observations (both surveys and catches), which are usually treated as missing values in SAM. This is done by introducing a detection limit for each fleet, and defining the likelihood of a zero observation to be the probability of obtaining a value less than the detection limit. The detection limit is set to 0.5 times the smallest positive observation by fleet. A special option is included to down-weight the influence of large jumps in log F on the estimated random walk variance due to periods where the fishery was closed. This option reduces the estimated log F process variance considerably).

The consumption amount of Norway pout by its main predators could be further evaluated in relation to production amount in the Norway pout stock under consideration of consumption and production of other prey species for those predators in the ecosystem. This has implications for setting of Blim levels. This issue was also addressed at the 2016 Benchmark issue, and

investigations on available data was provided in working documents to the benchmark assessment at that time. No further information has been compiled or is available at present to further address this questions, and the conclusions from the working documents and the answers to this point in the benchmark report of the 2016 benchmark assessment are still prevailing and this point will not be further addressed in the 2026 benchmark.

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10.12 Tables

Table 10.2.1. Norway pout in Subarea 4 and Division 3.a. Nominal landings (tonnes) from the North Sea and Skagerrak / Kattegat, ICES Subarea 27.4 and Division 27.3.a in the period 2004–2024, as officially reported to ICES. Bycatch of Norway pout in other (small-meshed) fishery included.

Subarea 4 and Division 3.a combined	Denmark	Faroe Is-lands	France	Norway	Sweden	Netherlands	Germany	UK	Lithuania	Total nominal landings	Unallocated catches***	ICES estimate of total landings (Subarea 4 + Division 3.a)
2004	11345	1159	0	4994	88	0	107	0	0	17693	–4193	13500
2005	941**	24	0	962	0	0	0	0	0	1927	-24	1903
2006	39943	0	0	13622	0	0	34	0	0	53599	–6973	46626
2007	83	0	<1	4712	0	0	0	0	0	4795	997	5792
2008	32558	0	<1	6650	10	0	0	0	0	39218	–3080	36138
2009	19820	0	0	37252	0	22	75	0	0	57170	–2645	54525
2010	71312	0	0	65634	10	18	0	29	0	137003	–11048	125955
2011	4072	0	0	3210	1	0	0	0	0	7282	–758	6524
2012	25558	0	0	4587	3	0	0	6	0	30154	–3081	27073
2013	38364	0	0	46601	4	0	0	0	0	84969	–2869	82100
2014	28448	0	0	18664	2	0	0	14	0	47128	–2958	44170
2015	13033	5270	0	44360	727	18	22	3	0	63434	–34	63400

Subarea 4 and Division 3.a combined	Denmark	Faroe Is-lands	France	Norway	Sweden	Netherlands	Germany	UK	Lithuania	Total nominal landings	Unallocated catches***	ICES estimate of total landings (Subarea 4 + Division 3.a)
2016	23507	3156	0	36052	2	8	27	30	0	62782	618	63400
2017	12467	0	0	21357	6	1	17	0	0	33848	85	33933
2018	10546	0	0	25504	5	2	3	<1	0	36060	87	36147
2019	37850	3034	0	59662	53	16	0	0	0	100615	-2961	97654
2020	65720	1554	<1	63777	48	88	4	86	23	131300	-1803	129497
2021	42002	0	8	29952	6	23	486	10	0	72486	-532	71954
2022	21212	0	61	14515	99	60	6	1	0	35954	-230	35724
2023	2839	0	2	24513	22	3	0	0	0	27379	-23	27356
2024*	1643	0	5	2256	34	6	2	0	0	3946	-47	3899

* Nominal landings are preliminary

** 781 tonnes from trial fishery (directed fishery); 160 tonnes from bycatches in other fisheries.

*** Difference between total nominal landings and ICES estimates of total landings including bycatch of other species.

Division 3.a	Denmark	Faroe Islands	Norway	Sweden	Germany	Total
2004	110	45	41	-	54	250
2005	-	-	-	-	-	0
2006	18	-	2	-	-	20
2007	24	-	-	-	-	24

Division 3.a	Denmark	Faroe Islands	Norway	Sweden	Germany	Total
2008	156	-	-	-	-	156
2009	-	-	209	-	-	209
2010	51	-	711	10	-	772
2011	2	-	-	-	-	2
2012	118	-	-	-	-	118
2013	6945	-	147	1	-	7093
2014	538	-	9	1	-	548
2015	2220	-	41	1	-	2262
2016	918	-	82	1	-	1001
2017	110	-	72	4	2	188
2018	159	-	6	1	-	166
2019	1125	-	6	18	-	1149
2020	5585	-	16	13	-	5614
2021	1942	-	1	2	-	1945
2022	802	-	1	3	-	806
2023	886	-	<1	2	-	888
2024*	11	-	<1	2	-	13

* Preliminary

Division 4.a	Denmark	Faroe Islands	France	Netherlands	Germany	Norway	Sweden	UK	Lithuania	Total
2004	10762	1085	-	-	27	4953	-	-	-	16827
2005	941**	24	-	-	-	962	-	-	-	1927
2006	39531	-	-	-	15	13618	-	-	-	53164
2007	59	-	-	-	-	4712	-	-	-	4771
2008	32158	-	-	-	-	6650	10	-	-	38818
2009	19226	-	-	22	-	36961	-	-	-	56209
2010	71032	-	-	18	-	64303	< 1	29	-	135382
2011	4038	-	-	-	-	3189	1	-	-	7227
2012	25431	-	-	-	-	4528	3	6	-	29968
2013	31375	-	-	-	-	45839	4	-	-	77218
2014	27894	-	-	-	-	18647	1	8	-	46550
2015	10760	5270	-	17	22	43742	12	3	-	59827
2016	21125	3156	-	8	27	35959	<1	12	-	60286
2017	12312	-	-	1	1	21275	<1	-	-	33589
2018	10367	-	-	2	-	25498	4	-	-	35871
2019	36152	3034	-	15	-	59546	32	-	-	98780
2020	59515	1554	<1	88	4	63726	35	82	23	125027

Division 4.a	Denmark	Faroe Islands	France	Netherlands	Germany	Norway	Sweden	UK	Lithuania	Total
2021	39871	-	8	23	486	29863	2	10	-	70263
2022	20365	2	61	59	6	14380	46	1	-	34920
2023	1923	-	2	3	-	24507	7	<1	-	26442
2024*	1626	-	5	6	2	2254	32	-	-	3925

* Preliminary

** 781 tonnes from trial fishery (directed fishery); 160 tonnes from bycatches in other fisheries.

Division 4.b	Denmark	Faroe Islands	France	Germany	Netherlands	Norway	Sweden	UK	Total
2004	473	29	-	26	-	-	88	-	616
2005	-	-	-	-	-	-	-	-	0
2006	394	-	-	19	-	2	-	-	415
2007	-	-	-	-	-	-	-	-	0
2008	244	-	-	-	-	-	-	-	244
2009	595	-	-	75	-	82	-	-	752
2010	229	-	-	-	-	620	-	-	849
2011	32	-	-	-	-	21	-	-	53
2012	9	-	-	-	-	59	-	-	68
2013	43	-	-	-	-	615	<1	-	658
2014	16	-	-	-	-	8	-	6	30

Division 4.b	Denmark	Faroe Islands	France	Germany	Netherlands	Norway	Sweden	UK	Total
2015	53	-	-	-	1	577	714	-	1345
2016	1463	-	-	-	<1	11	1	18	1493
2017	45	-	-	13	<1	10	2	-	71
2018	20	-	-	3	<1	-	-	-	23
2019	573	-	-	-	1	109	3	-	686
2020	620	-	-	-	<1	35	-	3	659
2021	189	-	-	-	-	88	2	-	278
2022	45	-	<1	-	-	133	51	-	229
2023	30	-	-	-	-	6	12	-	48
2024*	6	-	-	-	-	3	2	-	11

* Preliminary

Division 4.c	Denmark	France	UK	Total
2004	-	-	-	0
2005	-	-	-	0
2006	-	-	-	0
2007	-	< 1	-	<1
2008	-	< 1	-	<1

Division 4.c	Denmark	France	UK	Total
2009	-	-	-	0
2010	-	-	-	0
2011	-	-	-	0
2012	-	-	-	0
2013	-	-	-	0
2014	-	-	-	0
2015	-	-	-	0
2016	1	-	-	1
2017	-	-	-	0
2018	-	-	<1	<1
2019	-	-	-	0
2020	-	-	-	0
2021	-	-	-	0
2022	-	-	-	0
2023	-	-	-	0
2024*	-	-	-	0

* Preliminary

Table 10.2.2. Norway pout in Subarea 4 and Division 3.a. Annual landings ('000 t) in the North Sea and Skagerrak (not incl. Kattegat, 3.aS) by country, for 1961–2024 (Data provided by ICES WGNSSK Working Group members). Norwegian landing data include landings of bycatch of other species.

Year	Denmark		Faroes	Norway	Sweden	UK (Scotland)	Others	Total
	North Sea	Skagerrak						
1961	20.5	-	-	8.1	-	-	-	28.6
1962	121.8	-	-	27.9	-	-	-	149.7
1963	67.4	-	-	70.4	-	-	-	137.8
1964	10.4	-	-	51	-	-	-	61.4
1965	8.2	-	-	35	-	-	-	43.2
1966	35.2	-	-	17.8	-	-	+	53.0
1967	169.6	-	-	12.9	-	-	+	182.5
1968	410.8	-	-	40.9	-	-	+	451.7
1969	52.5	-	19.6	41.4	-	-	+	113.5
1970	142.1	-	32	63.5	-	0.2	0.2	238.0
1971	178.5	-	47.2	79.3	-	0.1	0.2	305.3
1972	259.6	-	56.8	120.5	6.8	0.9	0.2	444.8
1973	215.2	-	51.2	63	2.9	13	0.6	345.9
1974	464.5	-	85.0	154.2	2.1	26.7	3.3	735.8
1975	251.2	-	63.6	218.9	2.3	22.7	1	559.7
1976	244.9	-	64.6	108.9	+	17.3	1.7	437.4
1977	232.2	-	48.8	98.3	2.9	4.6	1	387.8
1978	163.4	-	18.5	80.8	0.7	5.5	-	268.9
1979	219.9	9	21.9	75.4	-	3	-	329.2
1980	366.2	11.6	34.1	70.2	-	0.6	-	482.7
1981	167.5	2.8	16.4	51.6	-	+	-	238.3
1982	256.3	35.6	12.3	88	-	-	-	392.2
1983	301.1	28.5	30.7	97.3	-	+	-	457.6
1984	251.9	38.1	19.11	83.8	-	0.1	-	393.01
1985	163.7	8.6	9.9	22.8	-	0.1	-	205.1
1986	146.3	4	2.5	21.5	-	-	-	174.3
1987	108.3	2.1	4.8	34.1	-	-	-	149.3
1988	79	7.9	1.3	21.1	-	-	-	109.3
1989	95.7	4.2	0.8	65.3	+	0.1	0.3	166.4
1990	61.5	23.8	0.9	77.1	+	-	-	163.3
1991	85	32	1.3	68.3	+	-	+	186.6
1992	146.9	41.7	2.6	105.5	+	-	0.1	296.8
1993	97.3	6.7	2.4	76.7	-	-	+	183.1
1994	97.9	6.3	3.6	74.2	-	-	+	182
1995	138.1	46.4	8.9	43.1	0.1	+	0.2	236.8
1996	74.3	33.8	7.6	47.8	0.2	0.1	+	163.8
1997	94.2	29.3	7.0	39.1	+	+	0.1	169.7
1998	39.8	13.2	4.7	22.1	-	-	+	79.8
1999	41	6.8	2.5	44.2	+	-	-	94.5
2000	127	9.3	-	48	0.1	-	+	184.4
2001	40.6	7.5	-	16.8	0.7	+	+	65.6
2002	50.2	2.8	3.4	23.6	-	-	-	80.0
2003	9.9	3.4	2.4	11.4	-	-	-	27.1
2004	8.1	0.3	-	5	-	-	0.1	13.5
2005	0.9*	-	-	1	-	-	-	1.9
2006	35.1	0.1	-	11.4	-	-	-	46.6
2007	2.0**	-	-	3.7	-	-	-	5.7
2008	30.4	-	-	5.7	+	-	+	36.1
2009	17.5	-	-	37.0	+	-	+	54.5
2010	64.9	0.2	-	60.9	+	+	+	126.0
2011	3.3	-	-	3.2	+	+	+	6.5
2012	22.3	0.1	-	4.6	+	+	+	27.0
2013	29.0	6.2	-	46.9	+	+	+	82.1
2014	25.0	0.5	-	18.7	+	+	+	44.2
2015	10.8	2.2	5.3	44.4	0.7	+	+	63.4
2016	23.2	0.9	3.2	36.1	+	+	+	63.4
2017	12.4	0.1	+	21.4	+	+	+	33.9
2018	10.5	0.2	+	25.5	+	+	+	36.2
2019	36.8	1.1	+	59.8	+	+	+	97.7
2020	60.1	5.6	+	63.8	+	+	+	129.5
2021	40.1	1.9	+	30.0	+	+	+	72.0
2022	20.4	0.8	+	14.5	+	+	+	35.7
2023	2.0	0.9	+	24.5	+	+	+	27.4
2024	1.6	+	+	2.3	+	+	+	3.9

* 781 t taken in a trial fishery; 160 t in by-catches in other (small meshed) fisheries.

** 681 t taken in trial fishery; 1300 t in by-catches in other (small meshed) fisheries.

"+" indicate catches below 50 t

Table 10.2.3. Norway pout in Subarea 4 and Division 3.a. National landings (tonnes) by quarter of year 2006–2025 and by area and country. Data provided by Working Group members. Norwegian landing data include landings of bycatch of other species. Bycatch of Norway pout in other (small-meshed) fisheries included.

Year	Quarter	Denmark									Norway		Total	
		Area	IIaN	IIaS	Div. IIa	IVaE	IVaW	IVb	IVc	Div. IV	Div. IV + IIaN	IVaE		Div. IV
2008	1		125	-	125	19	86	123	-	228	353	7	7	360
	2		-	-	-	-	-	30	-	30	30	1,803	1803	1,833
	3		-	-	-	-	6,102	-	-	6,102	6,102	3,582	3582	9,684
	4		-	-	-	-	22,686	1,239	-	23,925	23,925	336	336	24,261
	Total		125	-	125	19	28,874	1,392	-	30,285	30,410		5,728	36,138
2009	1		1	-	1	22	515	-	-	537	538	2	2	540
	2		-	-	-	-	-	-	-	-	-	4,026	4026	4,026
	3		2	-	2	-	11,567	-	-	11,567	11,569	31,251	31251	42,820
	4		-	-	-	-	5,399	4	-	5,403	5,403	1,736	1736	7,139
	Total		3	-	3	22	17,481	4	-	17,507	17,510	37,015	37,015	54,525
2010	1		-	-	-	-	194	-	-	194	194	104	104	298
	2		157	-	157	-	478	59	-	537	694	17,906	17906	18,600
	3		37	-	37	-	33,618	213	-	33,831	33,868	41,883	41883	75,751
	4		8	-	8	-	30,276	38	-	30,314	30,322	984	984	31,306
	Total		202	-	202	-	64,566	310	-	64,876	65,078	60,877	60,877	125,955
2011	1		-	-	-	-	-	-	-	-	-	-	0	-
	2		-	-	-	-	-	-	-	-	-	188	188	188
	3		-	-	-	-	456	5	-	461	461	3,004	3004	3,465
	4		-	-	-	-	2,853	-	-	2,853	2,853	18	18	2,871
	Total		-	-	-	-	3,309	5	-	3,314	3,314	3,210	3,210	6,524
2012	1		-	-	-	-	15	-	-	15	15	12	12	27
	2		-	-	-	-	-	-	-	-	-	280	280	280
	3		2	-	2	-	62	8	-	70	72	395	395	467
	4		125	-	125	-	22,204	-	-	22,204	22,329	3,900	3,900	26,229
	Total		127	-	127	-	22,281	8	-	22,289	22,416	4,587	4,587	27,003
2013	1		-	-	-	-	59	-	-	59	59	18	18	77
	2		6	-	6	-	409	-	-	409	415	10,045	10,045	10,460
	3		4,791	-	4,791	5	3,260	43	-	3,308	8,099	16,350	16,350	24,449
	4		1,366	-	1,366	-	25,211	-	-	25,211	26,577	20,537	20,537	47,114
	Total		6,163	-	6,163	5	28,939	43	-	28,987	35,150	46,950	46,950	82,100
2014	1		-	-	-	-	1,318	-	-	1,318	1,318	6	6	1,324
	2		62	-	62	-	-	2	-	2	64	3,146	3,146	3,210
	3		492	-	492	-	5,606	20	-	5,626	6,118	7,252	7,252	13,370
	4		-	-	-	-	18,006	-	-	18,006	18,006	8,260	8,260	26,266
	Total		554	-	554	-	24,930	22	-	24,952	25,506	18,664	18,664	44,170
2015	1		-	-	-	21	305	-	-	326	326	268	268	594
	2		2	-	2	-	549	-	-	549	551	6,812	6,812	7,363
	3		2,217	1	2,218	10	3,221	19	-	3,250	5,467	21,335	21,335	26,802
	4		-	-	-	-	6,689	-	-	6,689	6,689	15,945	15,945	22,634
	Total		2,219	1	2,220	31	10,764	19	-	10,814	13,033	44,360	44,360	57,393
2016	1		-	-	-	-	514	-	-	514	514	575	575	1,089
	2		244	1	245	-	267	-	-	267	511	8,296	8,296	8,807
	3		673	1	674	5	2,222	51	-	2,278	2,951	20,897	20,897	23,848
	4		-	-	-	3	20,135	-	-	20,138	20,138	6,286	6,286	26,424
	Total		917	2	919	8	23,138	51	-	23,197	24,114	36,054	36,054	60,168
2017	1		-	-	-	-	703	-	-	703	703	30	30	733
	2		5	-	5	-	-	-	-	-	5	3,470	3,470	3,475
	3		104	-	104	6	1,969	-	-	1,975	2,079	11,546	11,546	13,625
	4		-	-	-	68	9,597	2	-	9,667	9,667	6,433	6,433	16,100
	Total		109	-	109	74	12,269	3	-	12,345	12,454	21,479	21,479	33,933
2018	1		-	-	-	-	371	-	-	371	371	9	9	380
	2		2	-	2	-	3	-	-	3	5	4,138	4,138	4,143
	3		157	-	157	-	190	1	-	191	348	8,969	8,969	9,317
	4		-	-	-	-	9,921	-	-	9,921	9,921	12,386	12,386	22,307
	Total		159	-	159	-	10,485	1	-	10,486	10,645	25,502	25,502	36,147
2019	1		-	-	-	-	483	-	-	483	483	13	13	496
	2		178	-	178	-	2,166	-	-	2,166	2,344	8,832	8,832	11,176
	3		947	-	947	-	5,347	1	-	5,348	6,295	32,326	32,326	38,621
	4		-	-	-	-	28,208	567	-	28,775	28,775	18,586	18,586	47,361
	Total		1,125	-	1,125	-	36,204	568	-	36,772	37,897	59,757	59,757	97,654
2020	1		-	-	-	111	3,432	-	-	3,543	3,543	282	282	3,825
	2		2,343	-	2,343	-	1,288	-	-	1,288	3,631	7,333	7,333	10,964
	3		3,018	-	3,018	-	3,574	2	-	3,576	6,594	40,868	40,868	47,462
	4		225	-	225	129	51,570	33	-	51,732	51,957	15,289	15,289	67,246
	Total		5,586	-	5,586	240	59,864	35	-	60,139	65,725	63,772	63,772	129,497
2021	1		-	-	-	-	6,650	-	-	6,650	6,650	64	64	6,714
	2		1,036	-	1,036	-	606	-	-	606	1,642	12,787	12,787	14,429
	3		864	-	864	-	3,376	1	-	3,377	4,241	15,825	15,825	20,066
	4		42	-	42	44	29,382	1	-	29,427	29,469	1,276	1,276	30,745
	Total		1,942	-	1,942	44	40,014	2	-	40,060	42,002	29,952	29,952	71,954
2022	1		-	-	-	-	605	-	-	605	605	0	0	605
	2		218	-	218	-	-	-	-	218	218	3,431	3,431	3,649
	3		584	-	584	-	104	37	-	141	725	8,304	8,304	9,029
	4		-	-	-	-	19,656	8	-	19,664	19,664	2,779	2,779	22,443
	Total		802	-	802	-	20,365	45	-	20,410	21,212	14,514	14,514	35,726
2023	1		-	-	-	-	721	-	-	721	721	15	15	736
	2		299	-	299	-	-	-	-	-	299	6,694	6,694	6,993
	3		582	-	582	-	110	29	-	139	721	14,325	14,325	15,046
	4		-	-	-	-	1,091	-	-	1,091	1,091	3,478	3,478	4,569
	Total		881	-	881	-	1,922	29	-	1,951	2,832	24,512	24,512	27,344
2024	1		-	-	-	-	-	-	-	-	-	-	0	-
	2		-	-	-	-	-	-	-	-	-	173	173	173
	3		11	-	11	-	51	4	-	55	66	1,878	1,878	1,944
	4		-	-	-	-	1,575	2	-	1,577	1,577	205	205	1,782
	Total		11	-	11	-	1,626	6	-	1,632	1,643	2,256	2,256	3,899
2025	1		-	-	-	-	-	-	-	-	-	5	5	5
	2		-	-	-	-	-	1	-	1	1	503	503	504
	3		-	-	-	-	26	-	-	26	26	595	595	621

Table 10.2.3a. Norway pout in Subarea 4 and Division 3.a. Observed and SESAM model predicted total catches in tonnes by quarter.

Year	Observed	Predicted
1984.00	56790	64588
1984.25	56532	38131
1984.50	152291	118870
1984.75	110942	93692
1985.00	57467	49752
1985.25	15509	16046
1985.50	62489	68658
1985.75	92017	60433
1986.00	37773	26349
1986.25	7657	8837
1986.50	45085	36742
1986.75	89993	38167
1987.00	33883	29560
1987.25	15435	8124
1987.50	38729	35950
1987.75	60847	68970
1988.00	22181	22589
1988.25	3559	5602
1988.50	21793	17196
1988.75	61762	33055
1989.00	15379	11988
1989.25	13234	11686
1989.50	55066	40816
1989.75	82880	49131
1990.00	27984	25893
1990.25	39713	23916
1990.50	26156	29592
1990.75	45242	45473

Year	Observed	Predicted
1991.00	42722	32603
1991.25	20786	21407
1991.50	62518	56811
1991.75	64380	59391
1992.00	64218	53167
1992.25	27973	27657
1992.50	114122	89195
1992.75	96177	84886
1993.00	36214	48673
1993.25	29291	27909
1993.50	62290	62450
1993.75	53470	49612
1994.00	34575	28062
1994.25	15373	14982
1994.50	53799	45544
1994.75	79838	39979
1995.00	36942	27490
1995.25	28019	21098
1995.50	69763	61260
1995.75	97048	59462
1996.00	21888	25761
1996.25	13366	14823
1996.50	74631	66321
1996.75	46194	38773
1997.00	15320	15796
1997.25	8708	8767
1997.50	78809	65531
1997.75	54100	51929
1998.00	19502	18974

Year	Observed	Predicted
1998.25	11836	12146
1998.50	20866	29667
1998.75	22830	24719
1999.00	7827	7038
1999.25	12533	8219
1999.50	41445	27752
1999.75	30497	32867
2000.00	10207	11187
2000.25	11589	13298
2000.50	44173	42819
2000.75	119001	70979
2001.00	21400	16507
2001.25	11778	9256
2001.50	4630	12442
2001.75	26565	30964
2002.00	8553	7116
2002.25	6686	4394
2002.50	32922	19485
2002.75	28947	23995
2003.00	3190	3173
2003.25	3106	1980
2003.50	10833	11329
2003.75	7518	6967
2004.00	2040	1613
2004.25	667	584
2004.50	4018	5055
2004.75	6762	7114
2005.00	8	5
2005.25	8	5

Year	Observed	Predicted
2005.50	13	10
2005.75	13	12
2006.00	2205	1827
2006.25	2848	2493
2006.50	6551	8603
2006.75	34949	25522
2007.00	1428	701
2007.25	1100	1157
2007.50	2430	3434
2007.75	838	1695
2008.00	361	313
2008.25	1840	1591
2008.50	8532	5529
2008.75	24111	4870
2009.00	538	208
2009.25	2105	2971
2009.50	36661	22915
2009.75	6509	8552
2010.00	198	246
2010.25	40322	8664
2010.50	57487	29645
2010.75	33071	17180
2011.00	0	0
2011.25	222	1065
2011.50	3749	5391
2011.75	2872	5507
2012.00	29	42
2012.25	281	531
2012.50	469	1342

Year	Observed	Predicted
2012.75	26168	15152
2013.00	79	101
2013.25	10460	3238
2013.50	24444	14418
2013.75	47126	46367
2014.00	1324	502
2014.25	3212	3893
2014.50	13384	14802
2014.75	26244	19462
2015.00	594	522
2015.25	7364	6127
2015.50	26804	28478
2015.75	22655	26681
2016.00	1089	750
2016.25	8846	6878
2016.50	23849	24421
2016.75	26457	22017
2017.00	735	401
2017.25	3475	4696
2017.50	13623	18698
2017.75	16107	22383
2018.00	379	231
2018.25	4143	4849
2018.50	9316	12136
2018.75	22292	14970
2019.00	495	404
2019.25	11179	8237
2019.50	38621	25782
2019.75	47373	31292

Year	Observed	Predicted
2020.00	3808	1707
2020.25	10958	13345
2020.50	47467	39834
2020.75	67100	53939
2021.00	6724	2469
2021.25	14428	11663
2021.50	20060	22130
2021.75	30767	34237
2022.00	604	324
2022.25	3647	3861
2022.50	9031	10089
2022.75	22444	17050
2023.00	736	168
2023.25	6996	2174
2023.50	15048	8320
2023.75	4576	5246
2024.00	0	0
2024.25	173	254
2024.50	1944	2207
2024.75	1782	2004
2025.00	5	7
2025.25	504	231
2025.50	621	592

Table 10.2.6. Norway pout in Subarea 4 and Division 3.a. Mean weight at age in the stock, proportion mature, and natural mortality used in the assessment. From 2012 inter-benchmark and 2016 Benchmark Assessments.

Age	Weight (g)				Proportion mature	M Quarterly	
	Q1	Q2	Q3	Q4			
0	-	-	4	6	0	0.29	
1	9	12	25	25	0.2	0.29	
2	25	25	40	40	1	0.39	
3	40	50	60	58	1	0.44	

Table 10.2.7. Norway pout in Subarea 4 and Division 3.a. Danish fishing effort (number of fishing days) and catch per unit effort (CPUE in tonnes / fishing day) per year and quarter of year (1987–2025) for main Danish fishery with demersal vessels (métier) catching Norway pout. Data for fishing trips where the catch has consisted of at least 70% Norway pout.

[illegible]

Table 10.2.8. Norway pout in Subarea 4 and Division 3.a. Fishing effort (number of fishing days) and catch per unit effort (CPUE in ton / fishing day) per year (2011–2025) and quarter of year for main Norwegian fishery (métiers) catching Norway pout.

Year	Metier	Fishing days					CPUE (ton/fishing day)				
		Q1	Q2	Q3	Q4	Yearly	Q1	Q2	Q3	Q4	Yearly
2011	OTB_DEF_16-31_0_0			1	23	24		10.00	24.09		23.50
2011	OTB_DEF_16-31_2_40			5	75	80		20.20	29.21		28.65
2012	OTB_DEF_16-31_0_0				3	24			15.73	35.38	33.19
2012	OTB_DEF_16-31_2_40				0	74				38.91	38.91
2013	OTB_DEF_16-31_0_0			101	163	99	363	31.27	29.90	47.18	34.99
2013	OTB_DEF_16-31_2_40			224	341	227	792	30.74	31.06	60.80	39.49
2014	OTB_DEF_16-31_0_0			62	64	57	183	18.20	35.07	33.86	28.98
2014	OTB_DEF_16-31_2_40			41	123	143	307	25.96	34.73	38.19	35.17
2015	OTB_DEF_16-31_0_0	0	130	308	71	509		38.29	37.76	38.73	38.03
2015	OTB_DEF_16-31_2_40	5	38	235	192	470	28.66	40.99	42.53	55.65	47.62
2016	OTB_DEF_16-31_0_0	0	269	269	51	589		24.08	22.96	22.61	23.44
2016	OTB_DEF_16-31_2_40	23	37	357	80	497	24.94	23.50	38.64	45.78	38.03
2017	OTB_DEF_16-31_0_0			125	198	15	338	28.71	22.46	25.56	24.91
2017	OTB_DEF_16-31_2_40			1	105	87	193	8.80	37.77	51.21	43.67
2018	OTB_DEF_16-31_0_0			128	163	43	334	23.54	22.37	19.14	22.40
2018	OTB_DEF_16-31_2_40			17	112	233	362	27.83	35.27	44.99	41.17
2019	OTB_DEF_16-31_0_0			243	526	112	881	31.57	37.92	34.05	35.67
2019	OTB_DEF_16-31_2_40			44	272	220	536	36.09	40.45	53.99	45.65
2020	OTB_DEF_16-31_0_0	2	172	445	67	686	25.00	38.51	38.56	24.75	37.16
2020	OTB_DEF_16-31_2_40	6	24	474	131	635	24.33	40.54	37.93	79.27	46.43
2021	OTB_DEF_16-31_0_0	0	272	303	25	600		28.29	28.46	14.51	27.80
2021	OTB_DEF_16-31_2_40	4	171	182	8	365	16.33	30.31	25.11	12.53	27.17
2022	OTB_DEF_16-31_0_0			97	260	84	441	29.16	22.80	21.25	23.90
2022	OTB_DEF_16-31_2_40			15	26	1	42	26.70	21.64	7.50	23.11
2023	OTB_DEF_16-31_0_0	1	177	324	64	566	3.00	31.44	31.04	19.19	29.78
2023	OTB_DEF_16-31_2_40	0	7	33	3	43		11.59	21.77	7.35	19.10
2024	OTB_DEF_16-31_0_0			7	24	31		9.25	15.72		14.26
2025	OTB_DEF_16-31_0_0			3	4	7		14.82	29.11		22.98

Table 10.2.9. Norway pout in Subarea 4 and Division 3.a. Fishing effort (number of fishing days) and catch per unit effort (CPUE in ton per fishing day) per year and vessel horsepower (HP) class (1987–2025) for main Danish fishery with demersal vessels (métier) catching Norway pout.

[illegible]

Table 10.2.10. Norway pout in Subarea 4 and Division 3.a. Fishing effort (number of fishing days) and catch per unit effort (CPUE in ton / fishing day) per year (2011–2025) and quarter of year for main Norwegian fishery (métiers) catching Norway pout.

Year	Metier	Fishing days				Yearly	CPUE (ton/fishing day)					Yearly
		500-1000	1000-1500	1500-2000	> 2000		500-1000	1000-1500	1500-2000	> 2000		
2011	OTB_DEF_16-31_0_0		24		0	24		23.5				23.5
2011	OTB_DEF_16-31_2_40		20		60	80		18.3			32.1	28.6
2012	OTB_DEF_16-31_0_0	0	17	4	6	27		34.8	13.75		41.7	33.2
2012	OTB_DEF_16-31_2_40	19	28	0	27	74	21.2	26.9			63.8	38.9
2013	OTB_DEF_16-31_0_0		273	75	15	363		34.4	30.9		65.3	35.0
2013	OTB_DEF_16-31_2_40		162	130	500	792		23.2	34.10332		46.2	39.5
2014	OTB_DEF_16-31_0_0	0	142	16	25	183		25.5	16.6		56.4	29.0
2014	OTB_DEF_16-31_2_40	80	58	67	102	307	42.9	14.6	36.6		39.8	35.2
2015	OTB_DEF_16-31_0_0		228	106	175	509		33.7	42.7		40.8	38.0
2015	OTB_DEF_16-31_2_40		0	103	367	470			49.7		47.0	47.6
2016	OTB_DEF_16-31_0_0		207	136	246	589		25.5	21.0		23.0	23.4
2016	OTB_DEF_16-31_2_40		18	72	407	497		28.3	42.8		37.6	38.0
2017	OTB_DEF_16-31_0_0		123	107	108	338		24.7	21.4		28.6	24.9
2017	OTB_DEF_16-31_2_40		9	86	98	193		51.9	41.1		45.2	43.7
2018	OTB_DEF_16-31_0_0	40	121	107	66	334	20.9	20.2	22.1		27.8	22.4
2018	OTB_DEF_16-31_2_40	14	26	63	259	362	36.2	46.6	34.4		42.5	41.2
2019	OTB_DEF_16-31_0_0	144	232	171	334	881	27.3	29.5	32.4		45.3	35.7
2019	OTB_DEF_16-31_2_40	7	8	118	403	536	57.7	56.4	45.5		45.3	45.7
2020	OTB_DEF_16-31_0_0	146	133	118	289	686	28.1	33.9	31.8		45.5	37.2
2020	OTB_DEF_16-31_2_40	4	3	94	534	635	37.5	60.5	36.0		48.3	46.4
2021	OTB_DEF_16-31_0_0	128	122	141	209	600	25.9	26.2	23.6		32.7	27.8
2021	OTB_DEF_16-31_2_40	0	0	78	287	365			29.2		26.6	27.2
2022	OTB_DEF_16-31_0_0	217		84	140	441	23.6		17.9		28.0	23.9
2022	OTB_DEF_16-31_2_40	0		42	0	42			23.1			23.1
2023	OTB_DEF_16-31_0_0	266		93	207	566	24.5		22.7		39.7	29.8
2023	OTB_DEF_16-31_2_40	0		0	43	43					19.1	19.1
2024	OTB_DEF_16-31_0_0	4		6	21	31	7.5		9.6		16.9	14.3
2025	OTB_DEF_16-31_0_0	2			5	7	7.4				29.2	23.0

Table 10.2.11. Norway pout in Subarea 4 and Division 3.a. Research vessel indices (CPUE in catch in number per trawl hour) of abundance for Norway pout.

Year	IBTS/IYFS ¹ February (1 st Q)			EGFS ^{2,3} August			SGFS ⁴ August				IBTS 3 rd Quarter ¹				
	1-group	2-group	3-group	0-group	1-group	2-group	3-group	0-group	1-group	2-group	3-group	0-group	1-group	2-group	3-group
1971	1556	22	-	-	-	-	-	-	-	-	-	-	-	-	-
1972	2589	856	8	-	-	-	-	-	-	-	-	-	-	-	-
1973	4207	438	-	-	-	-	-	-	-	-	-	-	-	-	-
1974	25559	388	24	-	-	-	-	-	-	-	-	-	-	-	-
1975	5067	1850	36	-	-	-	-	-	-	-	-	-	-	-	-
1976	4422	328	35	-	-	-	-	-	-	-	-	-	-	-	-
1977	6122	238	44	-	-	-	-	-	-	-	-	-	-	-	-
1978	1480	565	56	-	-	-	-	-	-	-	-	-	-	-	-
1979	2737	316	76	-	-	-	-	-	-	-	-	-	-	-	-
1980	3274	552	30	-	-	-	-	-	1928	346	12	-	-	-	-
1981	1092	377	14	-	-	-	-	-	185	127	9	-	-	-	-
1982	4511	266	81	-	-	-	-	8	991	44	22	-	-	-	-
1983	2252	592	13	-	-	-	-	13	490	91	1	-	-	-	-
1984	5000	956	89	-	-	-	-	2	615	69	8	-	-	-	-
1985	2342	1401	98	-	-	-	-	5	636	173	5	-	-	-	-
1986	2066	386	19	-	-	-	-	38	389	54	9	-	-	-	-
1987	3171	475	63	-	-	-	-	7	338	23	1	-	-	-	-
1988	123	710	25	-	-	-	-	14	38	209	4	-	-	-	-
1989	2017	254	170	-	-	-	-	2	382	21	14	-	-	-	-

Year	IBTS/IYFS ¹ February (1 st Q)			EGFS ^{2,3} August			SGFS ⁴ August				IBTS 3 rd Quarter ¹				
	1-group	2-group	3-group	0-group	1-group	2-group	3-group	0-group	1-group	2-group	3-group	0-group	1-group	2-group	3-group
1990	1295	712	70	-	-	-	-	58	206	51	2	-	-	-	-
1991	2428	693	157	-	-	-	-	10	732	42	6	7523	515	486	6
1992	5060	860	33	2975	6116	1710	303	12	1715	221	24	2560	4106	740	151
1993	2574	2643	346	3706	3582	1706	108	2	580	329	20	4080	1506	921	92
1994	1532	374	99	9487	1148	147	25	136	387	106	6	3196	685	114	21
1995	5951	757	85	5478	8374	282	62	37	2438	234	21	2864	4106	860	134
1996	915	2626	233	8241	1326	378	9	127	412	321	8	4559	672	419	41
1997	9633	1557	674	441	6295	372	102	1	2154	130	32	490	3308	345	76
1998	1009	5332	268	1391	377	340	3	2628	938	127	5	2931	791	745	23
1999	3522	601	668	10985	1175	40	29	3603	1784	179	37	7854	2316	230	106
2000	8034	1563	98	2267	9730	264	2	2094	6656	207	23	1644	7556	590	14
2001	1306	2805	288	2243	1434	1344	31	759	727	710	26	2089	1164	938	57
2002	1784	812	864	4939	1137	58	18	2559	1192	151	123	1974	749	76	52
2003	1241	573	94	323	572	75	5	1767	779	126	1	1812	1015	193	8
2004	903	364	37	278	557	109	6	731	719	175	19	773	590	209	14
2005	698	123	38	3395	414	67	15	3073	343	132	18	2679	395	104	18
2006	3400	113	23	1813	1996	124	20	1127	1285	69	9	1391	1800	197	14
2007	1287	769	31	1610	1181	720	43	5003	1023	395	8	4151	1186	430	40
2008	2438	461	154	628	1340	411	104	3456	1263	263	57	3035	1610	267	98
2009	5553	1582	123	4871	3500	306	5	5835	1750	202	16	5899	2454	358	14
2010	4954	1439	143	103	4257	559	13	1449	5101	930	29	842	4780	812	37

Year	IBTS/IYFS ¹ February (1 st Q)			EGFS ^{2,3} August			SGFS ⁴ August				IBTS 3 rd Quarter ¹				
	1-group	2-group	3-group	0-group	1-group	2-group	3-group	0-group	1-group	2-group	3-group	0-group	1-group	2-group	3-group
2011	545	2126	347	290	555	1050	40	1895	226	935	38	1801	474	1114	64
2012	1002	327	527	3946	505	99	59	10067	1070	159	216	6416	829	217	139
2013	4469	508	102	498	2592	117	19	1754	2888	107	22	1317	2759	186	18
2014	818	936	48	10157	483	268	17	24896	537	149	0	10238	480	253	13
2015	6638	570	130	1415	4320	60	15	10208	6568	118	0	3511	3911	191	47
2016	2404	909	41	7199	1710	314	4	14830	1696	290	0	8965	1386	279	14
2017	4332	421	173	1280	5061	134	38	7478	1906	77	2	4235	2502	158	25
2018	1139	850	147	5096	586	144	12	20632	674	246	3	6115	578	201	7
2019	3892	303	55	4286	1308	68	8	17856	3888	86	3	6464	2204	134	19
2020	6078	1121	83	3126	5343	227	8	36298	3417	530	0	8463	3858	612	10
2021	3823	1535	165	428	2868	544	12	13785	2870	402	3	4163	1504	296	48
2022	1599	801	160	6770	952	252	40	12218	2580	816	74	9496	888	284	40
2023	2519	286	65	564	1162	100	30	12001	1881	178	5	3754	1817	183	15
2024	574	818	100	178	123	90	3	4096	158	72	1	2579	250	196	12
2025	1028	168	68	191	131	96	3	11578	448	62	3				

¹International Bottom Trawl Survey (IBTS), arithmetic mean catch in no./h in standard area. In general the quarter 1 (Q1) and quarter 3 (Q3) IBTS indices have been revised in 2012 and 2014 and 2015 and 2020 (see documentation on ICES DATRAS). The revised Q1 and Q3 IBTS survey indices introduced in 2020 are given, and used in the assessment. ²English groundfish survey (EGFS): Arithmetic mean catch no./h. Data for 1996, 2001, 2002, and 2003 have been revised compared to the 2003 assessment. In 2007, numbers for 1997 and 1998 as well as 2002 has been adjusted based on new automatic calculation and processing process has been introduced. In September 2015, the EGFS Survey index was for all years and ages radically revised in order to incorporate the relevant primes within the Norway pout index area following the ICES IBTS manual (2015). ³Minor GOV sweep changes in 2006 for the EGFS. ⁴Scottish groundfish surveys (SGFS), arithmetic mean catch no./h. Survey design changed in 1998 and 2000. The SGFS survey area changed slightly in 2009 and onwards, which is evaluated to have no main effect for the Norway pout indices as the indices are weighted by sub-area. SGFS data for the full area, i.e. indices based on all hauls, are included in the presented indices. In September 2019, the indices from 2013 onwards for all age groups were corrected with removal of a few invalid hauls (including also the Q3 2019 survey) resulting in very minor changes of the indices for all age groups not affecting the assessment. In 2025, only one station per ICES square was taken with the GOV trawl the North-Western area covered by the SGFS while the other station in each of those squares was taken with the new

JTS/BT238 trawl. That covers the 16 ICES statistical squares where Scotland was the sole surveying nation (and that requires Scotland to deploy the trawl twice), i.e. 16 stations out of totally 86 stations. Only stations taken with the GOV trawl have been included in the 2025 index until more data and longer time series are available for evaluating the effects of the new JTS gear.

Table 10.3.1. Norway pout in Subarea 4 and Division 3.a. Tuning fleets and stock indices and tuning fleets used in the final 2004 benchmark assessment, in the 2005–2015 assessments, as well as in the 2016–2025 assessments based on the 2016 benchmark assessment, compared to the 2003 assessment. (Changes from previous period marked with grey).

		2003 ASSESSMENT	2004, 2005, April 2006 ASSESSMENT	Sept. 2006 ASSESSMENT	2007-2015 ASSESSMENTS	2016-2025 ASSESSMENTS
Recruiting season		3rd quarter	2nd quarter (SXSA)	3rd quarter (SMS); 2nd quarter (SXSA)	2nd quarter (SXSA), autumn assessm.	3rd quarter SESAM (1984-2025)
Last season in last year		3rd quarter	2nd quarter (SXSA)	3rd quarter (SMS); 2nd quarter (SXSA)	2nd quarter (SXSA), autumn assessm.	3rd quarter SESAM (1984-2025)
Plus-group		4+	4+ (SXSA)	None (SMS); 4+ (SXSA)	4+ (SXSA)	3+ (SESAM) (1984-2025)
FLT01: comm Q1						
	Year range	1982-2003	1982-2004	1982-2004	1983-2004, 2006	1983-2004, 2006
	Quarter	1	1	1	1	1
	Ages	1-3	1-3	1-3	1-3	1-3
FLT01: comm Q2			NOT USED	NOT USED	NOT USED	NOT USED
	Year range	1982-2003				
	Quarter	2				
	Ages	1-3				
FLT01: comm Q3						
	Year range	1982-2003	1982-2004	1982-2004	1983-2004, 2006	1983-2004, 2006
	Quarter	3	3	3	3	3
	Ages	0-3	1-3	1-3	1-3	1-3
FLT01: comm Q4						
	Year range	1982-2003	1982-2004	1982-2004	1983-2004, 2006	1983-2004, 2006
	Quarter	4	4	4	4	4
	Ages	0-3	0-3	0-2 (SMS); 0-3 (SXSA)	0-3 (SXSA)	0-3 (SXSA)
FLT02: ibtsq1						
	Year range	1982-2003	1982-2006	1982-2006	1983-2015	1984-2025
	Quarter	1	1	1	1	1
	Ages	1-3	1-3	1-3	1-3	1-3
FLT03: egfs						
	Year range	1982-2003	1992-2005	1992-2005	1992-2015	1992-2025
	Quarter	3	Q3 -> Q2	Q3 -> Q2	Q3 -> Q2	3
	Ages	0-3	0-1	0-1	0-1	0-1
FLT04: sgfs						
	Year range	1982-2003	1998-2006	1998-2006	1998-2015	1998-2025
	Quarter	3	Q3 -> Q2	Q3 -> Q2	Q3 -> Q2	3
	Ages	0-3	0-1	0-1	0-1	0-1
FLT05: ibtsq3		NOT USED				
	Year range		1991-2005	1991-2005	1991-2014	1991-2024
	Quarter		3	3	Q3	3
	Ages		2-3	2-3	2-3	2-3

Table 10.3.2. Norway pout Subarea 4 and Division 3.aN (Skagerrak). Baseline run with SESAM seasonal stochastic assessment model. Settings and tuning fleets.

SURVIVORS ANALYSIS OF: Norway pout stock in September 2025

Run: September 2025 (NP_Sep2025_v1, www.stockassessment.org)

The following parameters were used:

Year range:	1984 - 2025
Seasons per year:	4
The last season in the last year is season:	3
Youngest age:	0
Oldest age:	2
Plus age:	3
Recruitment in season:	3
Spawning in season:	1

The following tuning fleets were included:

Fleet 2:	ibtsq1	(Age 1-3)
Fleet 3:	egfsq3	(Age 0-1)
Fleet 4:	sgfsq3	(Age 0-1)
Fleet 5:	ibtsq3	(Age 2-3)

Table 10.3.3. Norway pout Subarea 4 and Division 3.aN (Skagerrak). Baseline run with SESAM seasonal model September 2025. Estimated stock numbers in start of quarterly and yearly season.

Time\Age	0	1	2	3
1984	0	49189	11117	808
1984.25	0	33848	6337	473
1984.5	43360	23292	3709	275
1984.75	0	13291	1662	160
1985	0	23826	6904	812
1985.25	0	16234	3261	454
1985.5	31869	11536	1874	268
1985.75	0	6883	796	154
1986	0	17009	3639	412
1986.25	0	11782	1918	240

Time\Age	0	1	2	3
1986.5	54438	8624	1184	148
1986.75	0	5611	618	90
1987	0	32059	3273	378
1987.25	0	23586	1747	218
1987.5	11511	17561	1079	135
1987.75	0	12002	586	84
1988	0	5837	6888	303
1988.25	0	4555	4225	174
1988.5	52192	3624	2925	105
1988.75	0	2751	1880	65
1989	0	28955	1827	1107
1989.25	0	21225	1235	707
1989.5	53853	15524	832	453
1989.75	0	10477	492	293
1990	0	29013	6510	479
1990.25	0	21582	3896	299
1990.5	65695	15593	2313	184
1990.75	0	11010	1379	117
1991	0	35950	7382	884
1991.25	0	26215	4405	520
1991.5	118615	19692	2770	310
1991.75	0	14091	1634	197
1992	0	66257	9816	1074
1992.25	0	48213	6278	731
1992.5	63393	35503	4319	506
1992.75	0	23890	2715	322
1993	0	34059	15422	1939
1993.25	0	23837	8958	1240
1993.5	55142	16696	5438	798

Time\Age	0	1	2	3
1993.75	0	10593	2882	491
1994	0	28755	6539	1776
1994.25	0	20157	4009	1063
1994.5	158931	14523	2567	646
1994.75	0	10078	1565	401
1995	0	89309	6887	1277
1995.25	0	66570	4460	885
1995.5	63422	49143	2912	612
1995.75	0	34357	1853	396
1996	0	32052	23068	1496
1996.25	0	24291	14530	969
1996.5	132726	18044	9576	627
1996.75	0	13046	6090	396
1997	0	75039	9529	4324
1997.25	0	55164	6230	2790
1997.5	27199	43331	4120	1797
1997.75	0	31371	2512	1151
1998	0	15117	22649	2314
1998.25	0	11427	14310	1428
1998.5	48968	8502	9188	878
1998.75	0	6466	5840	565
1999	0	28710	4765	4199
1999.25	0	22074	3286	2701
1999.5	114193	16904	2181	1719
1999.75	0	12652	1324	1081
2000	0	67375	9448	1451
2000.25	0	51636	6587	909
2000.5	31461	40660	4544	564
2000.75	0	29677	2982	364

Time\Age	0	1	2	3
2001	0	16843	19869	2106
2001.25	0	12235	12451	1363
2001.5	32532	8892	7941	877
2001.75	0	6475	5461	564
2002	0	18847	4523	3895
2002.25	0	14063	2855	2405
2002.5	24775	10089	1892	1500
2002.75	0	6885	1205	951
2003	0	12740	4488	1291
2003.25	0	8908	2890	782
2003.5	9561	6199	1856	470
2003.75	0	4259	1124	299
2004	0	5331	3042	817
2004.25	0	3849	2052	523
2004.5	9506	2924	1387	334
2004.75	0	2092	886	214
2005	0	5358	1453	674
2005.25	0	3957	1023	454
2005.5	36824	2962	717	304
2005.75	0	2271	492	199
2006	0	20931	1801	459
2006.25	0	15488	1323	306
2006.5	27626	11757	942	202
2006.75	0	8909	616	130
2007	0	15454	6146	461
2007.25	0	11583	4106	322
2007.5	38699	8582	2742	226
2007.75	0	6397	1832	148
2008	0	22496	4945	1351

Time\Age	0	1	2	3
2008.25	0	17519	3514	884
2008.5	60884	13618	2437	581
2008.75	0	10698	1610	366
2009	0	39174	7971	1291
2009.25	0	29717	5471	808
2009.5	87467	23357	3707	503
2009.75	0	18004	2287	321
2010	0	53035	14242	1689
2010.25	0	42379	10994	1095
2010.5	7810	32044	7764	702
2010.75	0	23021	5142	447
2011	0	4614	16248	3624
2011.25	0	3559	10535	2274
2011.5	13185	2853	7258	1441
2011.75	0	2197	4943	921
2012	0	7776	1712	4091
2012.25	0	5937	1207	2738
2012.5	64940	4618	869	1836
2012.75	0	3697	614	1209
2013	0	38100	2686	1178
2013.25	0	29179	1936	753
2013.5	19483	20991	1351	478
2013.75	0	14453	866	307
2014	0	10689	9105	673
2014.25	0	7898	5724	423
2014.5	103496	5857	3580	264
2014.75	0	4350	2181	164
2015	0	56694	3025	1415
2015.25	0	40284	1974	914

Time\Age	0	1	2	3
2015.5	39759	28070	1285	588
2015.75	0	18141	741	368
2016	0	21414	11562	656
2016.25	0	14987	7370	417
2016.5	70563	10125	4592	261
2016.75	0	6348	2776	161
2017	0	38392	3867	1735
2017.25	0	26286	2523	1101
2017.5	22784	17995	1653	694
2017.75	0	11751	1020	442
2018	0	11639	7510	889
2018.25	0	8456	4621	533
2018.5	81343	6022	2801	315
2018.75	0	4357	1679	194
2019	0	44350	3018	1030
2019.25	0	32666	2122	649
2019.5	107180	23717	1356	399
2019.75	0	17009	835	248
2020	0	60426	11611	675
2020.25	0	44078	7615	432
2020.5	63647	32394	4839	272
2020.75	0	22514	2928	173
2021	0	33850	14733	1856
2021.25	0	24187	9171	1207
2021.5	25638	16952	5511	775
2021.75	0	11264	3346	493
2022	0	13978	7134	2206
2022.25	0	10802	4384	1392
2022.5	48250	8107	2731	872

Time\Age	0	1	2	3
2022.75	0	5764	1653	548
2023	0	25720	3900	1244
2023.25	0	19193	2508	780
2023.5	9453	13367	1603	489
2023.75	0	8979	989	315
2024	0	4625	6032	855
2024.25	0	2940	3723	542
2024.5	8851	2043	2331	343
2024.75	0	1481	1474	221
2025	0	4962	1091	1070
2025.25	0	3554	741	693
2025.5	10312	2576	486	445
2025.75	0	1917	325	286

Table 10.3.4. Norway pout in Subarea 4 and Division 3.a. Baseline run with SESAM seasonal model September 2025. Estimated fishing mortalities by quarter of year. (The last 2025 quarter 4 F-value is a projection of F based on the population estimate by end of 3rd quarter).

Time\Age	0	1	2	3+
1984	0.001	0.306	0.838	0.302
1984.25	0.001	0.292	0.675	0.297
1984.5	0.009	1.017	1.666	0.290
1984.75	0.155	1.388	1.785	0.052
1985	0.001	0.400	1.098	0.396
1985.25	0.000	0.178	0.412	0.181
1985.5	0.009	0.944	1.547	0.269
1985.75	0.150	1.337	1.719	0.050
1986	0.001	0.318	0.872	0.314
1986.25	0.000	0.114	0.264	0.116
1986.5	0.005	0.551	0.904	0.157
1986.75	0.100	0.890	1.144	0.033
1987	0.001	0.277	0.759	0.274

Time\Age	0	1	2	3+
1987.25	0.000	0.091	0.210	0.092
1987.5	0.004	0.423	0.694	0.121
1987.75	0.129	1.155	1.485	0.043
1988	0.001	0.198	0.543	0.196
1988.25	0.000	0.072	0.167	0.074
1988.5	0.003	0.294	0.481	0.084
1988.75	0.085	0.760	0.977	0.028
1989	0.000	0.142	0.389	0.140
1989.25	0.000	0.139	0.323	0.142
1989.5	0.004	0.429	0.703	0.122
1989.75	0.083	0.739	0.951	0.028
1990	0.001	0.196	0.538	0.194
1990.25	0.000	0.218	0.505	0.222
1990.5	0.003	0.323	0.529	0.092
1990.75	0.059	0.531	0.683	0.020
1991	0.001	0.223	0.612	0.221
1991.25	0.000	0.158	0.365	0.160
1991.5	0.003	0.337	0.552	0.096
1991.75	0.055	0.494	0.636	0.018
1992	0.001	0.199	0.547	0.197
1992.25	0.000	0.126	0.292	0.129
1992.5	0.003	0.348	0.570	0.099
1992.75	0.055	0.489	0.628	0.018
1993	0.000	0.167	0.457	0.165
1993.25	0.000	0.130	0.302	0.133
1993.5	0.004	0.418	0.685	0.119
1993.75	0.061	0.547	0.703	0.020
1994	0.001	0.178	0.487	0.176
1994.25	0.000	0.115	0.266	0.117

Time\Age	0	1	2	3+
1994.5	0.003	0.368	0.604	0.105
1994.75	0.041	0.362	0.466	0.013
1995	0.000	0.108	0.297	0.107
1995.25	0.000	0.100	0.231	0.102
1995.5	0.002	0.206	0.337	0.059
1995.75	0.033	0.291	0.374	0.011
1996	0.000	0.074	0.203	0.073
1996.25	0.000	0.060	0.139	0.061
1996.5	0.003	0.283	0.464	0.081
1996.75	0.028	0.255	0.328	0.009
1997	0.000	0.059	0.161	0.058
1997.25	0.000	0.037	0.086	0.038
1997.5	0.003	0.286	0.469	0.081
1997.75	0.034	0.303	0.389	0.011
1998	0.000	0.059	0.160	0.058
1998.25	0.000	0.058	0.133	0.059
1998.5	0.002	0.211	0.346	0.060
1998.75	0.032	0.285	0.367	0.011
1999	0.000	0.045	0.124	0.045
1999.25	0.000	0.071	0.165	0.072
1999.5	0.002	0.260	0.426	0.074
1999.75	0.038	0.342	0.440	0.013
2000	0.000	0.044	0.119	0.043
2000.25	0.000	0.051	0.118	0.052
2000.5	0.001	0.161	0.264	0.046
2000.75	0.045	0.406	0.522	0.015
2001	0.000	0.058	0.159	0.057
2001.25	0.000	0.048	0.111	0.049
2001.5	0.001	0.078	0.128	0.022

Time\Age	0	1	2	3+
2001.75	0.037	0.327	0.421	0.012
2002	0.000	0.057	0.156	0.056
2002.25	0.000	0.040	0.092	0.040
2002.5	0.002	0.217	0.356	0.062
2002.75	0.050	0.448	0.577	0.017
2003	0.000	0.032	0.088	0.032
2003.25	0.000	0.029	0.068	0.030
2003.5	0.002	0.183	0.300	0.052
2003.75	0.031	0.276	0.355	0.010
2004	0.000	0.024	0.066	0.024
2004.25	0.000	0.014	0.032	0.014
2004.5	0.001	0.132	0.217	0.038
2004.75	0.033	0.294	0.378	0.011
2005	0.000	0.000	0.000	0.000
2005.25	0.000	0.000	0.000	0.000
2005.5	0.000	0.000	0.001	0.000
2005.75	0.000	0.000	0.001	0.000
2006	0.000	0.017	0.064	0.021
2006.25	0.000	0.035	0.094	0.042
2006.5	0.001	0.101	0.205	0.039
2006.75	0.035	0.420	0.579	0.016
2007	0.000	0.004	0.015	0.005
2007.25	0.000	0.013	0.034	0.015
2007.5	0.000	0.023	0.047	0.009
2007.75	0.001	0.018	0.025	0.001
2008	0.000	0.002	0.006	0.002
2008.25	0.000	0.015	0.038	0.016
2008.5	0.000	0.047	0.096	0.018
2008.75	0.004	0.055	0.077	0.002

Time\Age	0	1	2	3+
2009	0.000	0.001	0.003	0.001
2009.25	0.000	0.014	0.037	0.016
2009.5	0.001	0.122	0.249	0.047
2009.75	0.004	0.055	0.077	0.002
2010	0.000	0.000	0.001	0.000
2010.25	0.000	0.026	0.068	0.029
2010.5	0.001	0.099	0.202	0.038
2010.75	0.006	0.088	0.123	0.003
2011	0.000	0.000	0.001	0.000
2011.25	0.000	0.004	0.011	0.005
2011.5	0.000	0.040	0.083	0.016
2011.75	0.006	0.083	0.116	0.003
2012	0.000	0.000	0.001	0.000
2012.25	0.000	0.006	0.016	0.007
2012.5	0.000	0.026	0.054	0.010
2012.75	0.025	0.348	0.489	0.013
2013	0.000	0.001	0.002	0.001
2013.25	0.000	0.025	0.065	0.028
2013.5	0.001	0.114	0.233	0.044
2013.75	0.034	0.488	0.685	0.018
2014	0.000	0.002	0.010	0.003
2014.25	0.000	0.024	0.062	0.026
2014.5	0.001	0.153	0.313	0.059
2014.75	0.023	0.322	0.453	0.012
2015	0.000	0.003	0.012	0.003
2015.25	0.000	0.031	0.080	0.034
2015.5	0.001	0.202	0.412	0.078
2015.75	0.020	0.278	0.390	0.010
2016	0.000	0.003	0.012	0.004

Time\Age	0	1	2	3+
2016.25	0.000	0.043	0.112	0.048
2016.5	0.001	0.196	0.401	0.076
2016.75	0.022	0.318	0.446	0.012
2017	0.000	0.002	0.006	0.002
2017.25	0.000	0.034	0.088	0.037
2017.5	0.001	0.161	0.329	0.062
2017.75	0.022	0.312	0.439	0.012
2018	0.000	0.002	0.008	0.002
2018.25	0.000	0.052	0.134	0.057
2018.5	0.001	0.152	0.311	0.059
2018.75	0.026	0.367	0.515	0.014
2019	0.000	0.005	0.019	0.006
2019.25	0.000	0.085	0.221	0.094
2019.5	0.001	0.201	0.411	0.078
2019.75	0.024	0.333	0.468	0.012
2020	0.000	0.011	0.045	0.014
2020.25	0.000	0.060	0.156	0.066
2020.5	0.001	0.184	0.376	0.071
2020.75	0.026	0.368	0.517	0.014
2021	0.000	0.007	0.028	0.009
2021.25	0.000	0.052	0.134	0.057
2021.5	0.001	0.145	0.296	0.056
2021.75	0.026	0.373	0.524	0.014
2022	0.000	0.002	0.008	0.003
2022.25	0.000	0.029	0.075	0.032
2022.5	0.001	0.122	0.249	0.047
2022.75	0.021	0.298	0.419	0.011
2023	0.000	0.001	0.006	0.002
2023.25	0.000	0.019	0.048	0.021

Time\Age	0	1	2	3+
2023.5	0.001	0.104	0.213	0.040
2023.75	0.007	0.103	0.145	0.004
2024	0.000	0.000	0.002	0.000
2024.25	0.000	0.004	0.010	0.004
2024.5	0.000	0.044	0.090	0.017
2024.75	0.005	0.075	0.105	0.003
2025	0.000	0.000	0.000	0.000
2025.25	0.000	0.007	0.017	0.007
2025.5	0.000	0.022	0.044	0.008
2025.75	0.005	0.075	0.105	0.003

Table 10.3.5. Norway pout in Subarea 4 and Division 3.a. Baseline run with SESAM seasonal model September 2025. Diagnostics of the SESAM baseline assessment. Estimated catchabilities by survey tuning fleet.

Fleet number	Age	Catchability	Low	High
2	1	0.10843	0.06312	0.18626
2	2	0.14983	0.07939	0.28276
2	3	0.13370	0.05618	0.31819
3	0	0.05456	0.03000	0.09925
3	1	0.14642	0.07889	0.27176
4	0	0.19217	0.10086	0.36614
4	1	0.16588	0.08536	0.32236
5	2	0.15488	0.07316	0.32788
5	3	0.07282	0.02942	0.18025

Table 10.3.6. Norway pout in Subarea 4 and Division 3.a. Stock Summary Table. Baseline run with SESAM September 2025. Estimated yearly and quarterly recruitment (millions), spawning-stock biomass SSB (t), total-stock biomass TSB (t) and fishing mortality for ages 1–2 (F_{1-2}).

Time	Recruits	Low	High	SSB	Low	High	TSB	Low	High	F_{12}	Low	High
1984				398760	228960	694484	752920	454630	1246925	0.996	0.531	1.868
1984.25				263296	148787	465933	588234	346059	999886			
1984.5	43360	22535	83428	281280	157939	500944	747125	432686	1290071			
1984.75				142238	73267	276136	408055	224117	742955			
1985				247961	133089	461983	419508	243963	721367	0.954	0.468	1.947

Time	Recruits	Low	High	SSB	Low	High	TSB	Low	High	F ₁₂	Low	High
1985.25				143174	73640	278363	299020	167929	532445			
1985.5	31869	16336	62168	148712	79184	279291	379428	215354	668505			
1985.75				75212	36177	156364	212878	112423	403096			
1986				138058	70622	269890	260522	144028	471241	0.632	0.308	1.298
1986.25				88231	43275	179889	201340	107687	376440			
1986.5	54438	27449	107962	99341	50524	195325	271816	147065	502389			
1986.75				57978	27742	121168	170207	88259	328243			
1987				154650	83469	286531	385474	213444	696156	0.637	0.299	1.355
1987.25				111184	58287	212084	337614	181211	629006			
1987.5	11511	5614	23601	139086	74715	258915	490302	264824	907758			
1987.75				88350	45803	170421	328406	173208	622666			
1988				194850	94794	400514	236874	122169	459276	0.437	0.223	0.853
1988.25				125248	58171	269669	168973	85503	333929			
1988.5	52192	26353	103369	141457	65799	304107	213944	110694	413502			
1988.75				92744	40935	210124	147763	74090	294696			
1989				142070	73816	273435	350544	190882	643755	0.477	0.242	0.939
1989.25				117170	59809	229544	320934	173117	594967			
1989.5	53853	27031	107289	138096	72305	263754	448575	242091	831174			
1989.75				89060	44824	176949	298601	157174	567283			
1990				234153	125438	437090	443045	248258	790667	0.440	0.225	0.861
1990.25				164127	85675	314416	371317	202844	679716			
1990.5	65695	32748	131789	181550	93536	352382	493419	264434	920690			
1990.75				117016	58148	235481	337225	176695	643599			
1991				284634	152328	531854	543474	301861	978479	0.422	0.214	0.831
1991.25				199020	103010	384515	450682	242765	836670			
1991.5	118615	59197	237674	227830	117989	439926	621678	331565	1165635			
1991.75				147264	73235	296126	429095	223162	825061			
1992				407615	219537	756818	884667	487323	1605993	0.400	0.202	0.793
1992.25				309221	160942	594112	772070	413748	1440712			
1992.5	63393	31976	125681	380637	196998	735463	1090701	580379	2049746			

Time	Recruits	Low	High	SSB	Low	High	TSB	Low	High	F ₁₂	Low	High
1992.75				246735	121203	502282	724537	373516	1405439			
1993				524415	266395	1032343	769638	413574	1432251	0.426	0.194	0.936
1993.25				343151	168001	700905	571987	300680	1088098			
1993.5	55142	27398	110977	348838	170590	713337	682758	360062	1294662			
1993.75				196745	89577	432128	408618	204626	815970			
1994				286283	141111	580806	493316	258247	942356	0.356	0.167	0.759
1994.25				201778	95661	425612	395288	201423	775745			
1994.5	158931	77706	325057	214063	102931	445180	504526	258116	986174			
1994.75				136251	61718	300792	337814	164957	691808			
1995				384007	197352	747199	1027031	532450	1981019	0.243	0.114	0.518
1995.25				315518	158363	628629	954588	484941	1879068			
1995.5	63422	30806	130571	398949	199442	798026	1381805	696090	2743014			
1995.75				268840	130259	554856	955995	469936	1944788			
1996				694231	338741	1422788	925008	474165	1804520	0.226	0.104	0.490
1996.25				470003	223788	987109	703198	356032	1388885			
1996.5	132726	63838	275953	510865	241235	1081861	871746	441137	1722688			
1996.75				331756	145677	755522	592686	283911	1237279			
1997				546231	266061	1121427	1086513	552875	2135223	0.224	0.101	0.497
1997.25				427647	204818	892900	957223	481286	1903808			
1997.5	27199	13013	56847	489257	239659	998801	1355881	682704	2692839			
1997.75				324092	151263	694389	951515	459242	1971465			
1998				686003	322113	1460977	794846	387025	1632402	0.202	0.092	0.443
1998.25				456583	210119	992145	566279	273766	1171334			
1998.5	48968	23871	100451	462684	210813	1015478	632721	307608	1301449			
1998.75				298700	128530	694171	428019	199327	919093			
1999				338775	156910	731429	545490	271200	1097196	0.234	0.105	0.519
1999.25				270168	123028	593284	482082	238456	974617			
1999.5	114193	55381	235463	274894	128390	588570	612985	307177	1223238			
1999.75				178892	80645	396830	431938	208707	893933			
2000				415524	206130	837628	900621	460817	1760173	0.211	0.093	0.479

Time	Recruits	Low	High	SSB	Low	High	TSB	Low	High	F ₁₂	Low	High
2000.25				334035	164188	679583	829744	419331	1641840			
2000.5	31461	15227	65001	418893	206108	851357	1232089	617230	2459446			
2000.75				288745	137234	607530	882283	429471	1812517			
2001				611304	285275	1309938	732577	355335	1510317	0.166	0.072	0.381
2001.25				408814	185835	899341	526273	251988	1099110			
2001.5	32532	15777	67079	414721	186642	921514	592558	284955	1232213			
2001.75				283518	123237	652260	413026	193779	880337			
2002				302811	137162	668509	438512	212107	906581	0.243	0.101	0.584
2002.25				225402	99765	509253	360408	172809	751664			
2002.5	24775	11717	52385	216114	98271	475268	417902	205269	850792			
2002.75				137790	60235	315201	275488	130953	579550			
2003				186781	86732	402240	278507	136396	568682	0.167	0.069	0.404
2003.25				132706	60844	289443	218222	106442	447387			
2003.5	9561	4535	20158	133437	61932	287502	257427	126977	521898			
2003.75				83562	37089	188268	168750	80162	355238			
2004				118320	54962	254716	156703	76048	322898	0.145	0.056	0.371
2004.25				86683	39494	190254	123629	59435	257156			
2004.5	9506	4497	20095	90126	41297	196690	148618	72250	305704			
2004.75				58274	25672	132280	100125	47149	212622			
2005				72912	33610	158173	111491	54153	229538	0.000	0.000	0.001
2005.25				57779	26225	127296	95770	46174	198637			
2005.5	36824	17480	77574	61693	28353	134238	120937	58846	248543			
2005.75				42576	19392	93480	88009	42459	182422			
2006				101042	51437	198486	251746	127309	497814	0.189	0.071	0.503
2006.25				85557	42910	170588	234241	117074	468670			
2006.5	27626	13014	58645	108563	54036	218112	343714	170342	693543			
2006.75				76712	37085	158681	254890	122935	528483			
2007				199889	93132	429021	311158	152489	634927	0.022	0.009	0.052
2007.25				146524	67657	317325	257717	125292	530106			
2007.5	38699	18211	82236	166180	76356	361673	337832	163315	698836			

Time	Recruits	Low	High	SSB	Low	High	TSB	Low	High	F ₁₂	Low	High
2007.75				113833	51152	253322	241780	114581	510185			
2008				218175	105580	450849	380150	189927	760890	0.042	0.018	0.098
2008.25				174070	82979	365158	342257	168526	695081			
2008.5	60884	28327	130860	200387	95107	422209	472755	230278	970555			
2008.75				139096	64401	300424	353054	167194	745523			
2009				321438	156042	662144	603487	301665	1207290	0.070	0.028	0.173
2009.25				248512	120101	514220	533795	264582	1076932			
2009.5	87467	41246	185484	295220	142681	610838	762359	375015	1549784			
2009.75				200079	93091	430027	560166	265705	1180956			
2010				519070	251234	1072444	900919	451597	1797298	0.076	0.032	0.181
2010.25				431297	204812	908232	838131	412461	1703100			
2010.5	7810	3608	16905	512897	240337	1094559	1153777	560467	2375164			
2010.75				346690	155781	771557	807123	379488	1716652			
2011				559463	258318	1211681	592683	277354	1266514	0.042	0.017	0.103
2011.25				385617	175138	849048	419786	194401	906474			
2011.5	13185	6248	27823	391085	175234	872815	448153	206978	970351			
2011.75				262081	113541	604949	306032	137565	680810			
2012				220455	97178	500121	276446	128423	595082	0.118	0.044	0.312
2012.25				181341	77664	423422	238337	109309	519667			
2012.5	64940	31058	135783	168019	72898	387262	260392	122932	551557			
2012.75				113164	48528	263894	187115	87775	398880			
2013				182847	89378	374066	457171	227327	919404	0.202	0.076	0.533
2013.25				156073	75980	320593	436188	215008	884899			
2013.5	19483	9278	40910	187689	92411	381201	607507	301117	1225654			
2013.75				124696	60192	258323	413754	202174	846758			
2014				273789	127225	589196	350751	170479	721652	0.167	0.067	0.418
2014.25				183194	85237	393728	259018	127091	527890			
2014.5	103496	48050	222921	188314	87771	404031	305455	151158	617256			
2014.75				118525	52554	267307	205528	97790	431965			
2015				234286	113581	483269	642484	313187	1318015	0.176	0.069	0.451

Time	Recruits	Low	High	SSB	Low	High	TSB	Low	High	F ₁₂	Low	High
2015.25				191751	92977	395458	578476	282180	1185892			
2015.5	39759	18387	85975	227043	111295	463169	788438	386863	1606858			
2015.75				141678	66946	299833	504504	240665	1057587			
2016				353809	167186	748753	507989	250896	1028524	0.191	0.074	0.494
2016.25				241052	113818	510518	384931	190838	776427			
2016.5	70563	32596	152754	249950	117259	532795	452462	224249	912921			
2016.75				152126	66516	347919	279099	131559	592105			
2017				235165	111372	496558	511585	250195	1046060	0.171	0.067	0.439
2017.25				181206	85689	383198	433552	212543	884374			
2017.5	22784	10792	48100	197748	95029	411498	557643	274868	1131325			
2017.75				125179	57930	270496	360207	172934	750284			
2018				244267	114613	520590	328069	161496	666452	0.193	0.077	0.484
2018.25				162464	76286	345993	243641	121028	490473			
2018.5	81343	39112	169172	161020	75238	344607	281458	140244	564862			
2018.75				100181	44774	224152	187336	90382	388296			
2019				196479	97327	396643	515802	260015	1023216	0.218	0.092	0.519
2019.25				163922	81430	329985	477515	240294	948922			
2019.5	107180	51237	224206	196788	97732	396245	671139	335500	1342556			
2019.75				132813	63450	278003	473000	228648	978489			
2020				426025	208549	870286	861089	435290	1703405	0.215	0.088	0.526
2020.25				317759	156107	646805	740905	373102	1471287			
2020.5	63647	29507	137289	371805	182426	757783	1019697	509149	2042193			
2020.75				239731	112323	511661	690010	332471	1432049			
2021				503466	235450	1076568	747188	365691	1526675	0.195	0.075	0.504
2021.25				347704	162654	743282	579903	284772	1180898			
2021.5	25638	11956	54975	351728	164471	752182	690768	339433	1405759			
2021.75				218740	97834	489063	444031	211403	932642			
2022				291748	132619	641813	392391	187046	823169	0.150	0.058	0.391
2022.25				205141	92663	454149	308838	147651	645992			
2022.5	48250	22118	105255	202120	92184	443162	364266	175955	754113			

Time	Recruits	Low	High	SSB	Low	High	TSB	Low	High	F ₁₂	Low	High
2022.75				126760	55776	288081	242047	113672	515405			
2023				193561	91142	411069	378743	184530	777358	0.080	0.032	0.197
2023.25				147766	69358	314812	332015	160080	688622			
2023.5	9453	4200	21275	160327	75878	338765	427674	205906	888292			
2023.75				102717	46979	224584	282300	132098	603287			
2024				193327	89315	418465	226628	107821	476348	0.041	0.016	0.107
2024.25				127245	57876	279754	155470	73411	329258			
2024.5	8851	3948	19845	124015	55707	276083	164871	77799	349392			
2024.75				79167	34382	182285	108792	50062	236418			
2025				78989	36456	171148	114718	55350	237764			
2025.25				61711	27769	137144	95828	45525	201714			
2025.5	10312	3314	32085	59029	26776	130132	110544	52178	234201			
2025.75				39767	17714	89272	78106	35721	170783			

Table 10.3.6 (cont.). Norway pout in Subarea 4 and Division 3.a. Stock Summary Table. Baseline run with SESAM September 2025. Long-term arithmetic means of yearly recruitment (millions), quarterly spawning-stock biomass SSB (t), quarterly total-stock biomass TSB (t) and yearly fishing mortality for ages 1–2 ($F_{bar}=F_{1-2}$) for the period 1984–2025. Numbers are given for start of the season.

	value
Avg. recruitment	51227.04
Geom. mean recruitment	38358.15
Avg SSB Q 1	307562.82
Avg SSB Q 2	223591.46
Avg SSB Q 3	245058.31
Avg SSB Q 4	158096.98
Avg TSB Q 1	521154.25
Avg TSB Q 2	432986.12
Avg TSB Q 3	566424.92
Avg TSB Q 4	382616.85
Avg. FBAR	0.26

Table 10.6.1. Norway pout in Subarea 4 and Division 3.a. Projected mean weight at age used in the forecast by quarter of year. Baseline run with SESAM September 2025.

Age/Quarter	1	2	3	4
0	6.292	8.352	6.182	6.911
1	10.946	15.909	27.484	27.417
2	26.977	31.087	39.969	39.666
3	40.446	41.611	48.889	47.688

Table 10.6.2. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled such that the fifth percentile of the SSB distribution one year ahead (1st October 2026) equals B_{lim} .

Basis:

F_{bar} (2024 up to Q4) = estimated from in year assessment 1st October 2025, $F_{(age1-2, quarter4\ 2024\ and\ quarter1,2,3\ 2025)} = 0.034$, average from Table 10.3.4 and Table 10.3.6..

SSB (2025 up to Q4) = estimated from in year assessment 1st October 2025 (start Q4) = 39 767 tonnes, Table 10.3.6;

$R(2025)$ = estimated / observed from in year assessment 1st July 2025 (age 0 in start of Q3) = 10 312 million, Table 10.3.6;

Biological parameters (2025–2026): Assume values for M , weight-at-age in the stock, and maturity-at-age for the projection period to be similar to the same parameter values used in the assessment. Assume projected mean weight at ages in the catches by quarter as given in Table 10.6.1.

F , R (Q4 2025 - Q4 2026): (i) Draw K samples from the joint posterior distribution of the states ($\log N$ and $\log F$) in the last year with data, and the recruitment in all years. (ii) Assume that $\log F_t = \log F_{t-4} + \log G_t$, for all future values of t where G_t is some chosen vector of multipliers of the F -process. If $G_t = 1$ for all t this corresponds to assuming the same level and quarterly pattern in F for all future time-steps as in the last data year. (iii) Create K forecasting trajectories starting from the samples of joint posterior distribution of the states. This is done by sampling K recruitments from the vector of historic recruitments obtained in step 2, and then projecting the states forward in time using the stock equation with randomly sampled process errors from their estimated distribution. (iv) Find G_t such that the fifth (or any other) percentile of the catches (total mass) in the projections equals some desired level such as B_{lim} (optional).

	F_{1-2}	SSB	SSB 5 th quantile	median catch
2025.75	0.0006	40782.6453	21024.4018	8.5113
2026	0.0000	66522.8986	33708.1460	0.0442
2026.25	0.0001	53070.4217	25005.5899	1.6653
2026.5	0.0002	59125.1488	25940.1419	5.0921
2026.75		42161.9223	16907.7391	
Sum				15.3129

Table 10.6.3. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to zero (no catch) for the period 1st October 2025 to 1st October 2026.

Basis: Same as above.

	F ₁₋₂	SSB	SSB 5 th quantile	median catch
2025.75	0.0000	40782.6453	21024.4018	0.0000
2026	0.0000	66528.0808	33717.3886	0.0000
2026.25	0.0000	53073.8172	25010.4507	0.0000
2026.5	0.0000	59134.6081	25943.4682	0.0000
2026.75		42170.6547	16910.4303	
Sum				0.0000

Table 10.6.4. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to F status quo for the previous year up to 1st October 2026.

Basis: Same as above.

	F ₁₋₂	SSB	SSB 5 th quantile	median catch
2025.75	0.0908	40782.6453	21024.4018	1250.0216
2026	0.0003	65335.4021	32493.5464	6.4372
2026.25	0.0121	52128.2609	24194.2802	240.7786
2026.5	0.0331	58012.7121	25275.3270	741.9662
2026.75		40800.5764	16309.1842	
Sum				2239.2036

Table 10.6.5. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled such that the median of the SSB distribution one year a head (1st October 2026) equals B_{lim}.

Basis: Same as above.

	F ₁₋₂	SSB	SSB 5 th quantile	median catch
2025.75	0.0006	40782.6453	21024.4018	8.5113
2026	0.0000	66522.8986	33708.1460	0.0442
2026.25	0.0001	53070.4217	25005.5899	1.6653
2026.5	0.0002	59125.1488	25940.1419	5.0921
2026.75		42161.9223	16907.7391	
Sum				15.3129

Table 10.6.6. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled such that SSB one year a head (1st October 2026) equals B_{pa} .

Basis: Same as above.

	F_{1-2}	SSB	SSB 5 th quantile	median catch
2025.75	0.0006	40782.6453	21024.4018	8.5113
2026	0.0000	66522.8986	33708.1460	0.0442
2026.25	0.0001	53070.4217	25005.5899	1.6653
2026.5	0.0002	59125.1488	25940.1419	5.0921
2026.75		42161.9223	16907.7391	
Sum				15.3129

Table 10.6.7. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to 0.2 ($F_{cap} = 0.2$) for the period 1st October 2025 to 1st October 2026.

Basis: Same as above.

	F_{1-2}	SSB	SSB 5 th quantile	median catch
2025.75	0.5366	40782.6453	21024.4018	7023.8222
2026	0.0018	59669.6068	27491.3118	34.6196
2026.25	0.0715	47992.2086	20671.6500	1303.6758
2026.5	0.1955	52603.2602	21368.8212	3995.8861
2026.75		35720.8521	13454.6225	
Sum				12358.0038

Table 10.6.8. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to 0.3 ($F_{cap} = 0.3$) for the period 1st October 2025 to 1st October 2026.

Basis: Same as above.

	F_{1-2}	SSB	SSB 5 th quantile	median catch
2025.75	0.8049	40782.6453	21024.4018	10216.8669
2026	0.0027	56644.1530	24918.2833	49.6476
2026.25	0.1072	45756.9813	18784.3005	1867.2640
2026.5	0.2932	50159.3723	19422.5165	5712.6936
2026.75		33011.5887	11723.6409	
Sum				17846.4721

Table 10.6.9. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to 0.4 ($F_{cap} = 0.4$) for the period 1st October 2025 to 1st October 2026.

Basis: Same as above.

	F_{1-2}	SSB	SSB 5 th quantile	median catch
2025.75	1.0732	40782.6453	21024.4018	13231.6672
2026	0.0037	54116.3553	22507.1819	63.2697
2026.25	0.1429	43899.4787	17022.4706	2390.6711
2026.5	0.3909	47729.0347	17740.7750	7271.8561
2026.75		30713.0710	10394.8104	
Sum				22957.4641

Table 10.6.10 Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to 0.5 ($F_{cap} = 0.5$) for the period 1st October 2025 to 1st October 2026.

Basis: Same as above.

	F_{1-2}	SSB	SSB 5 th quantile	median catch
2025.75	1.3415	40782.6453	21024.4018	16020.3835
2026	0.0046	51799.2847	20133.3853	75.7385
2026.25	0.1787	42074.2205	15682.2924	2877.0368
2026.5	0.4886	45720.4441	16218.8081	8731.1383
2026.75		28793.0944	9350.5080	
Sum				27704.2971

Table 10.6.11. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to 0.6 ($F_{cap} = 0.6$) for the period 1st October 2025 to 1st October 2026.

Basis: Same as above.

	F_{1-2}	SSB	SSB 5 th quantile	median catch
2025.75	1.6097	40782.6453	21024.4018	18710.4569
2026	0.0055	49671.9172	18215.8153	87.0865
2026.25	0.2144	40293.5849	14501.3572	3326.8231
2026.5	0.5864	43789.6303	14905.0293	10010.5999
2026.75		26958.1543	8372.4791	
Sum				32134.9664

Table 10.6.12. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to 0.7 ($F_{cap} = 0.7$) for the period 1st October 2025 to 1st October 2026.

Basis: Same as above

	F_{1-2}	SSB	SSB 5 th quantile	median catch
2025.75	1.8780	40782.6453	21024.4018	21174.3388
2026	0.0064	47657.3862	16872.4718	97.6395
2026.25	0.2502	38707.5048	13568.0869	3764.6531
2026.5	0.6841	41937.1452	13737.4646	11207.5520
2026.75		25271.7507	7515.2196	
Sum				36244.1834

Table 10.6.13. Norway pout in Subarea 4 and Division 3.a. The quarterly minima of the estimated SSB time-series (1984–2016) from the SESAM Benchmark Assessment Baseline Run from the Norway pout benchmark assessment under ICES WKPOUT 2016 with previous to 2020 IBTS Q1 and Q3 survey indices. The estimates are quarterly minima in tonnes estimated at the beginning of the season. The estimates are B_{loss} estimates which equals B_{lim} according to the ICES WKPOUT 2016 benchmark assessment which by 1st October is $B_{lim}=39\,450$ t.

SSB	Quarter	Year
72101.23	1	2005
55109.70	2	2005
57961.80	3	2005
39447.18	4	2005

Table 10.6.14. Norway pout in Subarea 4 and Division 3.a. The quarterly minima of the estimated SSB time-series (1984–2016) from the SESAM updated Benchmark Assessment Run (Run: NP_Sep17_fixC_Benchmark2016Data_NewIBTS, www.stockassessment.org) with new IBTS Q1 and Q3 survey indices made available in 2020. The estimates are quarterly minima in tonnes estimated at the beginning of the season. The estimates are B_{loss} estimates which equals B_{lim} according to the assessment run above which by 1st October is $B_{lim}=42\,573$ t.

SSB	Quarter	Year
77586	1	2005
59514	2	2005
62543	3	2005
42573	4	2005

10.13 Figures

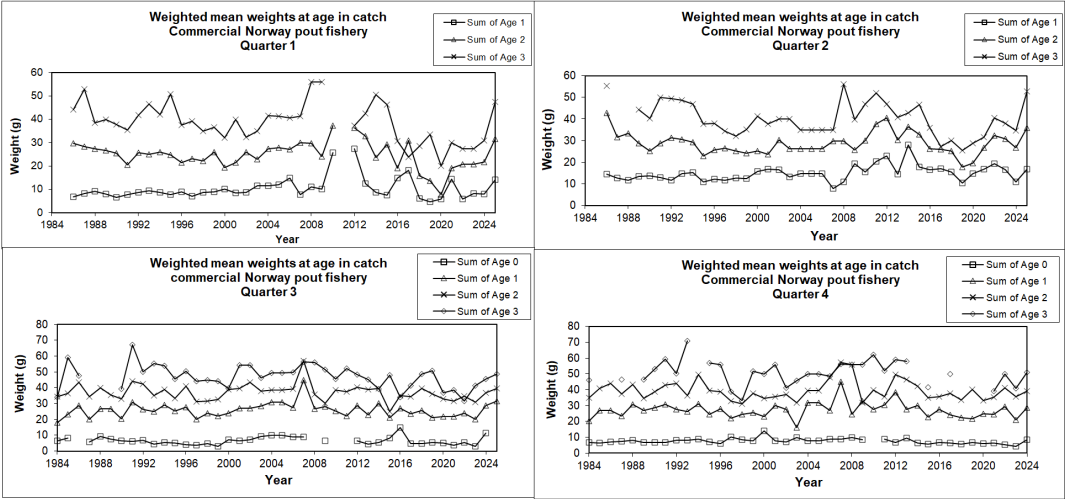


Figure 10.2.1. Norway pout in Subarea 4 and Division 3.a. Weighted mean weights at age in catch of the Danish and Norwegian commercial fishery for Norway pout by quarter of year during the period 1984–2025.

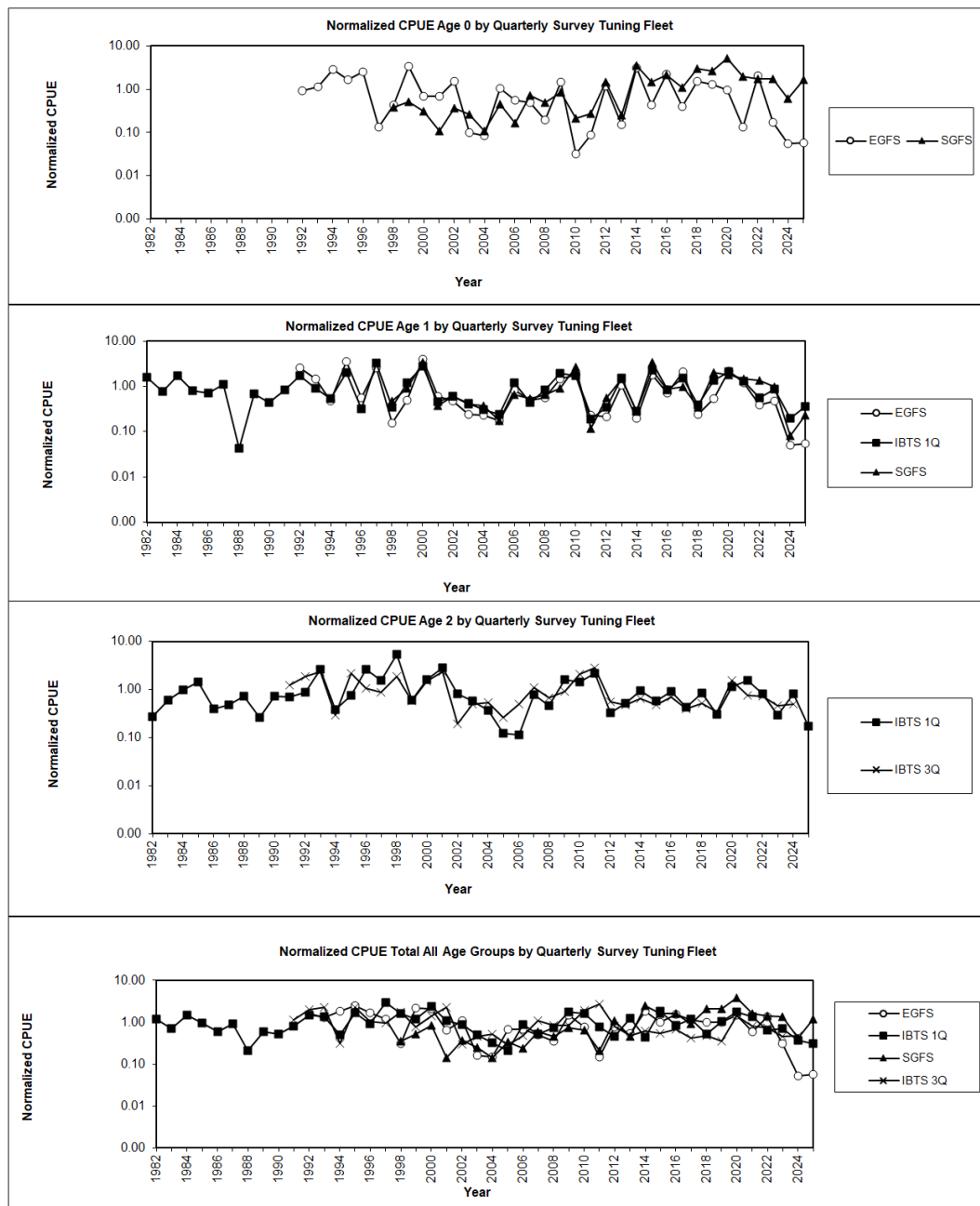


Figure 10.2.2. Norway pout in Subarea 4 and Division 3.a. Trends in CPUE (normalized to unit mean) by quarterly survey tuning fleet used in the Norway pout assessment for each age group and all age groups together.

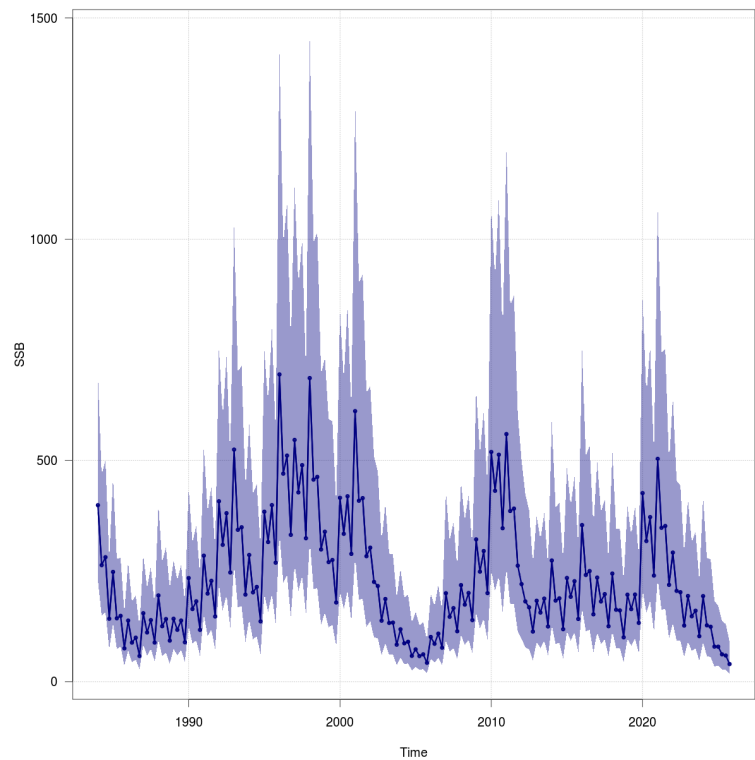


Figure 10.3.1. Norway pout in Subarea 4 and Division 3.a. Stock Summary Plots: SSB (t), quarterly. SESAM baseline run September 2025. Quarterly estimated SSB and confidence interval from SESAM (blue) where connecting lines are interpolations.

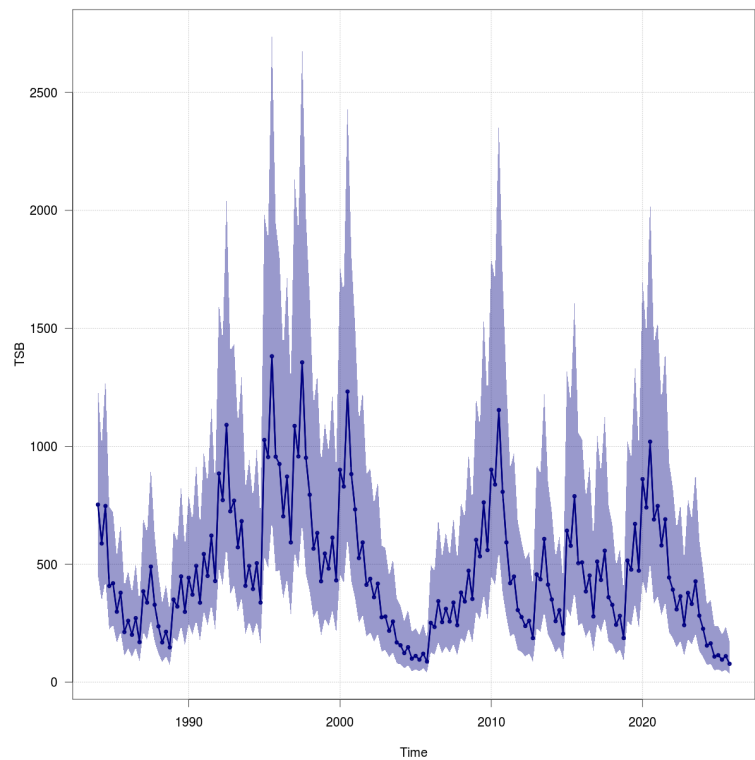


Figure 10.3.2. Norway pout in Subarea 4 and Division 3.a. Stock Summary Plots: TSB (t), quarterly. SESAM baseline run September 2025.

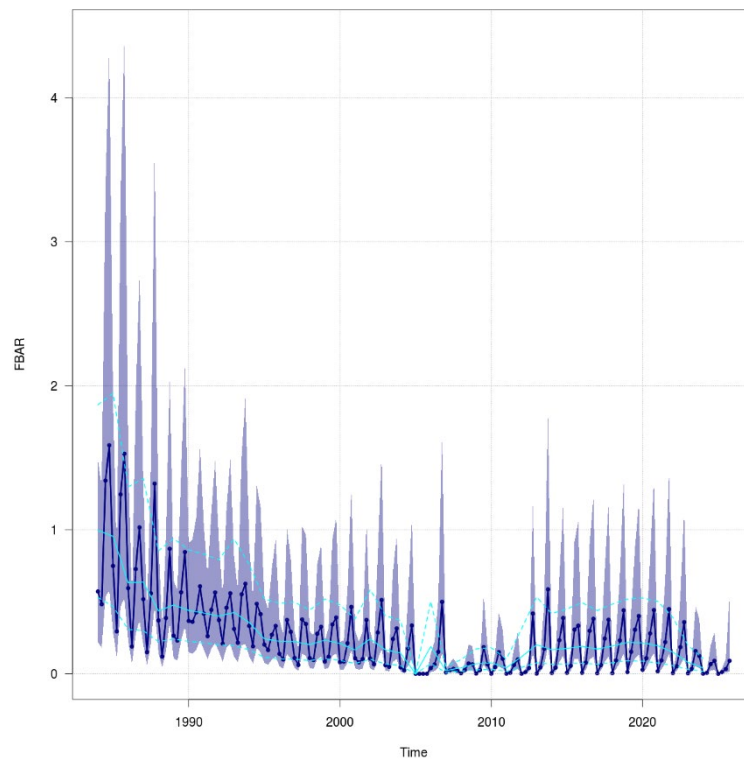


Figure 10.3.3. Norway pout in Subarea 4 and Division 3.a. Stock Summary Plots: $F_{bar} = F_{1-2}$, quarterly. SESAM baseline run September 2025. Blue is quarterly values from SESAM, cyan is the yearly average from SESAM.

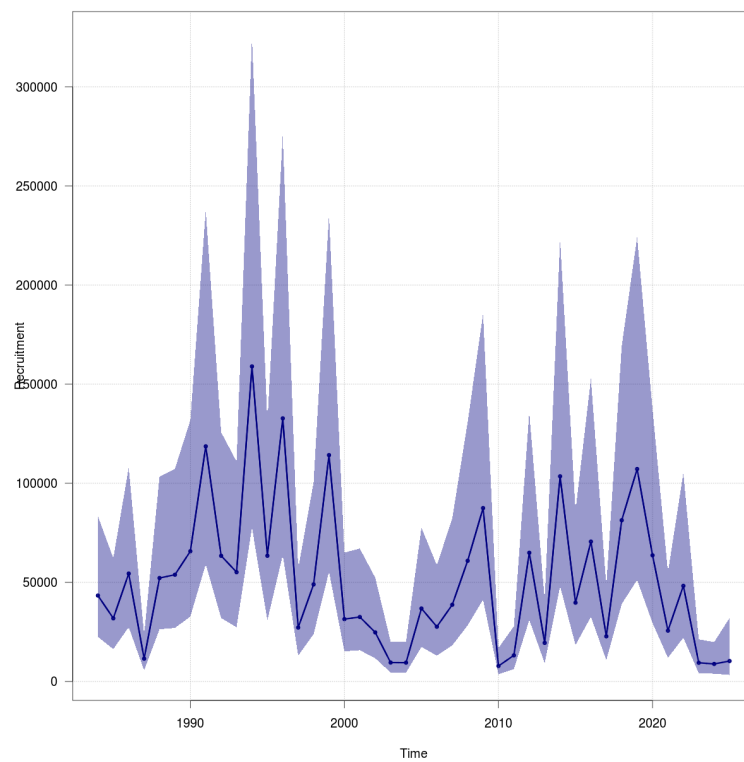


Figure 10.3.4. Norway pout in Subarea 4 and Division 3.a. Stock Summary Plots: Recruitment (millions), yearly. SESAM baseline run September 2025. Blue is SESAM.

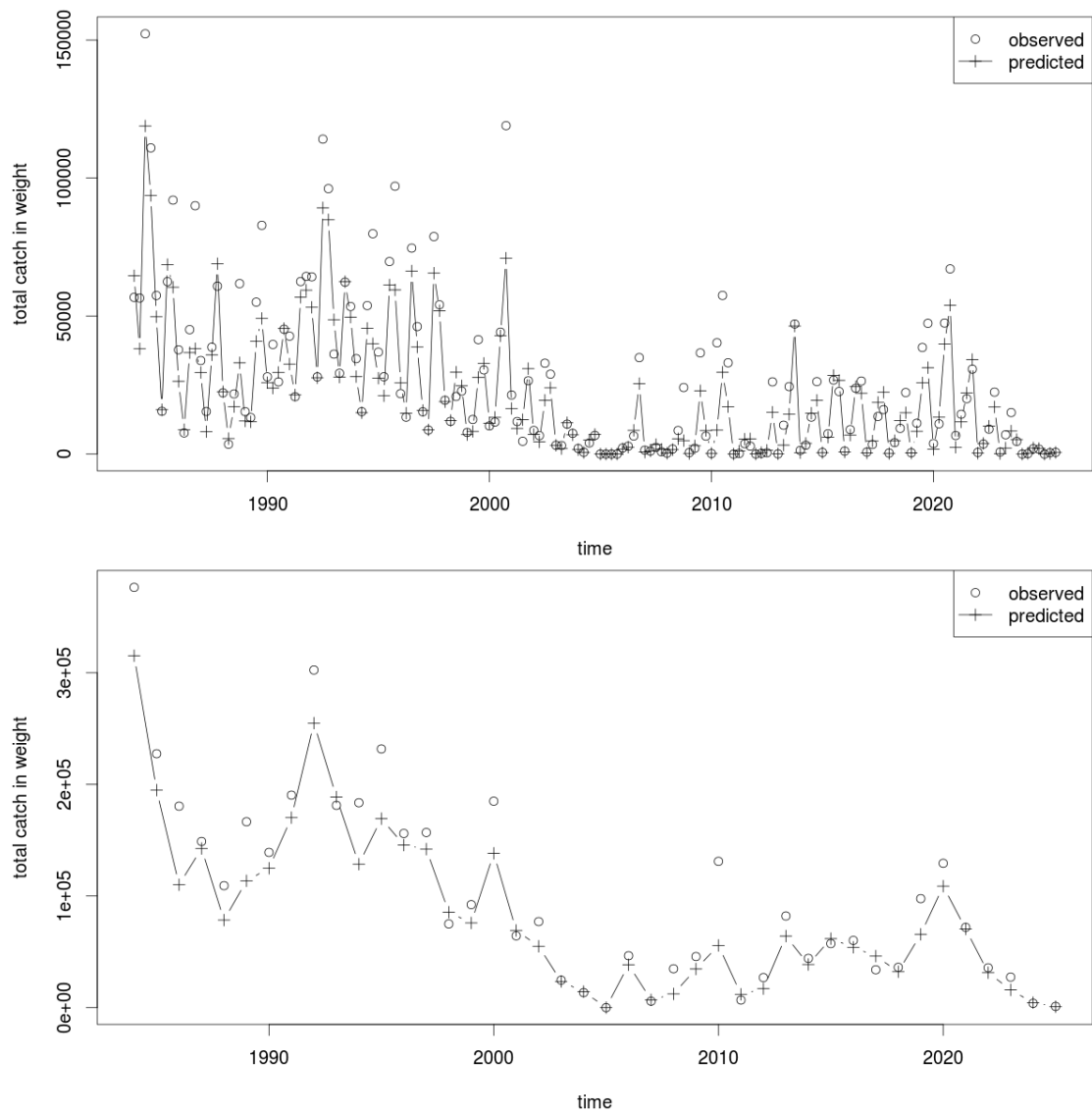


Figure 10.3.5. Norway pout in Subarea 4 and Division 3.a. Stock Summary Plots: Yield = Total Catch (t), quarterly and yearly. SESAM baseline run September 2025.

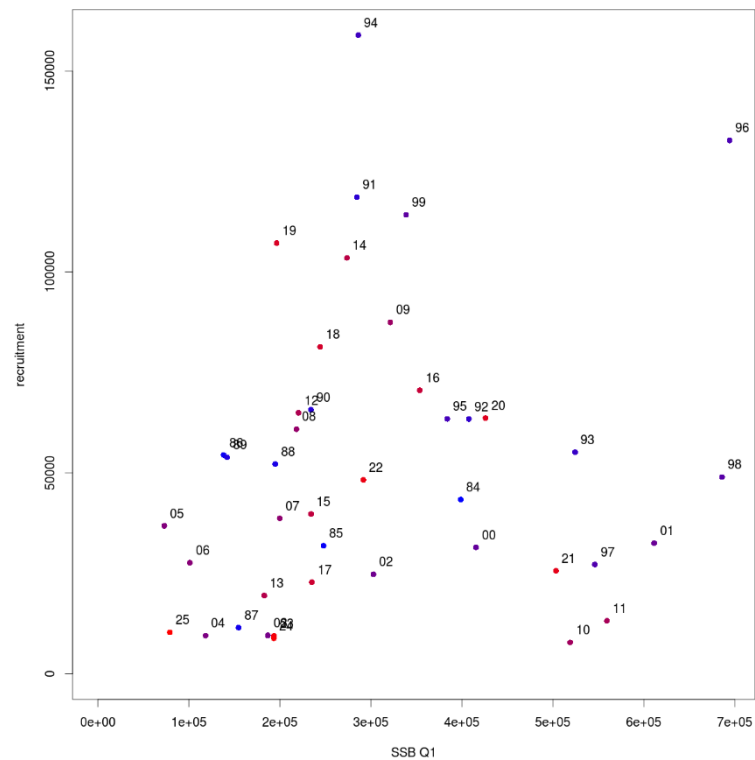


Figure 10.3.6. Norway pout in Subarea 4 and Division 3.a. Stock Summary Plots: Stock (SSB) – Recruitment Plot Quarter 1. SESAM baseline run September 2025.

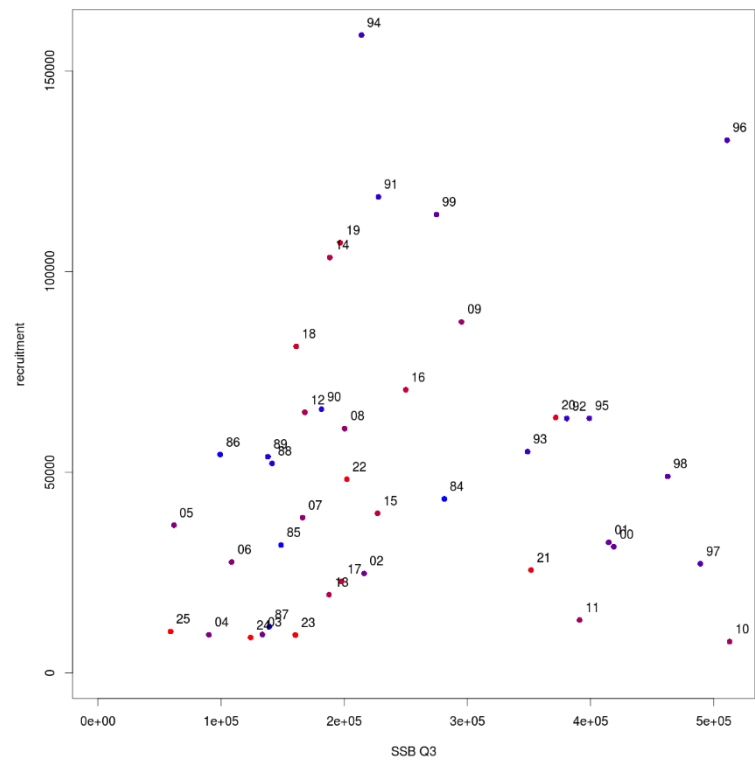


Figure 10.3.7. Norway pout in Subarea 4 and Division 3.a. Stock Summary Plots: Stock (SSB) – Recruitment Plot Quarter 3. SESAM baseline run September 2025.

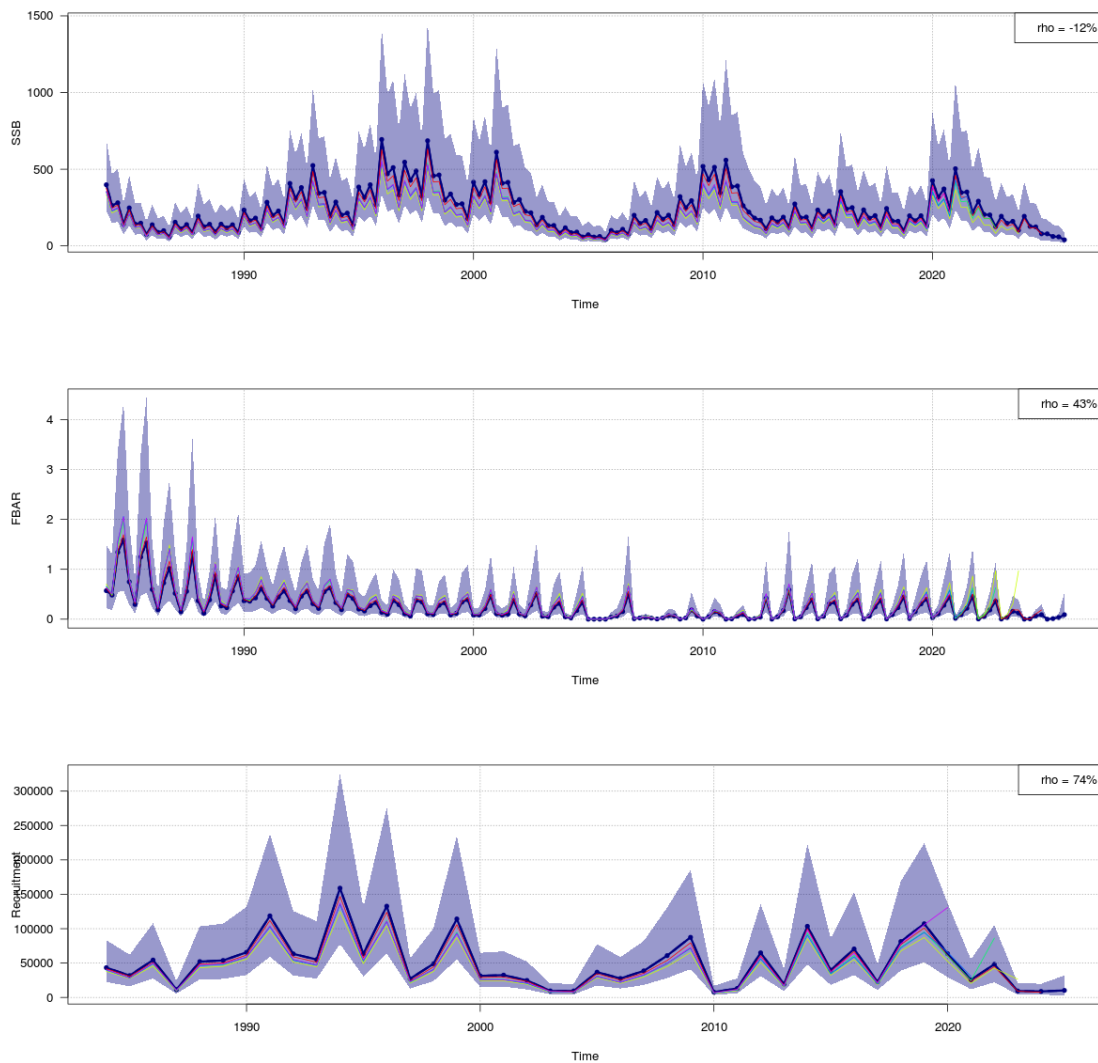


Figure 10.3.8. Norway pout in Subarea 4 and Division 3.a. Retrospective plots of baseline SESAM assessment September 2025, with terminal assessment year ranging from 2020–2025. Represent 5 year retrospective runs.

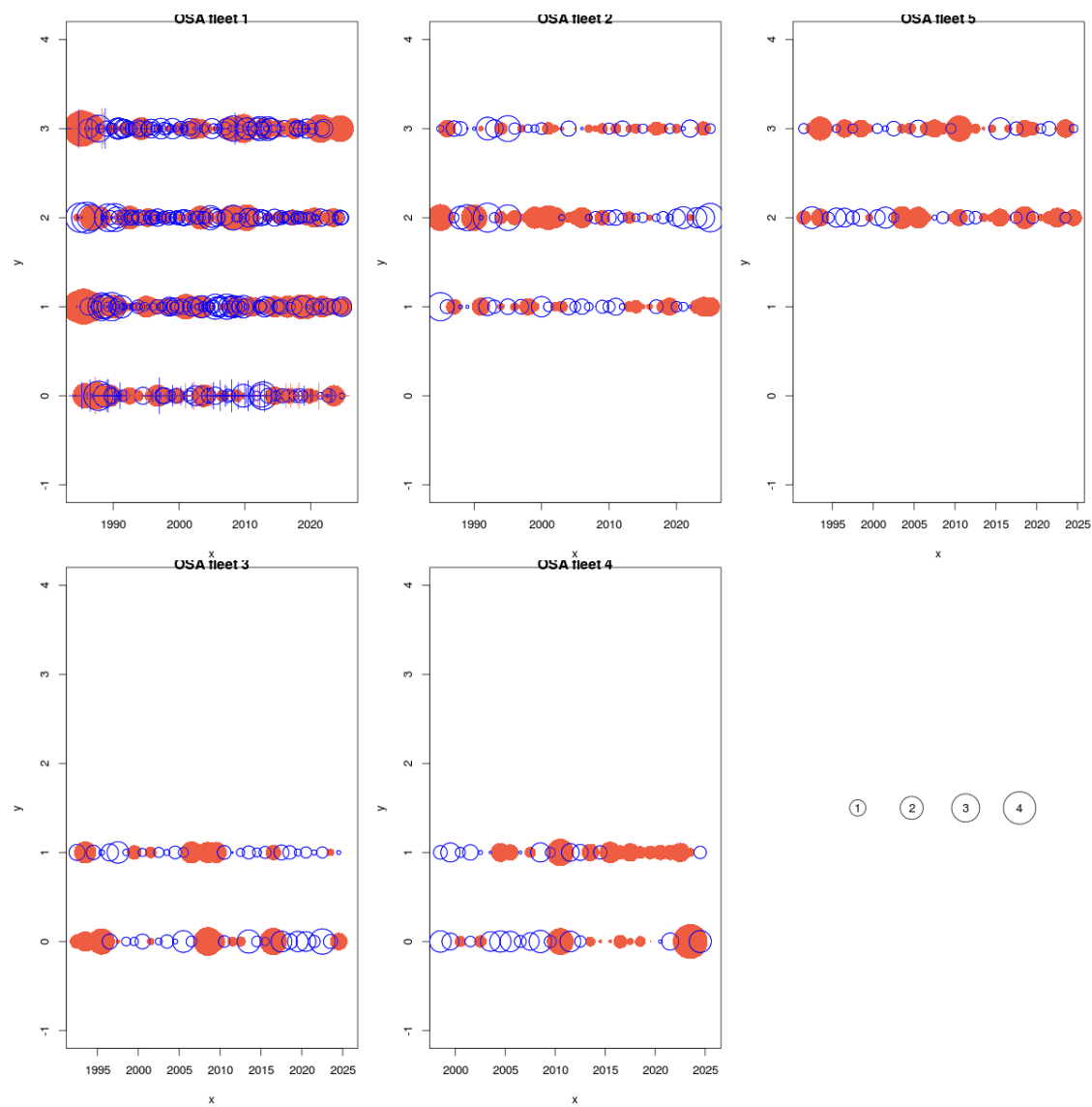


Figure 10.3.9. Norway pout in Subarea 4 and Division 3.a. Assessment Diagnostics Plots by fleet: One step ahead residuals (see Berg and Nielsen 2016). SESAM baseline run September 2025.

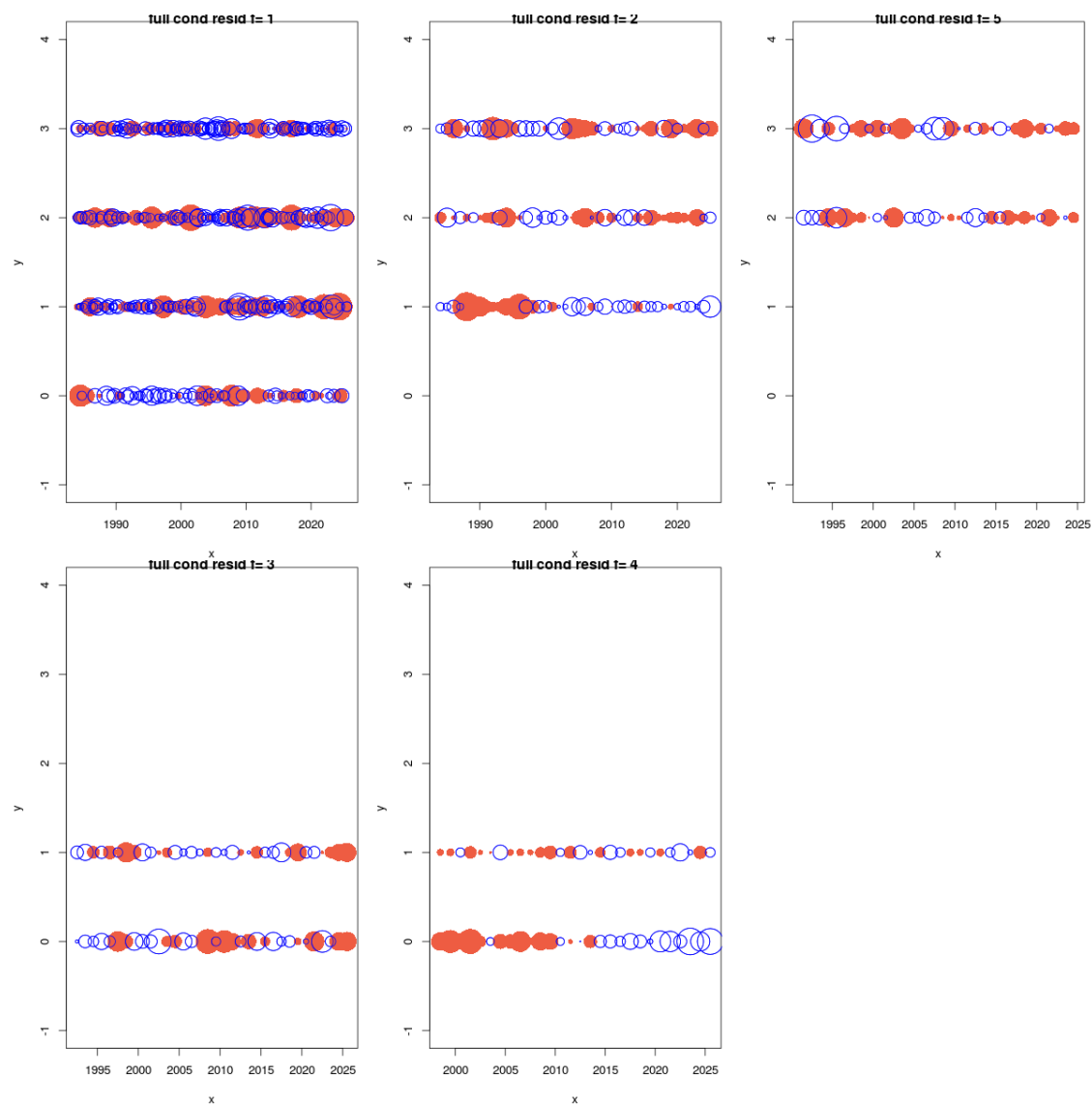


Figure 10.3.10. Norway pout in Subarea 4 and Division 3.a. Assessment Diagnostics Plots: Full conditional residuals or auxiliary residuals (see Berg and Nielsen 2016). SESAM baseline run September 2025.

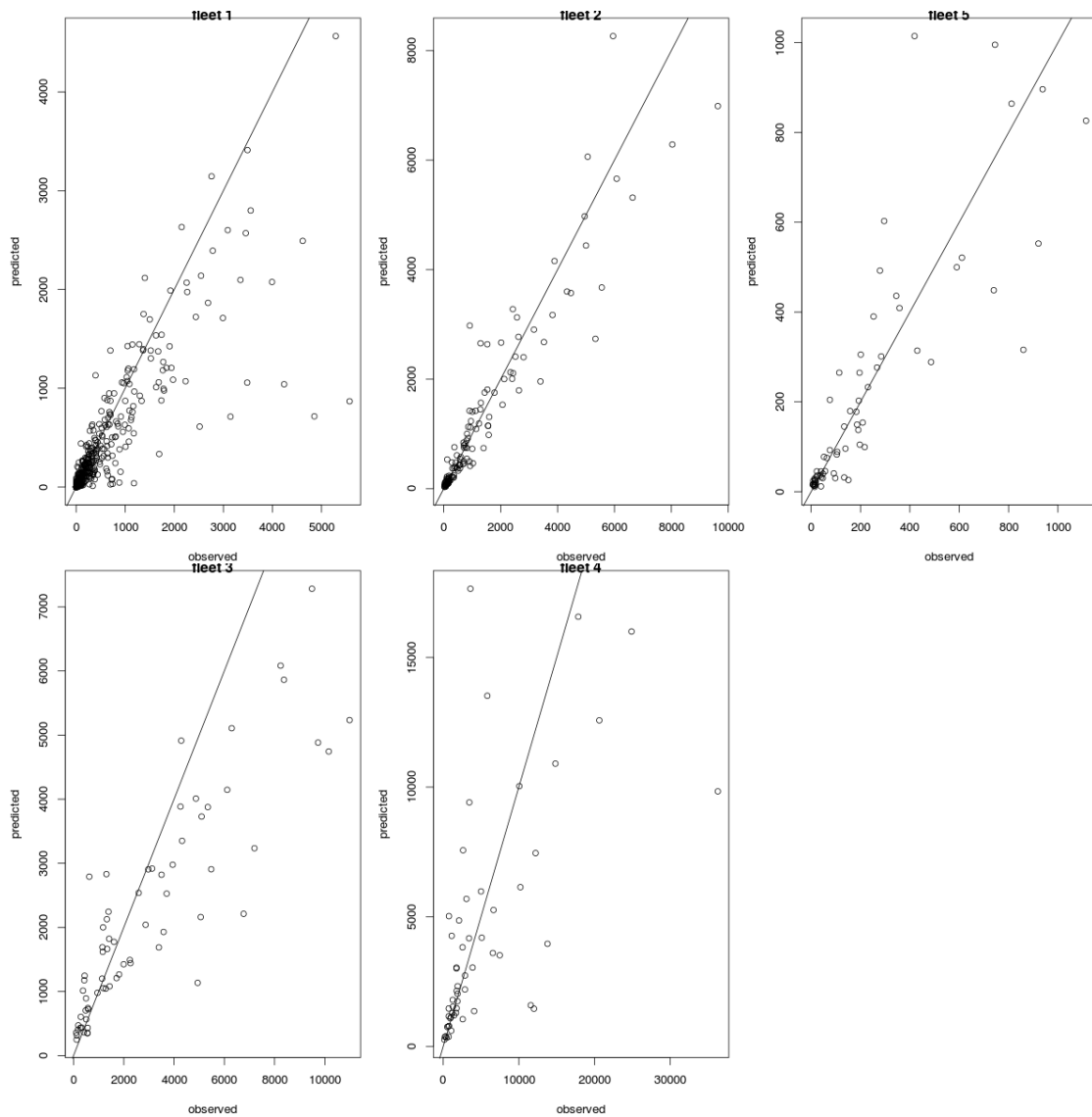


Figure 10.3.11. Norway pout in Subarea 4 and Division 3.a. Assessment Diagnostics Plots by fleet. SESAM baseline run September 2025.

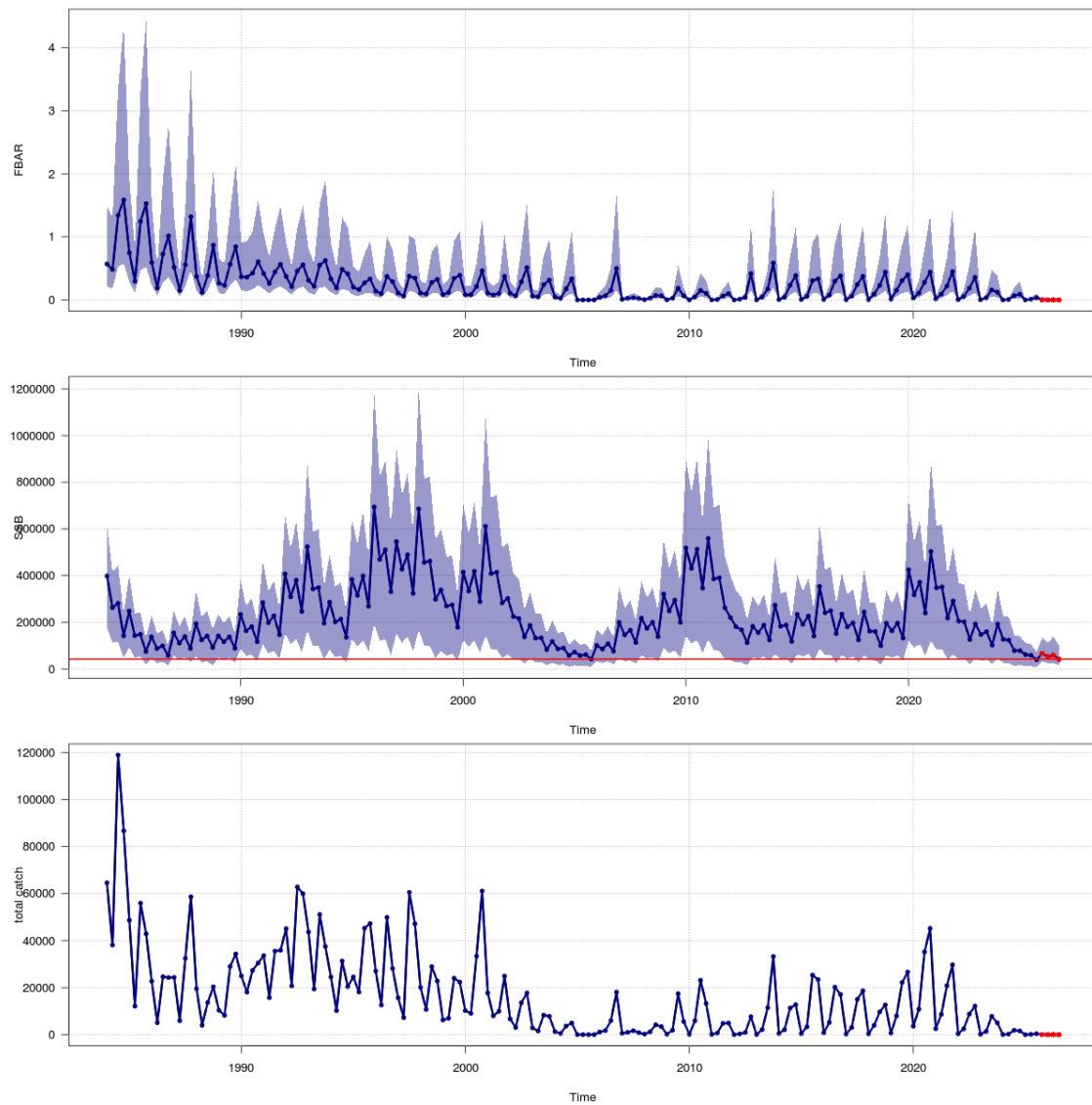


Figure 10.6.1. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled such that the fifth percentile of the SSB distribution one year ahead (1st October 2026) equals B_{lim} .

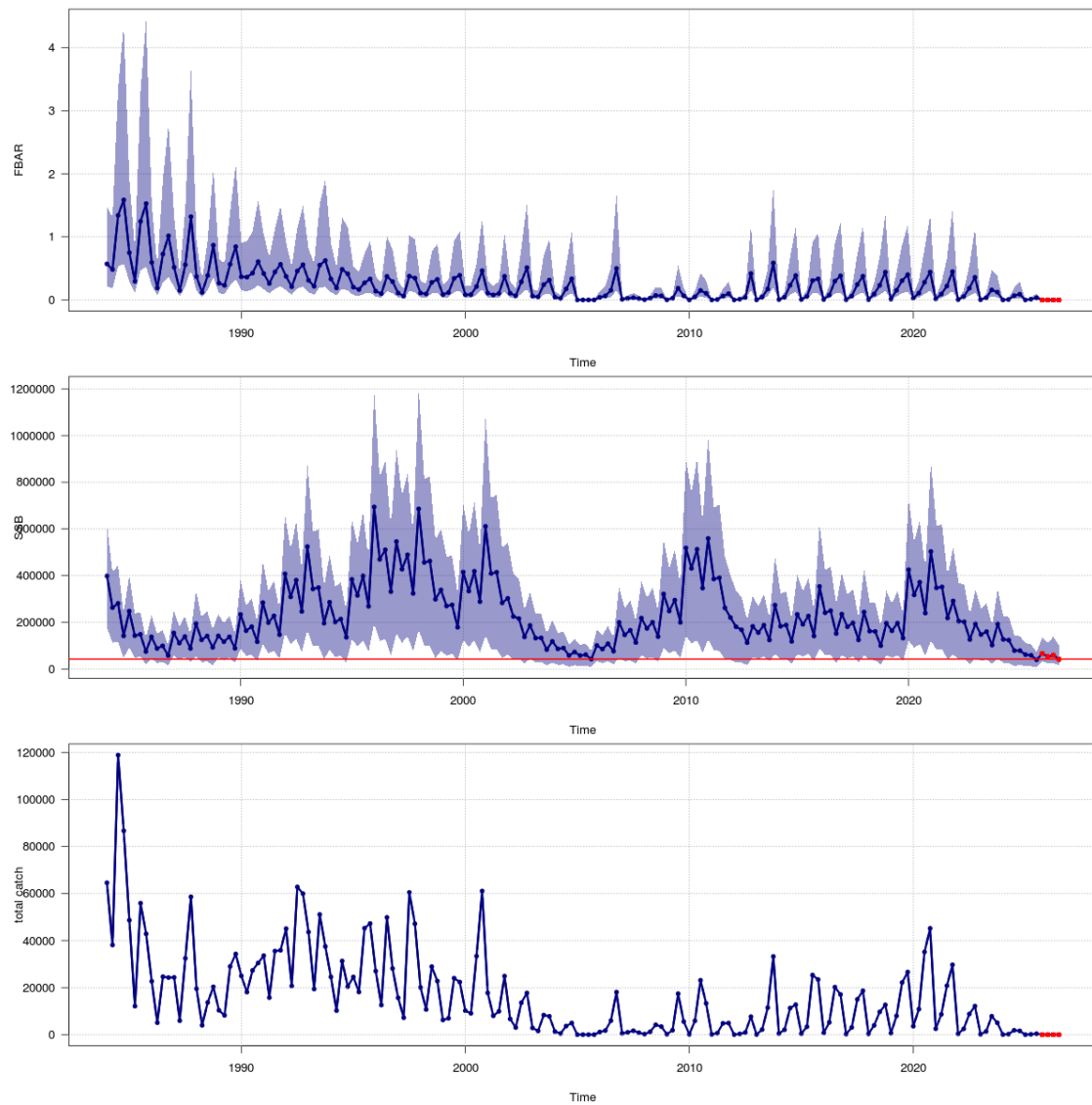


Figure 10.6.2. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to zero (no catch) for the period 1st October 2025 to 1st October 2026.

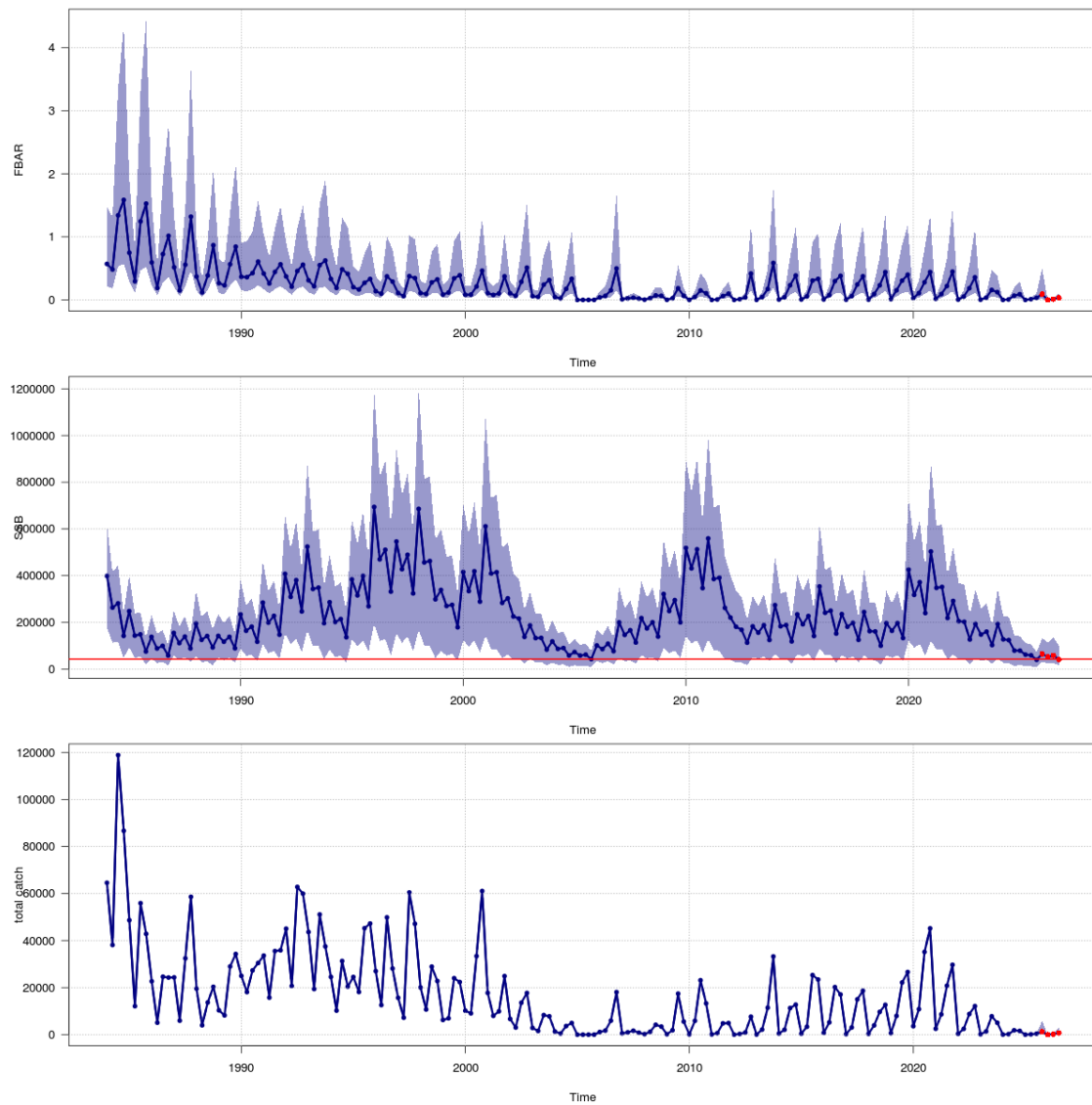


Figure 10.6.3. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to F status quo for the previous year to 1st October 2026.

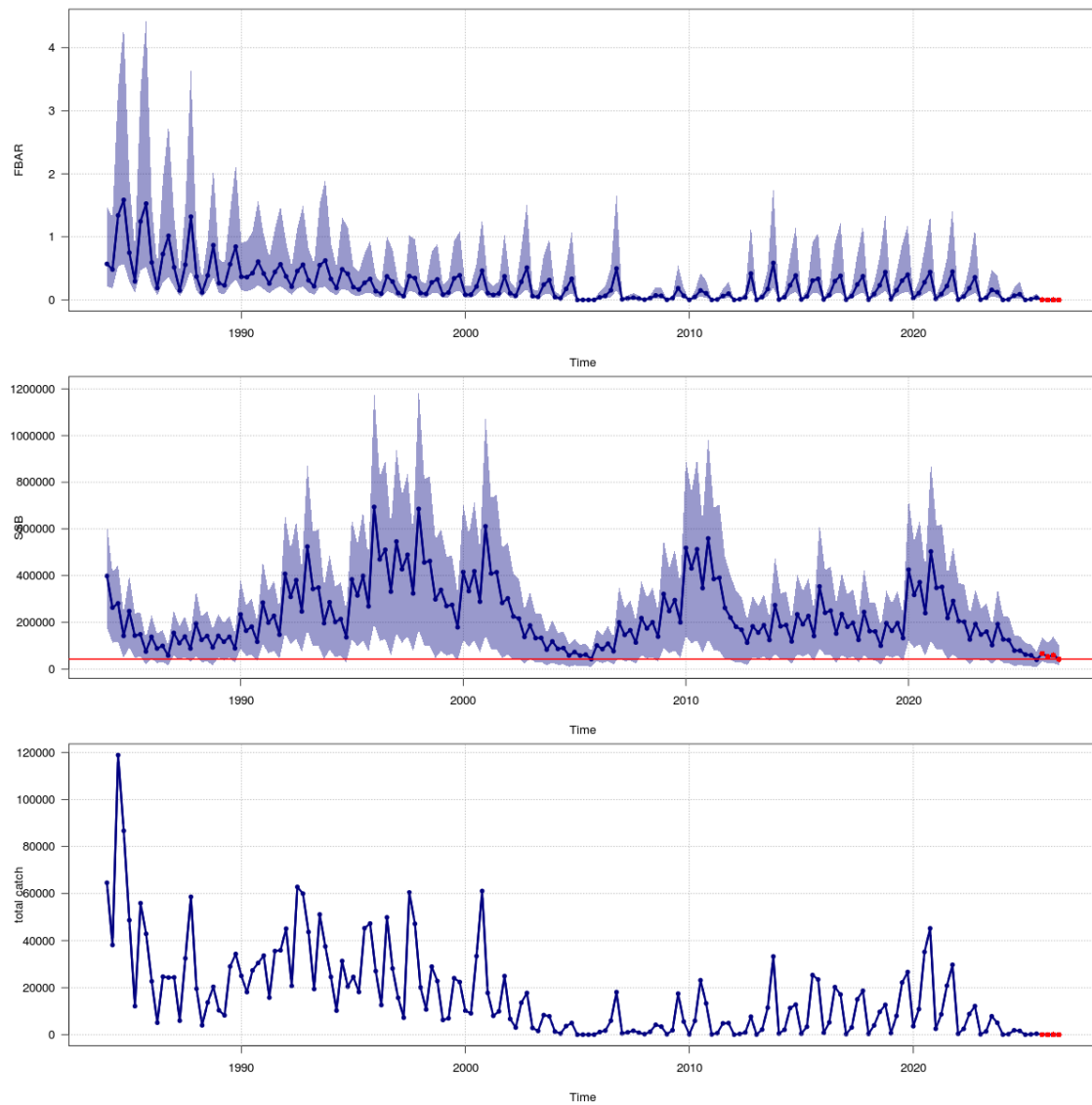


Figure 10.6.4. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled such that the median of the SSB distribution one year ahead (1st October 2026) equals B_{lim} .

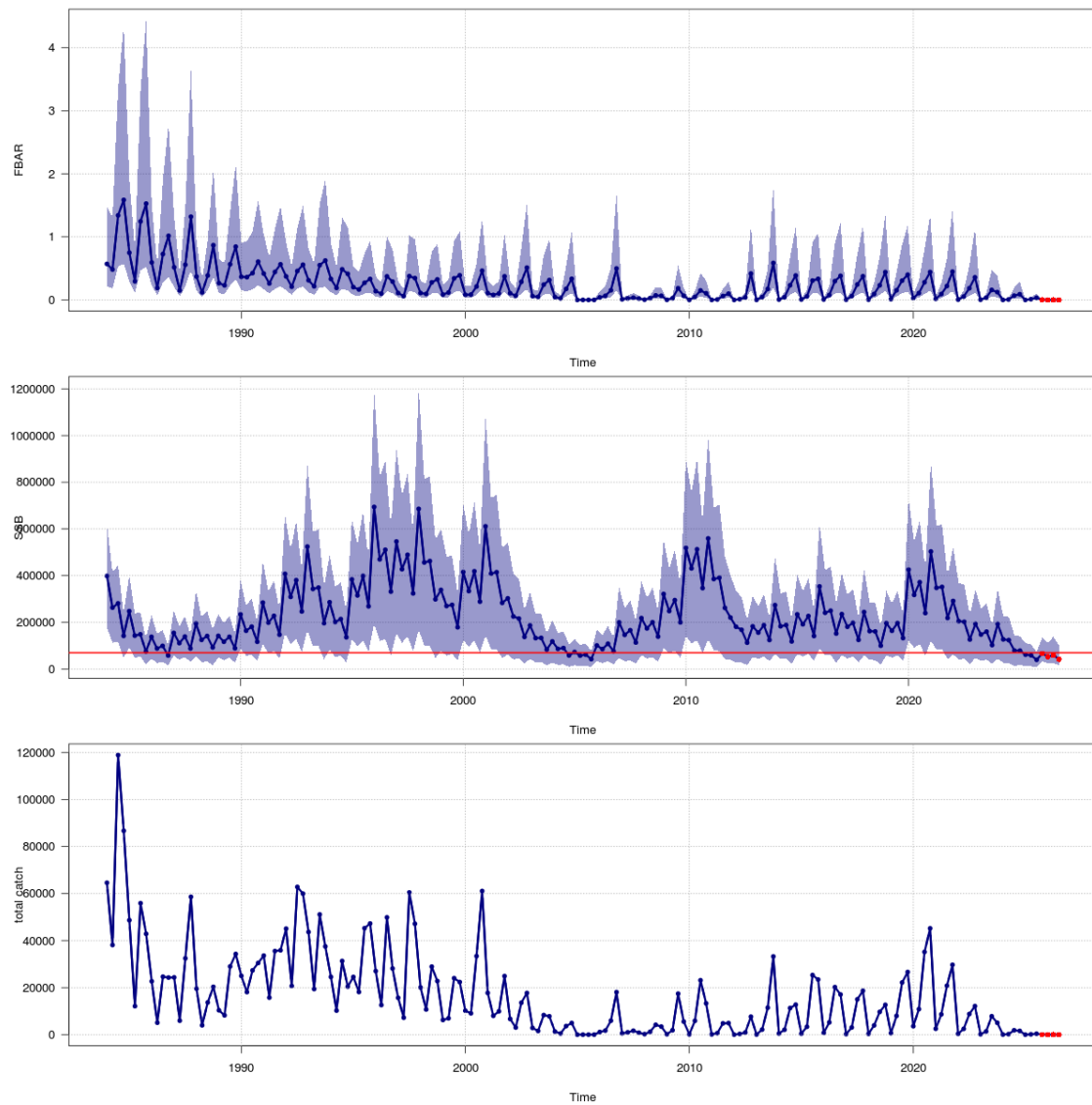


Figure 10.6.5. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled such that the SSB distribution one year ahead (1st October 2026) equals Bpa.

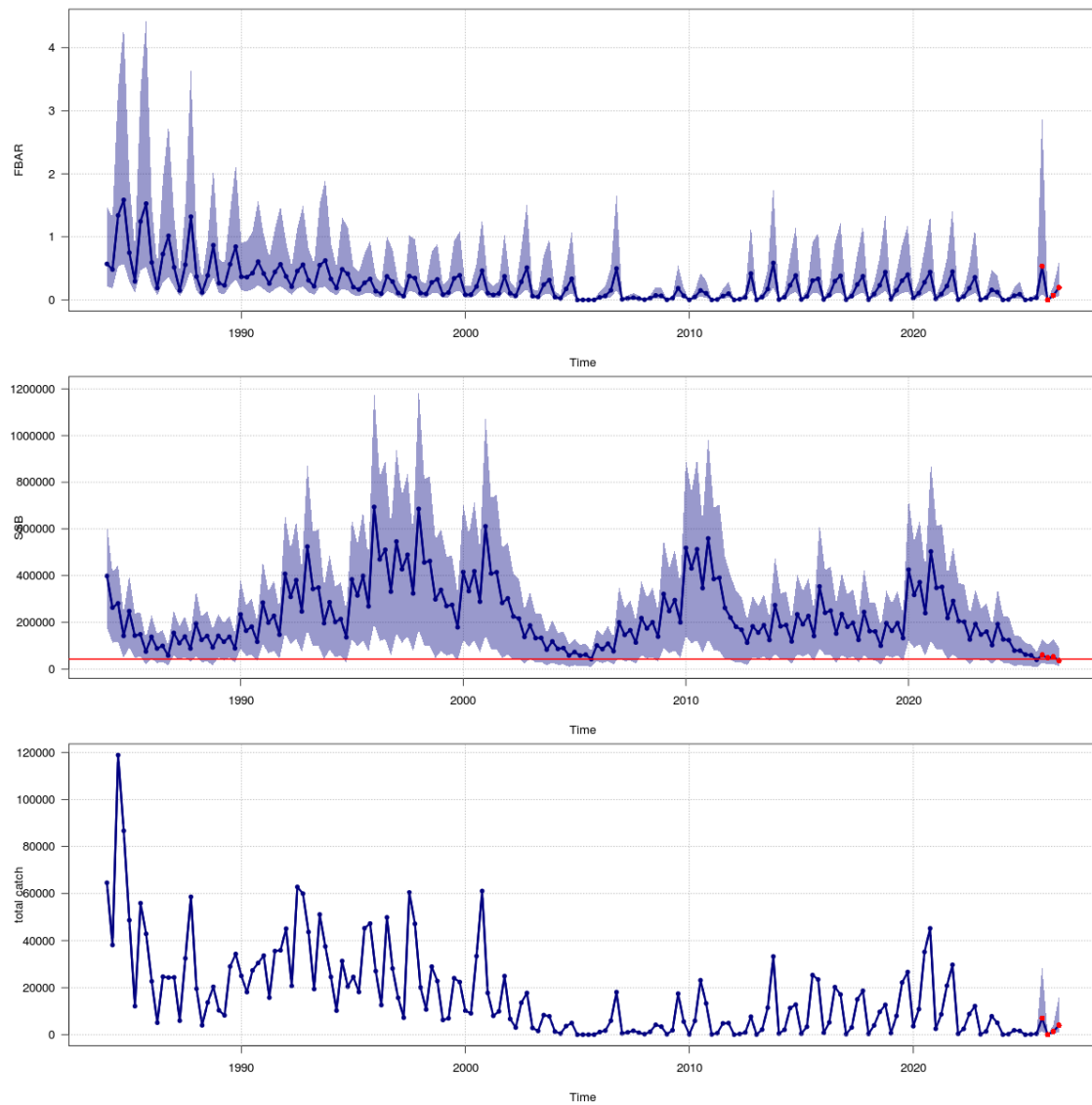


Figure 10.6.6. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to 0.2 ($F_{cap} = 0.2$) for the period 1st October 2025 to 1st October 2026.

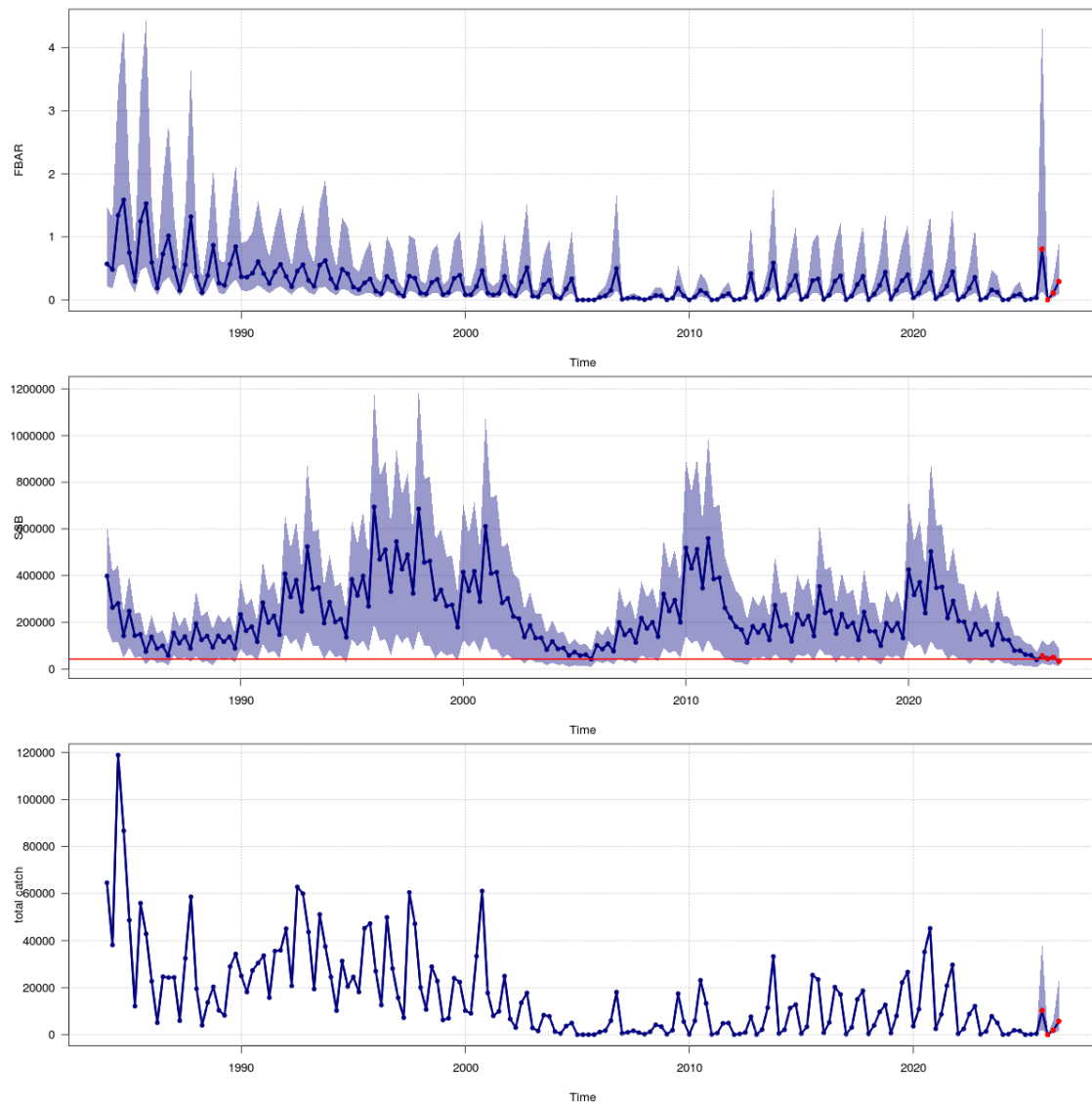


Figure 10.6.7. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to 0.3 ($F_{cap} = 0.3$) for the period 1st October 2025 to 1st October 2026.

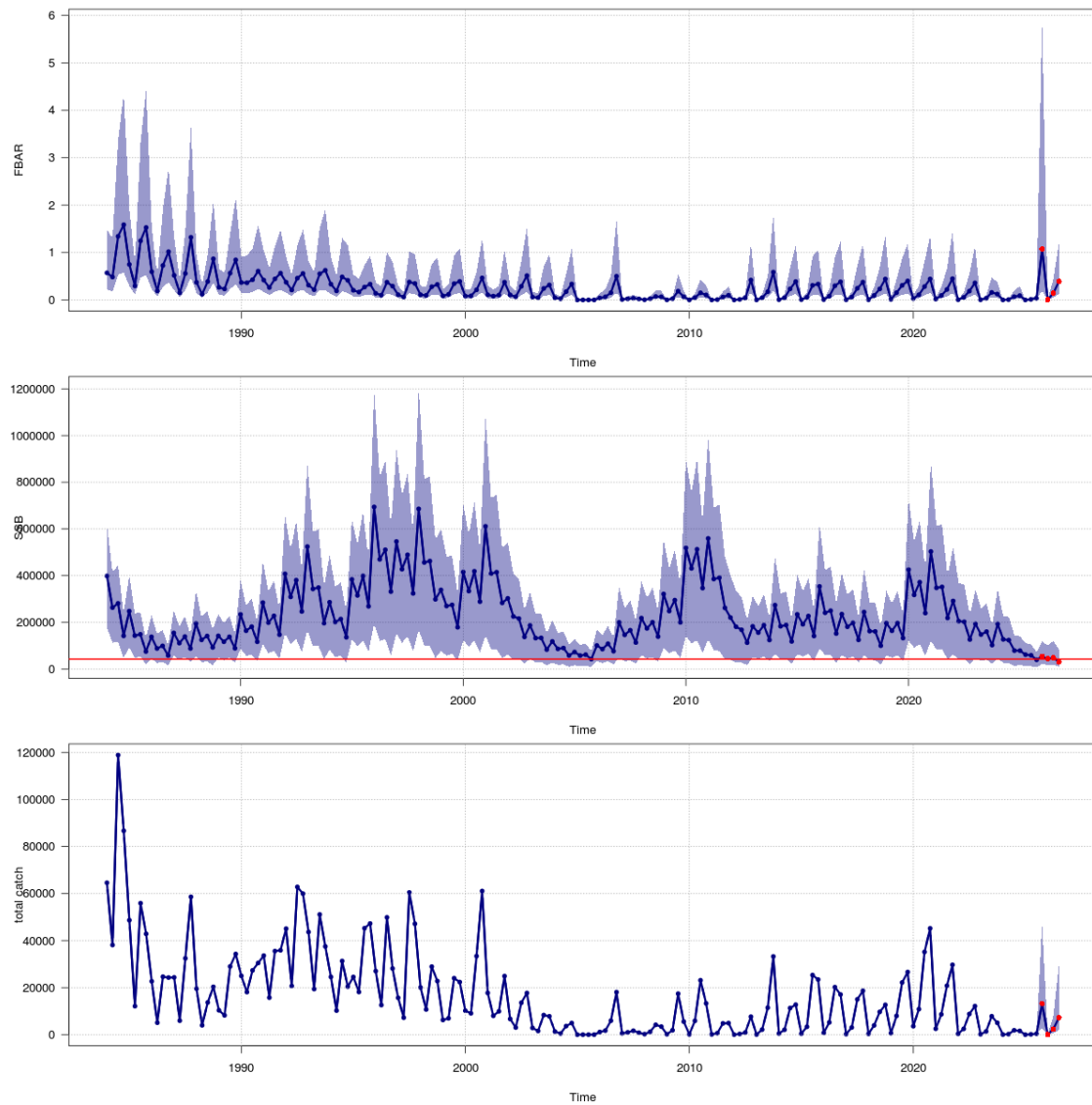


Figure 10.6.8. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to 0.4 ($F_{cap} = 0.4$) for the period 1st October 2025 to 1st October 2026.

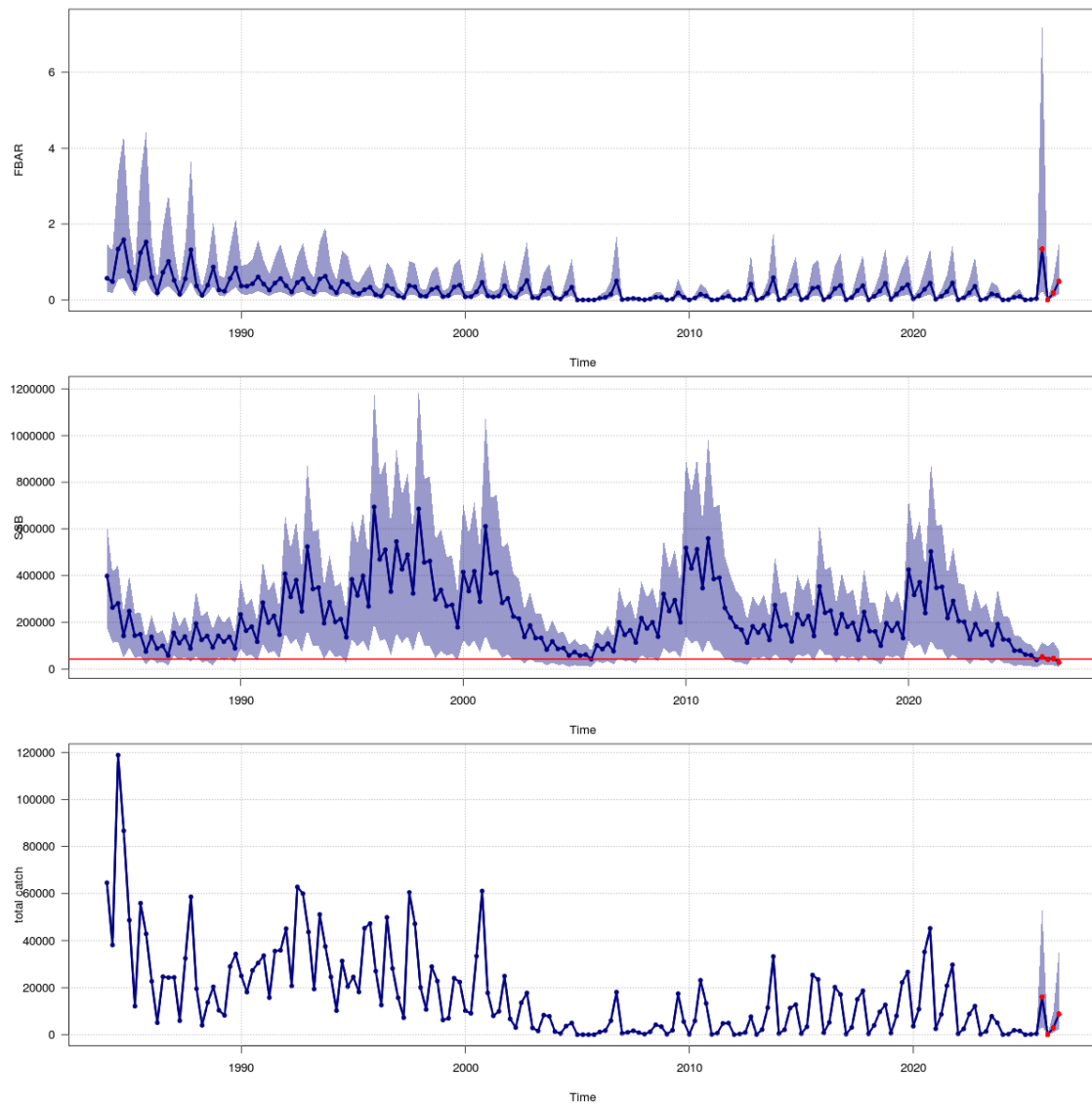


Figure 10.6.9. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to 0.5 ($F_{cap} = 0.5$) for the period 1st October 2025 to 1st October 2026.

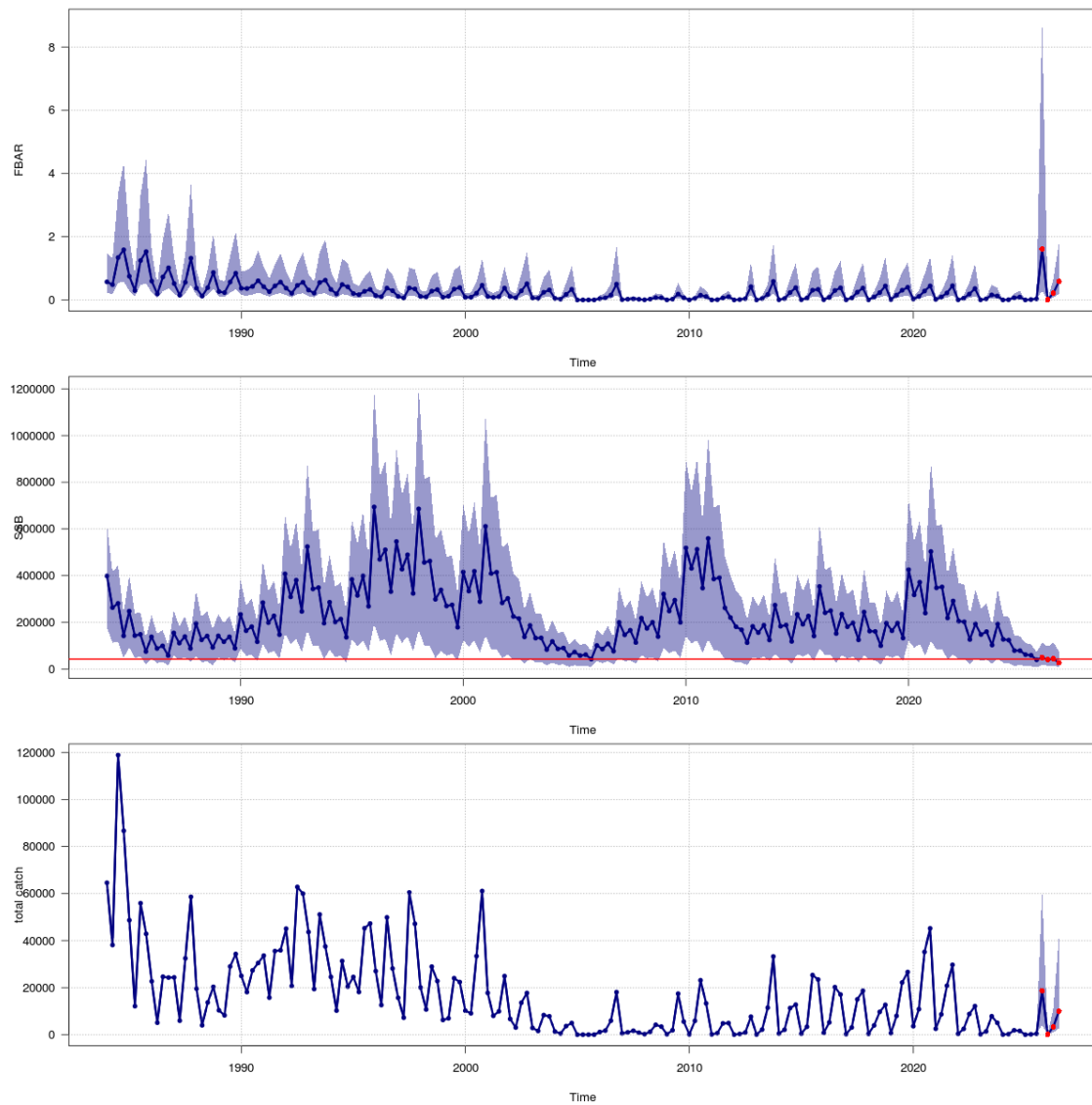


Figure 10.6.10. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to 0.6 ($F_{cap} = 0.6$) for the period 1st October 2025 to 1st October 2026.

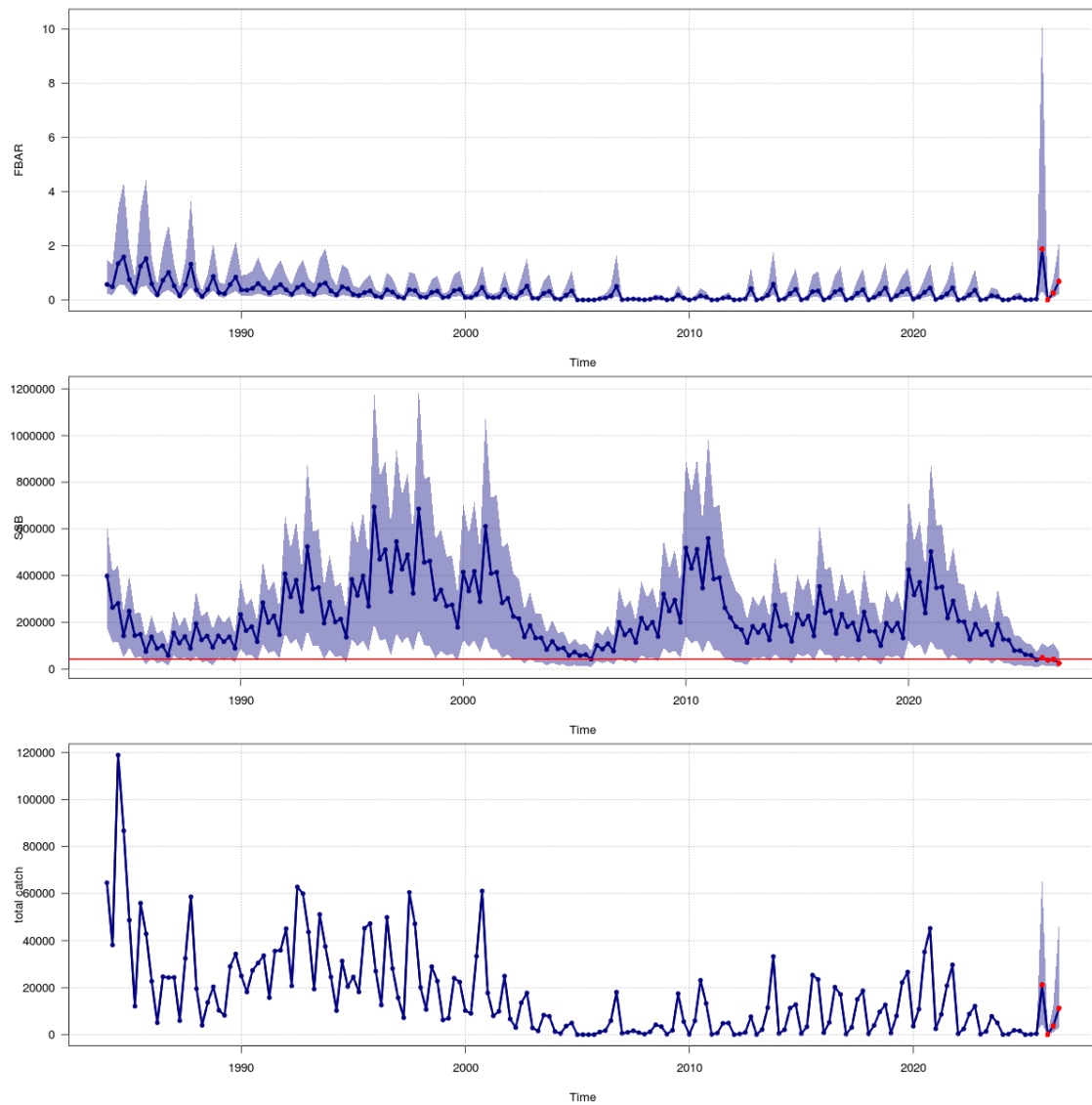


Figure 10.6.11. Norway pout in Subarea 4 and Division 3.a. Forecast of fishing mortality, SSB and median catch (t) with F scaled to 0.7 ($F_{cap} = 0.7$) for the period 1st October 2025 to 1st October 2026.

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